

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	20
Flg Calc [Int P]: Ch Cover	21
Flg Calc [Int P]: Ch.flange	24
Flg Calc [Int P]: Girth flg-1	33
Flg Calc [Int P]: girth flg.-2	42
Internal Pressure Calculations:	51
Element: Ch.shell	52
Element: Shell	54
Element: Shell -2	56
Element: head- front	58
Shop Pressure Test Results:	60
Field Pressure Test Results:	62
Ring MDMT Calcs:	64
ASME TS Calc: Pressure Test	65
ASME Fl-TS Calc: Pressure Test	75
Floating Head: Operating	83
External Pressure Calculations:	88
Element and Detail Weights:	89
Nozzle Flange MAWP:	92
Wind Load Calculation:	93
Earthquake Load Calculation:	95
Center of Gravity Calculation:	97
Saddle Calcs: Road Transport:	98
Saddle Calcs: Operating Case:	112
Saddle Calcs: Test Case:	127
Nozzle Calcs.: Tube inlet	141
Nozzle Calcs.: Tube outlet	152
Nozzle Calcs.: Shell inlet	163
Nozzle Calcs.: N4- Shell outle	174
Nozzle Calcs.: Drain	185
ASME TS Calc: DESIGN CASE	188
ASME Fl-TS Calc: DESIGN CASE	201
Floating Head: Operating	210



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\114-C-Pass.02-AK.pvdb

Date of Analysis : Sep 9, 2026 9:22am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 2 of 214

Vessel Design Summary:

Step: 33

9:23am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2320.000 mm ID
Vessel Design Length, Tangent to Tangent 24463.00 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 454 °C
Channel Side Design Temperature 454 °C
Shell Side Design Pressure 4550.000 KPa.
Channel Side Design Pressure 4550.000 KPa.
Shop Shell Side Test Pressure 7739.686 KPa.
Field Shell Side Test Pressure 5915.000 KPa.
Shop Channel Side Test Pressure 7146.339 KPa.
Field Channel Side Test Pressure 5915.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 22	2	...	K21590	
Shell	SA-336 F22	3	...	K21590	
Head	SA-336 F22	3	...	K21590	
Flange	SA-336 F22	3	...	K21590	
Nozzle	SA-336 F22	3	...	K21590	
Nozzle Flg	SA-182 F22	3	...	K21590	
Nozzle Flg	SA-182 F22	1	...	K21590	
Tubes	SA-213 T22	...	...	K21590	
Tubesheet	SA-336 F22	3	...	K21590	
Flo. Head	SA-387 22	2	...	K21590	
Bk. Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch. cover	4550.000	0.00	4731.000	3.0000	N/A	No
Ch.flange	4574.172	0.00	4808.000	3.0000	N/A	No
Ch.shell	4571.790	0.00	4810.000	3.0000	N/A	No
Shell	4571.790	0.00	4810.000	3.0000	N/A	No
Shell girth fln.-1	4550.000	0.00	4832.000	3.0000	N/A	No
Shell girth fln.-2	4550.000	0.00	4832.000	3.0000	N/A	No
Shell -2	4570.320	0.00	4811.000	3.0000	N/A	No
head- front	4572.280	0.00	4876.000	3.0000	No	No

Liquid Level: 2466.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 3 of 214

Vessel Design Summary:

Step: 33

9:23am

Sep 9,2026

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	299.5	295.0	303.0	295.0	289.4	...	1.00	1.00
Body Flg	754.5	455.0	363.0	355.0	348.1	...	1.00	1.00
Cylinder	3254.5	2500.0	48.0	48.0	45.5	...	1.00	1.00
Cylinder	22514.5	18920.0	48.0	48.0	45.5	...	1.00	1.00
Body Flg	22966.0	447.0	365.0	357.0	353.2	...	1.00	1.00
Body Flg	23413.0	447.0	365.0	357.0	353.2	...	1.00	1.00
Cylinder	24413.0	1000.0	48.0	48.0	45.5	...	1.00	1.00
Ellipse	24463.0	50.0	56.0	47.6	44.6	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	0.931	Shell	Passed
Tube outlet	WRC-297	0.939	Shell	Passed
Shell inlet	WRC-297	0.931	Shell	Passed
N4- Shell outl	WRC-297	0.939	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	735.000	mm
Saddle Bearing Angle	140.000	deg.
Centerline Dimension	1850.000	mm
Wear Pad Width	1110.000	mm
Wear Pad Thickness	16.000	mm
Wear Pad Bearing Angle	152.000	deg.
Distance from Saddle to Tangent	4000.000	mm
Baseplate Length	2350.000	mm
Baseplate Thickness	60.000	mm
Baseplate Width	760.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	16.000	mm
Web Thickness	16.000	mm
Height of Center Web	720.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

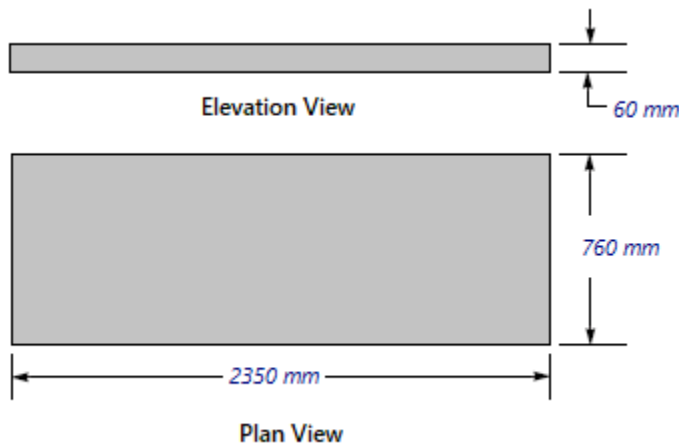
Page 4 of 214

Vessel Design Summary:

Step: 33

9:23am

Sep 9, 2026

**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	217079.16	Kgf
Maximum Transverse Saddle Shear Load	13033.32	Kgf
Maximum Longitudinal Saddle Shear Load	19257.84	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	184367.44	Kgf
Maximum Transverse Saddle Shear Load	893.53	Kgf
Maximum Longitudinal Saddle Shear Load	283.64	Kgf

Weights:

Fabricated - Bare w/o removable internals	232008.1	kg.
Shop Test - Fabricated + water (full)	324784.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	232008.1	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	233108.1	kg.
Empty - Fab. + intls. + details + wghts.	243142.5	kg.
Operating - Empty + operating liquid (no ca)	331896.8	kg.
Field Test - Empty weight + test medium (full)	334818.9	kg.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 5 of 214

Nozzle Summary:

Step: 31 9:23am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube inlet	4674	...	...	OK	13.57	OK	Passed
Tube outlet	4757	...	...	OK	17.04	OK	Passed
Shell inlet	4779	...	...	OK	16.97	OK	Passed
N4- Shell outle	4810	...	...	OK	17.04	OK	Passed
Drain	4810	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4779 Nozzle : Shell inlet [Shellside]
 Minimum MAWP Nozzles : 4674 Nozzle : Tube inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	Tube inlet	1754.500	90.000	1161.401
30	Tube outlet	1754.500	270.000	1536.000
40	Shell inlet	5094.500	90.000	1536.000
40	N4- Shell outle	20594.498	270.000	1536.000
40	Drain	4594.500	270.000	150.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 6 of 214

Nozzle Schedule:

Step: 30 9:23am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Drain	2.000 in	160	WNF	2.375	8.738	...	...	300.20	...
Tube inlet	24.000 in	40	WNF	24.000	17.450	...	...	533.60	600
Tube outlet	30.000 in Actual		None	31.624	20.620	...	...	692.68	...
Shell inlet	30.000 in Actual		None	31.624	20.620	...	...	692.68	...
N4- Shell outle	30.000 in Actual		None	31.624	20.620	...	...	692.68	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Drain	SA-336 F22	42.000	20.000	...	...	...
Tube inlet	SA-336 F22	42.000	15.000	...	...	...
Tube outlet	SA-336 F22	42.000	20.000	...	...	...
Shell inlet	SA-336 F22	42.000	20.000	...	...	...
N4- Shell o	SA-336 F22	42.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Drain	4594.499	270.0	250.00	0.00	Shell
Tube inlet	1754.500	90.0	400.00	0.00	Ch.shell
Tube outlet	1754.500	270.0	500.00	0.00	Ch.shell
Shell inlet	5094.500	90.0	500.00	0.00	Shell
N4- Shell outle	20594.498	270.0	500.00	0.00	Shell

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 7 of 214

Bill of Material:

Step: 29 9:23am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 303.0mm THK X 2880.0mm OD	SA-336 F22
1	WELD NECK FLANGE: 363.0mm THK X 2880.0mm OD	SA-336 F22
1	CYLINDER: 48.0mm THK X 2320.0mm ID X 2500.0mm	SA-387 22
2	WELD NECK FLANGE: 365.0mm THK X 2880.0mm OD	SA-336 F22
1	CYLINDER: 48.0mm THK X 2320.0mm ID X 1000.0mm	SA-336 F22
1	ELLIPTICAL HEAD: 2.0 X 1, 56.0mm THK X 2320.0mm ID X 50.0mm	SA-336 F22
1	INSULATION: 2500mm X 150mm THK	Hot Insulation
33	BAFFLES: 2150.0mm DIA X 10.0mm THK	SA-387 22
12	TIE RODS: 25mm DIA X 16000mm	SA-193 B7
1	TUBESHEET: 2416.0mm X 340.0mm THK	SA-336 F22
3250	TUBES: 19600.0mm X 25.4mm DIA X 2.1mm THK	SA-213 T22
1	FLOATING TUBESHEET: 2300.0mm X 340.0mm THK	SA-336 F22
36	FLOATING HEAD FLANGE BOLTS: 76.2mm DIA	SA-193 B7
72	NUTS FOR FLOATING HEAD FLANGE BOLTS: 76.2mm DIA	...
1	FLOATING HEAD FLANGE GASKET: 2046.0mm OD X 2020.0mm ID	...
2	SADDLE: 735.0mm X 140 DEG	SA-516 70
1	INSULATION: 18920mm X 150mm THK	Hot Insulation
1	INSULATION: 1000mm X 150mm THK	Hot Insulation
1	INSULATION: 617mm X 150mm THK	Hot Insulation
2	GASKET: 2419.0mm OD X 2349.0mm ID	...
72	BODY FLANGE BOLTS: 95.2mm DIA	SA-193 B7
144	NUTS FOR BODY FLANGE BOLTS: 95.2mm DIA	...
1	CYLINDRICAL SEGMENT 48.0mm THK X 2874.9mm X 18920.0mm	SA-387 22
1	CYLINDRICAL SEGMENT 48.0mm THK X 2854.6mm X 18920.0mm	SA-387 22
1	CYLINDRICAL SEGMENT 48.0mm THK X 1558.9mm X 18920.0mm	SA-387 22
1	NAMEPLATE	...



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 8 of 214

MDMT Summary:

Step: 32 9:23am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell girth f[11]	B	16	4		0.782	48.000	1.00	Yes
Bolting [21]		-40						
Shell girth f[11]	B	16	4		0.782	48.000	1.00	Yes
Shell [8]	A	35	26		0.832	48.000	1.00	Yes
Shell -2 [8]	B	16	-33		0.406	48.000	1.00	Yes
head- front [10]	B	16	6		0.824	47.600	1.00	Yes
head- front [7]	B	19	3		0.707	56.000	1.00	Yes
Left Saddle [24]	A	-11	-20		0.832	16.000	1.00	No
New Saddle [24]	A	-11	-20		0.832	16.000	1.00	No
Shell inlet [1]	A	35	25		0.810	48.000	1.00	No
N4- Shell outl[1]	A	35	25		0.813	48.000	1.00	No
Drain [1]	A	35	25		0.813	48.000	1.00	Yes
Tubesheet: SS[13]	B	28	17		0.801	85.000	1.00	No

Warmest MDMT: 35 26

Ch. cover [11]	B	25	15		0.819	73.750	1.00	No
Ch.flange [11]	B	16	3		0.766	48.000	1.00	Yes
Ch.shell [8]	A	35	25		0.815	48.000	1.00	Yes
Tube inlet [1]	A	35	25		0.810	48.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
Tube outlet [1]	A	35	25		0.813	48.000	1.00	No
Tubesheet: CS[14]	B	28	17		0.801	85.000	1.00	No
Bolting [21]		-40						

Warmest MDMT: 35 25

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	26.0	-12.0	Fail
Channel/Tube	25.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 9 of 214

MDMT Summary:

Step: 32 9:23am Sep 9, 2026

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 10 of 214

Warnings and Errors:

Step: 0 9:22am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 11 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4550	KPa.
Channel Side Design Pressure	4550	KPa.
Shell Side Design Temperature	454.0	°C
Channel Side Design Temperature	454.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 12 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 13 of 214

Input Echo:

Step: 1 9:22am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Ch. cover
Distance "FROM" to "TO"	295 mm
Flange Outside Diameter	2880 mm
Element Thickness	295 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	303 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	127.78 N./mm ²
Allowable Stress, Hydrotest	192.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	230.89 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Ch.flange
Distance "FROM" to "TO"	455 mm
Flange Inside Diameter	2320 mm
Element Thickness	355 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 14 of 214

Input Echo:

Step: 1 9:22am Sep 9, 2026

Internal Corrosion Allowance	3	mm
Nominal Thickness	363	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	454	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2466	mm
Liquid Density	0.0009984	kg./cm ³

Element#	3/8	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	Ch.shell	
Distance "FROM" to "TO"	2500	mm
Inside Diameter	2320	mm
Element Thickness	48	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	48	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	454	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Allowable Stress, Ambient	148	N./mm ²
Allowable Stress, Operating	127.78	N./mm ²
Allowable Stress, Hydrotest	192.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	230.89	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2220	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 15 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Tube inlet	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	24	in.
Nozzle Schedule	40	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	829.04	Kgf
Grade of Attached Flange	GR 1.10	
Nozzle Matl	SA-336 F22	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Tube outlet	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	809.04	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element#	4/8	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	18920	mm
Inside Diameter	2320	mm
Element Thickness	48	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	48	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	454	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	4000	mm
Width of Saddle	735	mm
Height of Saddle at Bottom	1850	mm
Saddle Contact Angle	140.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	1110	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.0	
Moment Factor	3.0	



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 16 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	15080	mm
Width of Saddle	735	mm
Height of Saddle at Bottom	1850	mm
Saddle Contact Angle	140.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	1110	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.3000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2220	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	18920	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell inlet	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1661.5	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N4- Shell outle	
Dist. from "FROM" Node / Offset dist	17000	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	901.14	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element From Node	40	
-------------------	----	--



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 17 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Detail Type	Nozzle
Detail ID	Drain
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F22

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	4000 mm
Miscellaneous Weight	550 Kgf
Offset from Element Centerline	1500 mm

Element From Node	40
Detail Type	Weight
Detail ID	Wght:[2 of 2]
Dist. from "FROM" Node / Offset dist	14800 mm
Miscellaneous Weight	550 Kgf
Offset from Element Centerline	1500 mm

Element#	5/8
Element From Node	50
Element To Node	60
Element Type	Flange
Description	Shell girth fln.-1
Distance "FROM" to "TO"	447 mm
Flange Inside Diameter	2320 mm
Element Thickness	357 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	365 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	127.78 N./mm ²
Allowable Stress, Hydrotest	192.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	230.89 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	6/8
Element From Node	60
Element To Node	70
Element Type	Flange



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 18 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Description	Shell girth fln.-2
Distance "FROM" to "TO"	447 mm
Flange Inside Diameter	2320 mm
Element Thickness	357 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	365 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	7/8
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell -2
Distance "FROM" to "TO"	1000 mm
Inside Diameter	2320 mm
Element Thickness	48 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2070 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm³

Element#	8/8
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	head- front
Distance "FROM" to "TO"	50 mm
Inside Diameter	2320 mm
Element Thickness	47.6 mm
Internal Corrosion Allowance	3 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 19 of 214

Input Echo:

Step: 1 9:22am Sep 9,2026

Nominal Thickness	56	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	454	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336	F22
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2270	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	617.5	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 20 of 214

XY Coordinate Calculations:

Step: 2 9:22am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Ch. cover		299.5	...	295	...
Ch.flange		754.5	...	455	...
Ch.shell		3254.5	...	2500	...
Shell		22514.5	...	18920	...
Shell girth fln		22966	...	447	...
Shell girth fln		23413	...	447	...
Shell -2		24413	...	1000	...
head- front		24463	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 21 of 214

Flg Calc [Int P]: Ch Cover

Flng: 9 9:22am Sep 9,2026

Flange Input Data Values

Description: Ch Cover :

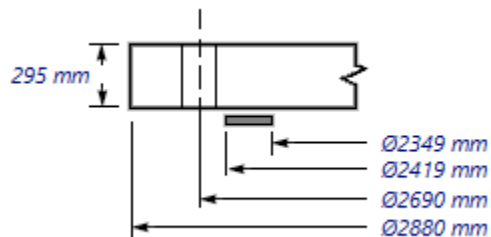
Ch. cover

Flange Type		Blind
Design Pressure	P	4550.00 KPa.
Design Temperature		454 °C
Internal Corrosion Allowance	ci	3.0000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	2880.000 mm
Flange Thickness	t	295.0000 mm
Flange Material		SA-336 F22
Flange Material UNS number		K21590
Flange Allowable Stress At Temperature	Sfo	127.78 N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	113.19 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	8450.884 mm
Diameter of Bolt Circle	C	2690.000 mm
Nominal Bolt Diameter	a	95.2500 mm
Type of Threads	TEMA Thread Series	
Number of Bolts		36
Flange Face Outside Diameter	Fod	2430.000 mm
Flange Face Inside Diameter	Fid	2320.000 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	2419.000 mm
Gasket Inside Diameter	Gi	2349.000 mm
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	4.5000 mm
Raised face depth	Fd	0.0000 mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	35.000 mm
Basic Gasket Width,	$bo = N/2$	17.500 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 22 of 214

Flg Calc [Int P]: Ch Cover

Flng: 9 9:22am Sep 9,2026

Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 10.542 mm
Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 2397.917 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2397.9167^2 \cdot 4550.0$$

$$= 2095395.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.5416 \cdot \pi \cdot 2397.9167 \cdot 3.0 \cdot 4550.0$$

$$= 221079.766 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2095396 + 221080 + 0, 0)$$

$$= 2316475.500 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 10.5416 \cdot \pi \cdot 2397.917 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 558335.750 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((2316476 + 0 + 4 \cdot 0 / 2397.92) / 113, 558336 / 159)$$

$$= 2006.931 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	2006.931	2347.660	
Radial Distance between Bolts and Edge:	88.900	95.000	
Circ. Spacing between the Bolts:	193.675	234.449	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 2347.66 \cdot 159.0 / (68.95 \cdot \pi (2419.0 + 2349.0))$$

$$= 36.141 \text{ mm} [\text{Note: Exceeds actual gasket width, 35.000 }]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 159.0 \cdot 2347.6599$$

$$= 3806236.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2316475.50 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2690.0 - 2397.9167) / 2$$

$$= 146.0416 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2316476 \cdot 146.0416 \cdot 1.0 / (292.0^2 \cdot 2397.9167) + 0.3 \cdot 1.0 \cdot 4550.0 \cdot 2397.9167^2 / 292.0^2$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 23 of 214

Flg Calc [Int P]: Ch Cover

Flng: 9 9:22am

Sep 9, 2026

= 122.89 N./mm²

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

= $1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$

= $1.9 \cdot 3806237 \cdot 146.0416 \cdot 1.0 / (292.0^2 \cdot 2397.9167)$

= 50.66 N./mm²

Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$

= $2316476 / 2347.6599$

= 96.77 N./mm²

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= $(558336 / 2347.6599)$

= 23.32 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	122.89	127.78	50.66	148.00
Bolting	96.77	113.19	23.32	159.00

Reqd. Blind Flange Thickness at Center 289.358 mm

Reqd. Blind Flange Thickness at Gasket 173.838 mm

Estimated M.A.W.P. (Operating) 4731 KPa.

Estimated Finished Weight of Flange at given Thk. 14894.5 kg.

Estimated Unfinished Weight of Forging at given Thk 14894.5 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.819, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C

Min Metal Temp. at Required thickness (UCS 66.1) 15 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	120.40	148.00	49.63	148.00
Bolting	96.77	159.00	23.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 266.090 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 24 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9,2026

Flange Input Data Values

Description: Ch.flange :

Ch.flange

Flange Type	Integral Weld Neck		
Design Pressure	P	4574.17	KPa.
Design Temperature		454	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2320.000	mm
Flange Outside Diameter	A	2880.000	mm
Flange Thickness	t	355.0000	mm
Thickness of Hub at Small End	go	48.0000	mm
Thickness of Hub at Large End	gl	72.0000	mm
Length of Hub	h	100.0000	mm
Flange Material	SA-336 F22		
Flange Material UNS number	K21590		
Flange Allowable Stress At Temperature	Sfo	127.78	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	113.19	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2690.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2430.000	mm
Flange Face Inside Diameter	Fid	2320.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	2419.000	mm
Gasket Inside Diameter	Gi	2349.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

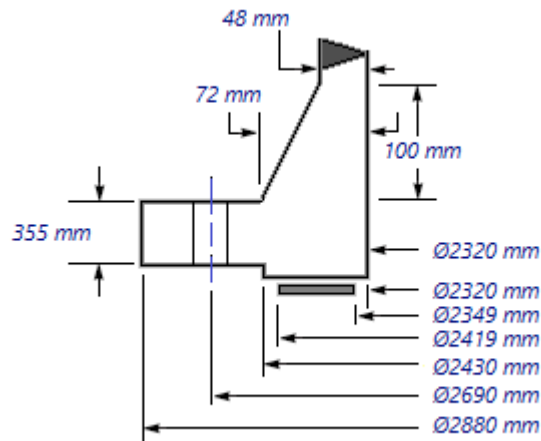
Page 25 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4574.17 \cdot (2320.0/2 + 3.0)) / (127.78 \cdot 1.0 - 0.6 \cdot 4574.17) + Ca$$

$$= 45.5489 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0)$$

$$= 4831.691 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0)$$

$$= 5975.319 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	352.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2326.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	69.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	45.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	113.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	35.000 mm
Basic Gasket Width, $b_o = N/2$	17.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	10.542 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2397.917 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2397.9167^2 \cdot 4574.172$$

$$= 2106527.500 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.5416 \cdot \pi \cdot 2397.9167 \cdot 3.0 \cdot 4574.17$$

$$= 222254.250 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2326.0^2 \cdot 4574.1719$$

$$= 1982067.125 \text{ Kg}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 26 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2106528 - 1982067$$

$$= 124460.352 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(2106528 + 222254 + 0, 0)$$

$$= 2328781.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 10.5416 \cdot \pi \cdot 2397.917 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 558335.750 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2328782 + 0 + 4 \cdot 0 / 2397.92) / 113, 558336 / 159)$$

$$= 2017.592 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 95.25 + 6 \cdot 352.0 / (3.0 + 0.5)$$

$$= 793.929 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2690.0 \cdot \sin(\pi / 36)$$

$$= 234.449 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 234.449 / (2 \cdot 95.25 + 352.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	2017.592	2347.660	
Radial Distance between Hub and Bolts:	112.713	113.000	
Radial Distance between Bolts and Edge:	88.900	95.000	
Circ. Spacing between the Bolts:	193.675	234.449	793.929

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2347.66 \cdot 159.0 / (68.95 \cdot \pi (2419.0 + 2349.0))$$

$$= 36.141 \text{ mm} [\text{Note: Exceeds actual gasket width, 35.000 }]$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 2347.6599$$

$$= 3806236.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 2328782 - 2106528$$

$$= 222254.25 \text{ Kg}$$

Moment Arm Calculations:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 27 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2690.0 - 2397.9167) / 2$$

$$= 146.0416 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2690.0 - 2326.0) / 2 + 146.0416)$$

$$= 164.0208 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2690.0 - 2326.0 - 69.0) / 2$$

$$= 147.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1982067.	147.4999	1.0000	292361.
Face Pressure, Mt	124460.	164.0208	1.0000	20414.
Gasket Load, Mg	222254.	146.0416	1.0000	32459.
Gasket Seating, Matm	3806236.	146.0416	1.0000	555880.

Total Moment for Operation, Mo 345234. Kg-m.

Total Moment for Gasket seating, Mg 555880. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 323.527 mm

Hub Ratio, h/h0 = HL / H0 0.309

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.533

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.879

Factor V per 4.16.5 0.361

Factor f per 4.16.5 1.196

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2880.0 / 2326.0$$

$$= 1.238$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.238^2 \cdot \log_{10}(1.238))) / (1.238^2 - 1)) / (1.238 - 1)$$

$$= 9.211$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.238^2 (1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / ((1.0472 + 1.9448 \cdot 1.238^2) \cdot (1.238 - 1))$$

$$= 1.823$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.238^2 (1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / (1.36136 (1.238^2 - 1) \cdot (1.238 - 1))$$

$$= 10.122$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.238^2 + 1) / (1.238^2 - 1)$$

$$= 4.752$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 10.122 \cdot 45.0^2 \cdot 323.527 / 0.361$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 28 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026

$$= 18379178.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.879/323.527$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (352.0 \cdot 0.003 + 1) / 1.823 + 352.0^3/18379178$$

$$= 3.446$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.1957 \cdot 345234/2326.0) / (3.4458 \cdot 69.0^2)$$

$$= 106.09 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.1957 \cdot 555880/2326.0) / (3.4458 \cdot 69.0^2)$$

$$= 170.82 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2747 \cdot 345234/2326.0) / (3.4458 \cdot 352.0^2)$$

$$= 7.75 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.2747 \cdot 555880/2326.0) / (3.4458 \cdot 352.0^2)$$

$$= 12.49 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.2112 \cdot 345234 / (352.0^2 \cdot 2326.0)) - 4.7518 \cdot 8$$

$$= 71.36 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.2112 \cdot 555880 / (352.0^2 \cdot 2326.0)) - 4.7518 \cdot 12$$

$$= 114.90 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (106 + \max(8, 71)) / 2$$

$$= 88.72 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (171 + \max(12, 115)) / 2$$

$$= 142.86 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 2328782/2347.6599$$

$$= 97.28 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (558336/2347.6599)$$

$$= 23.32 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 29 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9,2026

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	106.09	191.67	170.82	222.00
Radial Flange	7.75	127.78	12.49	148.00
Tangential Flange	71.36	127.78	114.90	148.00
Maximum Average	88.72	127.78	142.86	148.00
Bolting	97.28	113.19	23.32	159.00

Minimum Required Flange Thickness 348.082 mm

Estimated M.A.W.P. (Operating) 4832 KPa.

Estimated Finished Weight of Flange at given Thk. 6640.4 kg.

Estimated Unfinished Weight of Forging at given Thk 8065.3 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	522.06	527.74	0.989	522.06	619.99	0.842
Tangential Flange	St	352.87	395.81	0.892	352.87	464.99	0.759
Tangential Flange	Sto	151.41	263.87	0.574	151.41	309.99	0.488
Combined	Sh/f + St	791.61	791.61	1.000	791.61	929.98	0.851
Combined	Sr + St	391.22	791.61	0.494	391.22	929.98	0.421
Combined	Sh + Sto	673.47	791.61	0.851	673.47	929.98	0.724
Radial Flange	Sr	38.35	527.74	0.073	38.35	619.99	0.062
Maximum Stress Ratio				1.000			0.851



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 30 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10 9:22am Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.4986/1.2594)

= 1.1899

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4143/1.2005) if $h/h0 < 0.35$, else max (0.25; 0.2756/0.7666)

= 0.3451

In this case, the hub ratio $h/h0$, 0.3091 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	488.3 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 31 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026

Root Area of a Single Bolt	Abs	65.2128 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.9 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.563 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2419.0^2 - 2349.0^2)$$

$$= 2621.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 2621.345 / (36 \cdot 65.213)$$

$$= 232.2 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(232.248; 507.5)$$

$$= 232.2 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(232.248; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 488.304)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \geq 156.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 2621.345 + \pi/4 \cdot 4574.172 \cdot 2349.0^2) / (0.7 \cdot 65.213 \cdot 36)$$

$$:: 290.0 \geq 275.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 276.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \leq 308.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 290.0 \leq 488.304(1.0/0.563)$$

$$:: 290.0 \leq 866.6 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.26 \cdot 65.213 \cdot 95.25$$

$$= 4775.8 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 75 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 113 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 32 of 214

Flg Calc [Int P]: Ch.flange

Flng: 10

9:22am

Sep 9, 2026

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.361 / (3.298 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.749 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 345234.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (3.298 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.549 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.361 / (3.446 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.717 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 345234.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (3.446 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.525 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.766, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	103.58	222.00	165.19	222.00
Radial Flange	8.17	148.00	13.04	148.00
Tangential Flange	68.05	148.00	108.52	148.00
Maximum Average	85.81	148.00	136.85	148.00
Bolting	97.28	159.00	23.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 339.141 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 33 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9,2026

Flange Input Data Values

Description: Girth flg-1 :

Shell girth fln.-1

Flange Type	Integral Weld Neck		
Design Pressure	P	4550.00	KPa.
Design Temperature		454	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2320.000	mm
Flange Outside Diameter	A	2880.000	mm
Flange Thickness	t	357.0000	mm
Thickness of Hub at Small End	go	48.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	90.0000	mm
Flange Material	SA-336 F22		
Flange Material UNS number	K21590		
Flange Allowable Stress At Temperature	Sfo	127.78	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	113.19	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2690.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2430.000	mm
Flange Face Inside Diameter	Fid	2320.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	2419.000	mm
Gasket Inside Diameter	Gi	2349.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

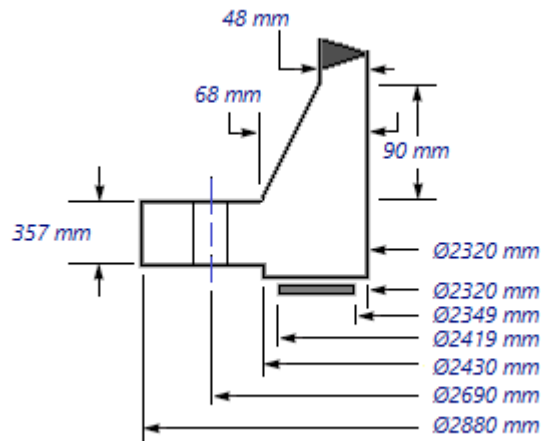
Page 34 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4550.0 \cdot (2320.0/2 + 3.0)) / (127.78 \cdot 1.0 - 0.6 \cdot 4550.0) + C_a$$

$$= 45.3192 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0)$$

$$= 4831.691 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0)$$

$$= 5975.319 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	354.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2326.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	65.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	45.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	117.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	35.000 mm
Basic Gasket Width, $b_o = N/2$	17.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	10.542 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2397.917 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2397.9167^2 \cdot 4550.0$$

$$= 2095395.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.5416 \cdot \pi \cdot 2397.9167 \cdot 3.0 \cdot 4550.0$$

$$= 221079.766 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2326.0^2 \cdot 4550.0$$

$$= 1971593.000 \text{ Kg}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 35 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2095396 - 1971593$$

$$= 123802.633 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(2095396 + 221080 + 0, 0)$$

$$= 2316475.500 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 10.5416 \cdot \pi \cdot 2397.917 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 558335.750 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2316476 + 0 + 4 \cdot 0 / 2397.92) / 113, 558336 / 159)$$

$$= 2006.931 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 95.25 + 6 \cdot 354.0 / (3.0 + 0.5)$$

$$= 797.357 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2690.0 \cdot \sin(\pi / 36)$$

$$= 234.449 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 234.449 / (2 \cdot 95.25 + 354.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	2006.931	2347.660	
Radial Distance between Hub and Bolts:	112.713	117.000	
Radial Distance between Bolts and Edge:	88.900	95.000	
Circ. Spacing between the Bolts:	193.675	234.449	797.357

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2347.66 \cdot 159.0 / (68.95 \cdot \pi (2419.0 + 2349.0))$$

$$= 36.141 \text{ mm} [\text{Note: Exceeds actual gasket width, 35.000 }]$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 2347.6599$$

$$= 3806236.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 2316476 - 2095396$$

$$= 221079.88 \text{ Kg}$$

Moment Arm Calculations:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 36 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2690.0 - 2397.9167) / 2$$

$$= 146.0416 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2690.0 - 2326.0) / 2 + 146.0416)$$

$$= 164.0208 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2690.0 - 2326.0 - 65.0) / 2$$

$$= 149.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1971593.	149.4999	1.0000	294759.
Face Pressure, Mt	123803.	164.0208	1.0000	20307.
Gasket Load, Mg	221080.	146.0416	1.0000	32287.
Gasket Seating, Matm	3806236.	146.0416	1.0000	555880.

Total Moment for Operation, Mo 347353. Kg-m.
Total Moment for Gasket seating, Mg 555880. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 323.527 mm
Hub Ratio, h/h0 = HL / H0 0.278
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.444

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.888
Factor V per 4.16.5 0.389
Factor f per 4.16.5 1.151

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2880.0 / 2326.0$$

$$= 1.238$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1)) / (K - 1))$$

$$= (0.66845 + (5.7169(1.238^2 \cdot \log_{10}(1.238)) / (1.238^2 - 1)) / (1.238 - 1))$$

$$= 9.211$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.238^2(1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / ((1.0472 + 1.9448 \cdot 1.238^2) \cdot (1.238 - 1))$$

$$= 1.823$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.238^2(1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / (1.36136(1.238^2 - 1) \cdot (1.238 - 1))$$

$$= 10.122$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.238^2 + 1) / (1.238^2 - 1)$$

$$= 4.752$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 10.122 \cdot 45.0^2 \cdot 323.527 / 0.389$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 37 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

$$= 17031036.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.888/323.527$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (354.0 \cdot 0.003 + 1) / 1.823 + 354.0^3/17031036$$

$$= 3.686$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.151 \cdot 347353/2326.0) / (3.6859 \cdot 65.0^2)$$

$$= 108.24 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.151 \cdot 555880/2326.0) / (3.6859 \cdot 65.0^2)$$

$$= 173.22 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (2.2951 \cdot 347353/2326.0) / (3.6859 \cdot 354.0^2)$$

$$= 7.28 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (2.2951 \cdot 555880/2326.0) / (3.6859 \cdot 354.0^2)$$

$$= 11.65 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (9.2112 \cdot 347353 / (354.0^2 \cdot 2326.0)) - 4.7518 \cdot 7$$

$$= 73.07 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (9.2112 \cdot 555880 / (354.0^2 \cdot 2326.0)) - 4.7518 \cdot 12$$

$$= 116.93 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (108 + \max(7, 73)) / 2$$

$$= 90.65 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (173 + \max(12, 117)) / 2$$

$$= 145.08 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 2316476/2347.6599$$

$$= 96.77 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (558336/2347.6599)$$

$$= 23.32 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 38 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9,2026

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	108.24	191.67	173.22	222.00
Radial Flange	7.28	127.78	11.65	148.00
Tangential Flange	73.07	127.78	116.93	148.00
Maximum Average	90.65	127.78	145.08	148.00
Bolting	96.77	113.19	23.32	159.00

Minimum Required Flange Thickness 353.212 mm

Estimated M.A.W.P. (Operating) 4832 KPa.

Estimated Finished Weight of Flange at given Thk. 6630.4 kg.

Estimated Unfinished Weight of Forging at given Thk 7923.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	506.63	527.74	0.960	506.63	619.99	0.817
Tangential Flange	St	346.13	395.81	0.875	346.13	464.99	0.744
Tangential Flange	Sto	134.88	263.87	0.511	134.88	309.99	0.435
Combined	Sh/f + St	791.61	791.61	1.000	791.61	929.98	0.851
Combined	Sr + St	380.60	791.61	0.481	380.60	929.98	0.409
Combined	Sh + Sto	641.51	791.61	0.810	641.51	929.98	0.690
Radial Flange	Sr	34.47	527.74	0.065	34.47	619.99	0.056
Maximum Stress Ratio				1.000			0.851



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 39 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.3798/1.2133)

= 1.1373

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3471/1.1463) if $h/h0 < 0.35$, else max (0.25; 0.252/0.7756)

= 0.3028

In this case, the hub ratio $h/h0$, 0.2782 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	470.6 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 40 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

Root Area of a Single Bolt	Abs	65.2128 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.9 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.548 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2419.0^2 - 2349.0^2)$$

$$= 2621.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 2621.345 / (36 \cdot 65.213)$$

$$= 232.2 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(232.248; 507.5)$$

$$= 232.2 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(232.248; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 470.648)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \geq 156.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 2621.345 + \pi/4 \cdot 4550.0 \cdot 2349.0^2) / (0.7 \cdot 65.213 \cdot 36)$$

$$:: 290.0 \geq 274.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 275.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \leq 307.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetaamax})$$

$$:: 290.0 \leq 470.648(1.0/0.548)$$

$$:: 290.0 \leq 859.0 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.26 \cdot 65.213 \cdot 95.25$$

$$= 4775.8 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 75 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 113 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 41 of 214

Flg Calc [Int P]: Girth flg-1

Flng: 11

9:22am

Sep 9, 2026

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.597 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.742 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347352.9 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.597 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.546 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.686 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.724 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347352.9 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.686 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$$

$$= 0.533 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.782, Temperature Reduction per Fig. UCS 66.1 = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	105.89	222.00	167.88	222.00
Radial Flange	7.70	148.00	12.21	148.00
Tangential Flange	69.70	148.00	110.49	148.00
Maximum Average	87.80	148.00	139.19	148.00
Bolting	96.77	159.00	23.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 344.856 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 42 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9,2026

Flange Input Data Values

Description: girth flg.-2 :

Shell girth fln.-2

Flange Type	Integral Weld Neck		
Design Pressure	P	4550.00	KPa.
Design Temperature		454	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2320.000	mm
Flange Outside Diameter	A	2880.000	mm
Flange Thickness	t	357.0000	mm
Thickness of Hub at Small End	go	48.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	90.0000	mm
Flange Material		SA-336 F22	
Flange Material UNS number		K21590	
Flange Allowable Stress At Temperature	Sfo	127.78	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	113.19	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2690.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2430.000	mm
Flange Face Inside Diameter	Fid	2320.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2419.000	mm
Gasket Inside Diameter	Gi	2349.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

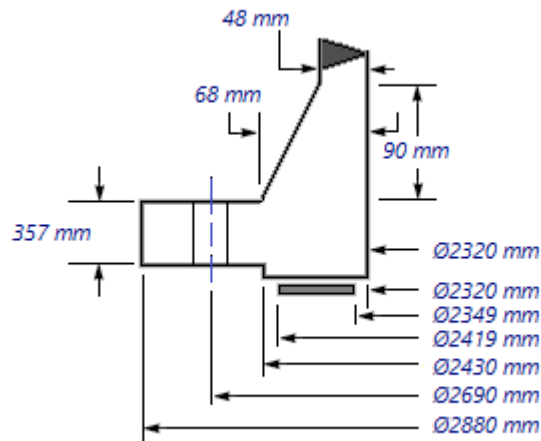
Page 43 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4550.0 \cdot (2320.0/2 + 3.0)) / (127.78 \cdot 1.0 - 0.6 \cdot 4550.0) + Ca$$

$$= 45.3192 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0)$$

$$= 4831.691 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0)$$

$$= 5975.319 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	354.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2326.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	65.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	45.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	117.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	35.000 mm
Basic Gasket Width, $bo = N/2$	17.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	10.542 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2397.917 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2397.9167^2 \cdot 4550.0$$

$$= 2095395.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.5416 \cdot \pi \cdot 2397.9167 \cdot 3.0 \cdot 4550.0$$

$$= 221079.766 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2326.0^2 \cdot 4550.0$$

$$= 1971593.000 \text{ Kg}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 44 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2095396 - 1971593$$

$$= 123802.633 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(2095396 + 221080 + 0, 0)$$

$$= 2316475.500 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 10.5416 \cdot \pi \cdot 2397.917 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 558335.750 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2316476 + 0 + 4 \cdot 0 / 2397.92) / 113, 558336 / 159)$$

$$= 2006.931 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 95.25 + 6 \cdot 354.0 / (3.0 + 0.5)$$

$$= 797.357 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2690.0 \cdot \sin(\pi / 36)$$

$$= 234.449 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 234.449 / (2 \cdot 95.25 + 354.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	2006.931	2347.660	
Radial Distance between Hub and Bolts:	112.713	117.000	
Radial Distance between Bolts and Edge:	88.900	95.000	
Circ. Spacing between the Bolts:	193.675	234.449	797.357

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2347.66 \cdot 159.0 / (68.95 \cdot \pi (2419.0 + 2349.0))$$

$$= 36.141 \text{ mm} [\text{Note: Exceeds actual gasket width, 35.000 }]$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 2347.6599$$

$$= 3806236.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 2316476 - 2095396$$

$$= 221079.88 \text{ Kg}$$

Moment Arm Calculations:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 45 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9, 2026

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2690.0 - 2397.9167) / 2$$

$$= 146.0416 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2690.0 - 2326.0) / 2 + 146.0416)$$

$$= 164.0208 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2690.0 - 2326.0 - 65.0) / 2$$

$$= 149.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1971593.	149.4999	1.0000	294759.
Face Pressure, Mt	123803.	164.0208	1.0000	20307.
Gasket Load, Mg	221080.	146.0416	1.0000	32287.
Gasket Seating, Matm	3806236.	146.0416	1.0000	555880.

Total Moment for Operation, Mo 347353. Kg-m.

Total Moment for Gasket seating, Mg 555880. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 323.527 mm

Hub Ratio, h/h0 = HL / H0 0.278

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.444

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.888

Factor V per 4.16.5 0.389

Factor f per 4.16.5 1.151

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2880.0 / 2326.0$$

$$= 1.238$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.238^2 \cdot \log_{10}(1.238))) / (1.238^2 - 1)) / (1.238 - 1)$$

$$= 9.211$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.238^2 (1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / ((1.0472 + 1.9448 \cdot 1.238^2) \cdot (1.238 - 1))$$

$$= 1.823$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.238^2 (1 + 8.55246 \cdot \log_{10}(1.238)) - 1) / (1.36136 (1.238^2 - 1) \cdot (1.238 - 1))$$

$$= 10.122$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.238^2 + 1) / (1.238^2 - 1)$$

$$= 4.752$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 10.122 \cdot 45.0^2 \cdot 323.527 / 0.389$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 46 of 214

Flg Calc [Int P]: girth flg.-2 Flng: 12 9:22am Sep 9, 2026

$$= 17031036.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.888/323.527 \\ &= 0.003 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (354.0 \cdot 0.003 + 1) / 1.823 + 354.0^3/17031036 \\ &= 3.686 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g^{1^2}) \\ &= (1.151 \cdot 347353/2326.0) / (3.6859 \cdot 65.0^2) \\ &= 108.24 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g^{1^2}) \\ &= (1.151 \cdot 555880/2326.0) / (3.6859 \cdot 65.0^2) \\ &= 173.22 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2951 \cdot 347353/2326.0) / (3.6859 \cdot 354.0^2) \\ &= 7.28 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2951 \cdot 555880/2326.0) / (3.6859 \cdot 354.0^2) \\ &= 11.65 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (9.2112 \cdot 347353 / (354.0^2 \cdot 2326.0)) - 4.7518 \cdot 7 \\ &= 73.07 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (9.2112 \cdot 555880 / (354.0^2 \cdot 2326.0)) - 4.7518 \cdot 12 \\ &= 116.93 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (108 + \max(7, 73)) / 2 \\ &= 90.65 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (173 + \max(12, 117)) / 2 \\ &= 145.08 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 2316476/2347.6599 \\ &= 96.77 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (558336/2347.6599) \\ &= 23.32 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 47 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9,2026

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	108.24	191.67	173.22	222.00
Radial Flange	7.28	127.78	11.65	148.00
Tangential Flange	73.07	127.78	116.93	148.00
Maximum Average	90.65	127.78	145.08	148.00
Bolting	96.77	113.19	23.32	159.00

Minimum Required Flange Thickness 353.212 mm

Estimated M.A.W.P. (Operating) 4832 KPa.

Estimated Finished Weight of Flange at given Thk. 6630.4 kg.

Estimated Unfinished Weight of Forging at given Thk 7923.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	506.63	527.74	0.960	506.63	619.99	0.817
Tangential Flange	St	346.13	395.81	0.875	346.13	464.99	0.744
Tangential Flange	Sto	134.88	263.87	0.511	134.88	309.99	0.435
Combined	Sh/f + St	791.61	791.61	1.000	791.61	929.98	0.851
Combined	Sr + St	380.60	791.61	0.481	380.60	929.98	0.409
Combined	Sh + Sto	641.51	791.61	0.810	641.51	929.98	0.690
Radial Flange	Sr	34.47	527.74	0.065	34.47	619.99	0.056
Maximum Stress Ratio				1.000			0.851



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 48 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.3798/1.2133)

= 1.1373

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3471/1.1463) if $h/h0 < 0.35$, else max (0.25; 0.252/0.7756)

= 0.3028

In this case, the hub ratio $h/h0$, 0.2782 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	470.6 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 49 of 214

Flg Calc [Int P]: girth flg.-2

Flng: 12

9:22am

Sep 9, 2026

Root Area of a Single Bolt	Abs	65.2128 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.9 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.548 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2419.0^2 - 2349.0^2)$$

$$= 2621.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 2621.345 / (36 \cdot 65.213)$$

$$= 232.2 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(232.248; 507.5)$$

$$= 232.2 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(232.248; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 470.648)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \geq 156.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 2621.345 + \pi/4 \cdot 4550.0 \cdot 2349.0^2) / (0.7 \cdot 65.213 \cdot 36)$$

$$:: 290.0 \geq 274.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 275.0(2621.345 / (36 \cdot 65.213))$$

$$:: 290.0 \leq 307.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 290.0 \leq 470.648(1.0/0.548)$$

$$:: 290.0 \leq 859.0 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.26 \cdot 65.213 \cdot 95.25$$

$$= 4775.8 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 75 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 113 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.597 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$
 $= 0.742$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 347352.9 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.597 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$
 $= 0.546$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 555880 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.686 \cdot 210987 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$
 $= 0.724$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 347352.9 / 1.0 \cdot 9806.637 \cdot 0.389 / (3.686 \cdot 178967 \cdot 45.0^2 \cdot 323.527 \cdot 0.3)$
 $= 0.533$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.782, Temperature Reduction per Fig. UCS 66.1 = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	105.89	222.00	167.88	222.00
Radial Flange	7.70	148.00	12.21	148.00
Tangential Flange	69.70	148.00	110.49	148.00
Maximum Average	87.80	148.00	139.19	148.00
Bolting	96.77	159.00	23.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 344.856 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 51 of 214

Internal Pressure Calculations: Step: 7 9:22am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Ch. cover		4550	303	3	2880	127.78	1
Ch.flange		4574.2	363	3	2320	127.78	2
Ch.shell		4571.8	48	3	2320	127.78	3
Shell		4571.8	48	3	2320	127.78	4
Shell girth fln.		4550	365	3	2320	127.78	5
Shell girth fln.		4550	365	3	2320	127.78	6
Shell -2		4570.3	48	3	2320	127.78	7
head- front		4572.3	56	3	2320	127.78	8

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Ch. cover		4550	4731.01	5592.83	295	289.358	1
Ch.flange		4550	4807.52	5975.32	355	348.082	2
Ch.shell		4550	4809.9	5975.32	48	45.5263	3
Shell		4550	4809.9	5975.32	48	45.5263	4
Shell girth fln.		4550	4831.69	5975.32	357	353.212	5
Shell girth fln.		4550	4831.69	5975.32	357	353.212	6
Shell -2		4550	4811.37	5975.32	48	45.5123	7
head- front		4550	4875.64	6047.82	47.6	44.6241	8

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4675 KPa.
 Shell Side MAPnc = 5975 KPa.
 Channel Side MAWP = 4580 KPa.
 Channel Side MAPnc = 5519 KPa.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

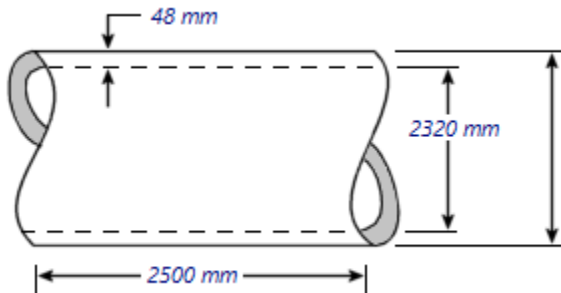
Page 52 of 214

Element: Ch.shell

Step: 8 9:22am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Ch.shell

Cylindrical Shell Design Information:					
Design Pressure	P	4571.790 KPa.	Int. Design Temperature	454 °C	
Material		SA-387 22	Ext. Design Temperature	25 °C	
Length		2500.000 mm	External Pressure Chart	CS-3	
			UNS Number	K21590	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	127.8 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	2320.000 mm	Actual Stress	120.9 N./mm ²	
Spec. Min. Thk t,e, or sa		48.0000 mm	Spec. Nominal Thk.	48.000 mm	
Surface Area		189752.2 cm ²	SG of Contents	1.000	
Element Volume		10570212.0 cm ³	Weight of Contents	10405.4 Kg	
Empty Weight		6919.0 Kg	Operating Weight	17324.4 Kg	
			Comp. All. Ope. Stress	76.0 N./mm ²	
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4571.79 \cdot 1163.0) / (127.78 \cdot 1.0 - 0.6 \cdot 4571.79) \\
 &= 42.5263 + 3.0000 = 45.5263 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 21.790 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0) \\
 &= 4832 - 22 = 4810 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0) \\
 &= 5975 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4571.79 \cdot (1163.0 + 0.6 \cdot 45.0)) / (1.0 \cdot 45.0) \\
 &= 120.905 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 53 of 214

Element: Ch.shell

Step: 8

9:22am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.027 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 48.0$, $t_r = 36.672$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.815$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 35 °C

Min Metal Temp. at Required thickness (UCS 66.1) 25 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4579.784 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	48.000 mm	Basic MDMT	35 °C
Stress/Thk. Ratio	0.815	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	25 °C
		Joint Efficiency E^*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

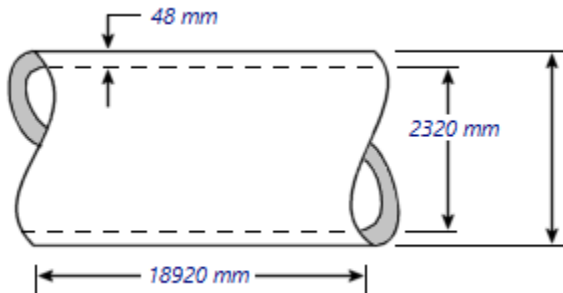
Page 54 of 214

Element: Shell

Step: 9 9:22am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	4571.790 KPa.	Int. Design Temperature	454 °C
Material	SA-387 22	Ext. Design Temperature	25 °C
Length	18920.002 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	127.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	2320.000 mm	Actual Stress	120.9 N./mm ²
Spec. Min. Thk t,e, or sa	48.0000 mm	Spec. Nominal Thk.	48.000 mm
Surface Area	1436044.9 cm ²	SG of Contents	1.000
Element Volume	48832348.0 cm ³	Weight of Contents	48802.6 Kg
Empty Weight	52362.9 Kg	Operating Weight	101165.5 Kg
		Comp. All. Ope. Stress	76.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4571.79 \cdot 1163.0) / (127.78 \cdot 1.0 - 0.6 \cdot 4571.79) \\
 &= 42.5263 + 3.0000 = 45.5263 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 21.790 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0) \\
 &= 4832 - 22 = 4810 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0) \\
 &= 5975 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4571.79 \cdot (1163.0 + 0.6 \cdot 45.0)) / (1.0 \cdot 45.0) \\
 &= 120.905 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 55 of 214

Element: Shell

Step: 9

9:22am

Sep 9, 2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.027 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 48.0$, $t_r = 37.452$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.832$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 35 °C

Min Metal Temp. at Required thickness (UCS 66.1) 26 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4675.361 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	48.000 mm	Basic MDMT	35 °C
Stress/Thk. Ratio	0.832	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	26 °C
		Joint Efficiency E^*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

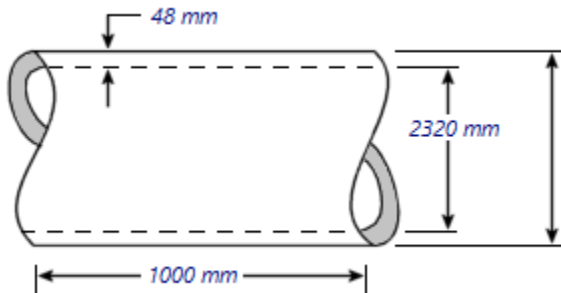
Page 56 of 214

Element: Shell -2

Step: 10 9:22am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell -2

Cylindrical Shell Design Information:			
Design Pressure P	4570.320 KPa.	Int. Design Temperature	454 °C
Material	SA-336 F22	Ext. Design Temperature	25 °C
Length	1000.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	127.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	2320.000 mm	Actual Stress	120.9 N./mm ²
Spec. Min. Thk t,e, or sa	48.0000 mm	Spec. Nominal Thk.	48.000 mm
Surface Area	75900.9 cm ²	SG of Contents	1.000
Element Volume	3308177.2 cm ³	Weight of Contents	3306.2 Kg
Empty Weight	2767.6 Kg	Operating Weight	6073.8 Kg
		Comp. All. Ope. Stress	76.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4570.32 \cdot 1163.0) / (127.78 \cdot 1.0 - 0.6 \cdot 4570.32) \\
 &= 42.5123 + 3.0000 = 45.5123 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 20.320 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (127.78 \cdot 1.0 \cdot 45.0) / (1163.0 + 0.6 \cdot 45.0) \\
 &= 4832 - 20 = 4811 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 48.0) / (1160.0 + 0.6 \cdot 48.0) \\
 &= 5975 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4570.32 \cdot (1163.0 + 0.6 \cdot 45.0)) / (1.0 \cdot 45.0) \\
 &= 120.867 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 57 of 214

Element: Shell -2

Step: 10

9:22am

Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 48.0, tr = 18.256, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.406$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4675.361 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	48.000 mm	Basic MDMT	16 °C
Stress/Thk. Ratio	0.406	Temperature Difference	49 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		Joint Efficiency E*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 58 of 214

Element: head- front

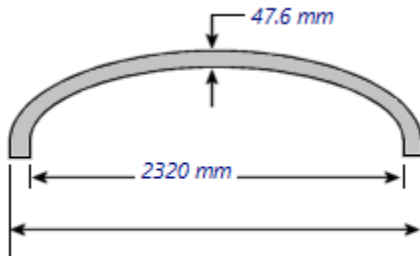
Step: 11

9:22am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: head- front

Elliptical Head Design Information:			
Design Pressure P	4572.280 KPa.	Int. Design Temperature	454 °C
Material	SA-336 F22	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	127.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	2320.000 mm	Actual Stress	119.3 N./mm ²
Spec. Min. Thk t,e, or sa	47.6000 mm	Spec. Nominal Thk.	56.000 mm
Surface Area	67933.8 cm ²	SG of Contents	1.000
Element Volume	1846263.6 cm ³	Weight of Contents	1842.6 Kg
Empty Weight	3094.6 Kg	Operating Weight	4937.2 Kg
		Comp. All. Ope. Stress	76.0 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 29.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4572.28 \cdot 2326.0 \cdot 0.997) / (2 \cdot 127.78 \cdot 1.0 - 0.2 \cdot 4572.28)$$

$$= 41.6241 + 3.0000 = 44.6241 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.280 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 127.78 \cdot 1.0 \cdot 44.6) / (0.997 \cdot 2326.0 + 0.2 \cdot 44.6)$$

$$= 4898 - 22 = 4876 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 148.0 \cdot 1.0 \cdot 47.6) / (1.0 \cdot 2320.0 + 0.2 \cdot 47.6)$$

$$= 6048 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 59 of 214

Element: head- front

Step: 11

9:22am

Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4572.28 \cdot (0.997 \cdot 2326.0 + 0.2 \cdot 44.6)) / (2 \cdot 1.0 \cdot 44.6)$$

$$= 119.284 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4572.28 \cdot 1163.0) / (127.78 \cdot 1.0 - 0.6 \cdot 4572.28) + 3.0$$

$$= 45.531 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 22.280 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (127.78 \cdot 1.0 \cdot 53.0) / (1163.0 + 0.6 \cdot 53.0)$$

$$= 5668 - 22 = 5646 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2326.0 / (2 \cdot 583.0))^2) / 6$$

$$= 0.996574$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 47.6, tr = 36.732, c = 3.0 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.824, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	6 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 56.0, tr = 37.452, c = 3.0 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.707, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	3 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4675.361 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	56.000 mm	Basic MDMT	19 °C
Stress/Thk. Ratio	0.707	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 29.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.500 mm

[PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.](#)



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 60 of 214

Shop Pressure Test Results: Step: 12 9:22am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6077.970 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7739.686 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5142.897 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4675.361 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5953.719 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7146.339 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5037.762 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4579.784 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell	177.6	192.4	0.923	7169.08
Shell	192.3	192.4	0.999	7762.43
Shell -2	192.3	192.4	0.999	7762.43
head- front	190.0	192.4	0.987	7762.43

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	148.00	127.78	1.158
Tube outlet	Nozzle	148.00	127.78	1.158
Shell inlet	Nozzle	148.00	127.78	1.158
N4- Shell outle	Nozzle	148.00	127.78	1.158
Drain	Nozzle	148.00	127.78	1.158

Minimum 1.158

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch. cover	148.00	127.78	1.158
Ch.flange	148.00	127.78	1.158
Ch.shell	148.00	127.78	1.158
Shell	148.00	127.78	1.158
Shell girth fln.-1	148.00	127.78	1.158
Shell girth fln.-2	148.00	127.78	1.158
Shell -2	148.00	127.78	1.158
head- front	148.00	127.78	1.158

Minimum 1.158



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 61 of 214

Shop Pressure Test Results:

Step: 12

9:22am

Sep 9, 2026

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	148.00	127.78	1.158
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	148.00	127.78	1.158
Tube Material	118.00	111.78	1.056
Tubesheet Material	148.00	127.78	1.158

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	122.36	192.40	0.636
Tube outlet	136.77	192.40	0.711
Shell inlet	148.09	192.40	0.770
N4- Shell outle	148.09	192.40	0.770
Drain	23.69	192.40	0.123

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 62 of 214

Field Pressure Test Results: Step: 13 9:22am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6077.970 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7739.686 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5142.897 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4675.361 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5953.719 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7146.339 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5037.762 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4579.784 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4550.0 • 1.0

= 5915.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4550.0 • 1.0

= 5915.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch. shell	157.0	192.4	0.816	5937.74
Shell	157.0	192.4	0.816	5937.74
Shell -2	157.0	192.4	0.816	5937.74
head- front	154.9	192.4	0.805	5937.74

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	148.00	127.78	1.158
Tube outlet	Nozzle	148.00	127.78	1.158
Shell inlet	Nozzle	148.00	127.78	1.158
N4- Shell outle	Nozzle	148.00	127.78	1.158
Drain	Nozzle	148.00	127.78	1.158

Minimum

1.158

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 63 of 214

Field Pressure Test Results: Step: 13 9:22am Sep 9,2026

Ch. cover	148.00	127.78	1.158
Ch.flange	148.00	127.78	1.158
Ch.shell	148.00	127.78	1.158
Shell	148.00	127.78	1.158
Shell girth fln.-1	148.00	127.78	1.158
Shell girth fln.-2	148.00	127.78	1.158
Shell -2	148.00	127.78	1.158
head- front	148.00	127.78	1.158

Minimum 1.158

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	148.00	127.78	1.158
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	148.00	127.78	1.158
Tube Material	118.00	111.78	1.056
Tubesheet Material	148.00	127.78	1.158

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	101.35	192.40	0.527
Tube outlet	113.28	192.40	0.589
Shell inlet	113.28	192.40	0.589
N4- Shell outle	113.28	192.40	0.589
Drain	18.12	192.40	0.094

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 64 of 214

Ring MDMT Calcs:

Step: 14

9:22am

Sep 9, 2026

MDMT Calculations for Stiffening Rings:

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 65 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		5915.00 KPa.
Shell Thickness	ts	48.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2320.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		5915.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2320.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3250
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	19600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18920.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	987.701 mm
Largest Center-to-Center Tube Distance	U1	31.7500 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 66 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

Tubesheet TYPE: Stationary, Integral both Sides, Conf. a
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	148.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2416.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

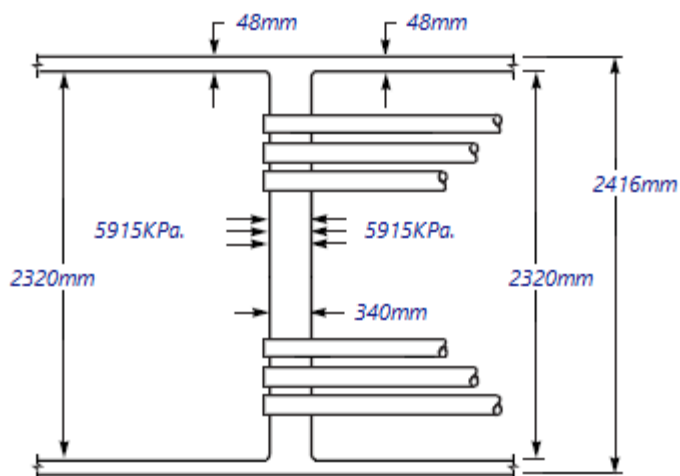
Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Junction Stress Reduction option	Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Tubesheet Integral w/Shell and Channel



Configuration a:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 67 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(5915.0 \cdot (25.4/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 5915.0)$
= $0.6241 + 0.0000 = 0.6241$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emapw):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	18920.00	12.04	50.0000	0.0075908	122.73

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.731) / (3 \cdot 12.0379) = 13593.0391$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0852	25.40	18920.00	23.41	50.0000	0.0020080	103.85

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 103.8465) / (3 \cdot 23.405) = 5915.562$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all. 1.0852 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 68 of 214

ASME TS Calc: Pressure Test

Case: 1

9:22a Sep 9, 2026

Allowable Internal Pressure at Corroded thickness	20998.74 KPa.
Required Internal Design Pressure	5915.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	5915.00 KPa.
Required Thickness due to Shell Side pressure	1.0852 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Stationary Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

= l_{tx} / h (modified for corrosion if present)

= 0/340.0

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(25.4 - 2 \cdot 2.11 (0.21097E+09/.21097E+09) \cdot (117/147) \cdot (0); 25.4 - 2 \cdot 2.11)$

= 25.4000 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 987.701 + 25.4$

= 2000.802 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(31.75 - 25.4) / 31.75$

= 0.200

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= 2000.802/2

= 1000.4011 mm

Effective tube pitch [p^*]:

= $p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$

= $31.75 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100.0; 4 \cdot 2000.802 \cdot 31.75) / (\pi \cdot 2000.802^2) }$

= 31.7500 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(31.75 - 25.4) / 31.75$

= 0.2000

Ratio [ρ_s]:

= a_s / a_o

= 1160.0/1000.4011

= 1.1595

Ratio [ρ_c]:

= a_c / a_o

= 1160.0/1000.4011

= 1.1595

Compute parameter [x_t]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 69 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

$$= 1 - 3250 \left((25.4 - 2 \cdot 2.11) / (2 \cdot 1000.4011) \right)^2$$

$$= 0.6358$$

Determine parameter [xs]:

$$= 1 - Nt \left(dt / (2 \cdot ao) \right)^2$$

$$= 1 - 3250 \left(25.4 / (2 \cdot 1000.4011) \right)^2$$

$$= 0.4762$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Shell coefficient [β_s]:

$$= \left((12(1 - v_s^2))^{1/4} \right) / \left((Ds + ts)ts \right)$$

$$= \left((12(1 - 0.3^2))^{1/4} \right) / \left((2320.0 + 48.0)48.0 \right)$$

$$= 0.0054 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 0.21097E+09 \cdot 48.0^3 / (6(1 - 0.3^2))$$

$$= 23041232896.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2320.0 + 48.0) \cdot 0.23041E+11 / 340.0^3 \cdot (1 + 340.0 \cdot 0.005 + (340.0 \cdot 0.005)^2 / 2)$$

$$= 37595132.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot v_s / (2(Ds + ts)))$$

$$= 2320.0^2 / (4 \cdot 0.21097E+09 \cdot 48.0) \cdot (1 - 2320.0 \cdot 0.3 / (2(2320.0 + 48.0)))$$

$$= 0.11334E+00 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= \left((12(1 - v_c^2))^{1/4} \right) / \left((Dc + tc)tc \right)$$

$$= \left((12(1 - 0.3^2))^{1/4} \right) / \left((2320.0 + 48.0)48.0 \right)$$

$$= 0.0054 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - v_c^2))$$

$$= 0.005 \cdot 0.21097E+09 \cdot 48.0^3 / (6(1 - 0.3^2))$$

$$= 23041232896.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2320.0 + 48.0 / \cos(0)) \cdot 0.23041E+11 / 340.0^3 \cdot (1.0 + 340.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (340.0 \cdot 0.005)^2 / 2 \cdot 1.0)$$

$$= 37595132.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot E_c \cdot Tc) \cdot (1 - Dc \cdot v_c / (2(Dc + tc)))$$

$$= 2320.0^2 / (4 \cdot 0.21097E+09 \cdot 48.0) \cdot (1 - 2320.0 \cdot 0.3 / (2(2320.0 + 48.0)))$$

$$= 0.11334E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.9.4 Step 3:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 70 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

$h/p = 10.708662$; $\mu^* = 0.200000$; $E^* = 32774746$ KPa.
 $E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (10.709) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.459^2)3250 \cdot 0.21097E+09 \cdot 2.11 \right. \right.$$

$$\left. \left(25.4 - 2.11 \right)1000.4011^2 \right) / (32774746 \cdot 18920.0 \cdot 340.0^3) \right)^{1/4}$$

$$= 2.2629$$

Values from Table 4.18.3

$Z_d = 0.117981$; $Z_v = 0.137663$; $Z_m = 0.586475$
 $Z_a = 0.109820E+01$; $Z_w = 0.137663$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$= A / Do$$

$$= 2416.0/2000.8021$$

$$= 1.2075$$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 32774746 \cdot (37595132 + 37595132 +$$

$$0.21097E+09 \cdot \ln(1.21))$$

$$= 1.8993$$

Compute parameter $[\phi]$:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4586)1.8993$$

$$= 2.7703$$

Compute parameter $[Q_1]$:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.1595 - 1 - 2.7703 \cdot 0.1377) / (1 + 2.7703 \cdot 0.5865)$$

$$= -0.084516$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$= 0 \text{ (config a)}$$

Calculate parameter $[\omega_s]$:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.1595 \cdot 0.23041E+11 \cdot 0.0054 \cdot 0.113341(1 + 340.0 \cdot 0.0054)$$

$$= 46262.9609 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1000.401^2(1.16^2 - 1) \cdot (1.16 - 1) / 4 - 46262.961$$

$$= -32511.1211 \text{ mm}^2$$

Parameter $[\omega_c]$:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.16 \cdot 0.23041E+11[0.113341(1.0 + 0.005 \cdot 1.0 \cdot 340.0) -$$

$$0(1.0/340.0 + 1.0 \cdot 0.005/2)]$$

$$= 46262.9609 \text{ mm}^2$$

Parameter $[\omega^*c]$:

$$= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$

$$= 1000.40106^2[(1.15954^2 + 1) \cdot (1.15954 - 1) / 4 - (1.15954 - 1) / 2] - 46262.96094$$

$$= -32511.1211 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 71 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 5915.0 - 5915.0 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega_s * P * s - \omega_c * P * c)) + W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$$

$$= [((-32511.121 * 5915.0 - -32511.121 * 5915.0) - (46262.961 * 0 - 46262.961 * 0)) * 0. + 0 * 0 / (2 * \pi)] / (1 + 2.77028 * 0.58648)$$

$$= 0.0000 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * 0 / (0.2 * (340.0 - 0)^2)$$

$$= 0.0000 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 * \beta_T * S_y = 1.5 * 0.95 * 309.99 = 441.74 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 0.0000 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 3.81 + 0 + 0$$

$$= 3.8100 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 * \mu * h / a_o) * (0.8 * \beta_T * S_y)$$

$$= (2 * 0.2 * 340.0 / 1000.401) * 235.596$$

$$= 32.0282 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (1 / (2 * \mu)) * (a_o / h) * P_e$$

$$= (1 / (2 * 0.2)) * (1000.401 / 3.81) * 0$$

$$= 0.0000 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 3.81 + 0 + 0$$

$$= 3.8100 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(3.81 ; 3.81)$$

$$= 3.8100 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 72 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

$$F_{tmin} = -1.6503, F_{tmax} = 1.805$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmin} / a_o^2) / (X_t - X_s) \\ &= ((5915.0 \cdot 0.4762 - 5915.0 \cdot 0.6358) - 2 \cdot 0 \cdot -1.65 / 1000.401^2) / (0.6358 - 0.4762) \\ &= -5.9153 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmax} / a_o^2) / (X_t - X_s) \\ &= ((5915.0 \cdot 0.4762 - 5915.0 \cdot 0.6358) - 2 \cdot 0 \cdot 1.805 / 1000.401^2) / (0.6358 - 0.4762) \\ &= -5.9153 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 5.915 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as } P_e \text{ is 0.0)}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4 \\ &= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4 \\ &= 8.2680 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{y,t}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.21097E+09 / 204) } \\ &= 139.8902 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0 \cdot 0 / 8.268 \\ &= 0.0000 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ &= (204 / 1.25) \cdot (1 - 0 / (2 \cdot 139.89)) \\ &= 194.746 \text{ N./mm}^2 \text{ (Never greater than } S_{ot}) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 5.92 \cdot 1.5438 \\ &= 93.12 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.11) 2.11 \\ &= 1.544 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (25.4 - 2.11) 194.746 \\ &= 3065.793 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 73 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 194.75$$

$$= 3000.1914 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 194.746 / 194.746$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 194.75$$

$$= 1941.3005 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 3065.79 is $\geq$ tube strength 3065.79 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.9.4 Step 10:

Note:
For a given Shell thickness of 48.0 mm, the minimum Shell length adjacent to the tubesheet should be 600.672 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 $\leq$	441.7 N./mm ²	Pass
Tubesheet Shear Stress	0.0 $\leq$	235.6 N./mm ²	Pass
Maximum Tube Stress	5.9 $\leq$	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-5.9 $\leq$	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	93.1 $\leq$	3065.8 Kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	340.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses		Case		Pass/ Fail
		Bend	Allwd	Shear	Allwd	Type	
D1uc	143.351	...	183	442	44	236	Ps+Pt-Th D1 Ok
D2uc	143.351	...	183	442	44	236	Ps+Pt-Th D2 Ok
D3uc	3.810	...	...	442	...	236	Ps+Pt-Th D3 Ok



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 74 of 214

ASME TS Calc: Pressure Test

Case: 1 9:22a Sep 9, 2026

Max: 143.3510 ... mm 0.414 0.185 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses						Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	26	194	-3	-194	417	3065	...	441	...	441	...	441	Ok
D2uc	32	194	-32	-194	510	3065	...	441	...	441	...	441	Ok
D3uc	6	194	-6	-194	93	3065	...	441	...	441	...	441	Ok
Max RATIO	0.166			0.166		0.166		...				...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	143.351	340.000	Ok
Tube thickness :	1.085	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (48.000) adj. to tubesheet: 600.672 mm

Note: This is a full strength Tube to Tubesheet Joint.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 75 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		5915.00 KPa.
Shell Thickness	ts	48.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2320.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		5915.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2320.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3250
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	19600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18920.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	987.701 mm
Largest Center-to-Center Tube Distance	U1	31.7500 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 76 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9,2026

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	148.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2300.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 0.000 mm
Tube End condition corresponding to Span (l) k 0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2300.000 mm
Flange Inside Diameter	B	2014.000 mm
Flange Face Outside Diameter	Fod	2049.000 mm
Flange Face Inside Diameter	Fid	2014.000 mm
Gasket Outside Diameter	Go	2046.000 mm
Gasket Inside Diameter	Gi	2020.000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.0000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2144.000 mm
Nominal Bolt Diameter	dB	76.2000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Extended as Flange Yes
Thickness of Extended Portion of Tubesheet Tf 324.0000 mm
Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



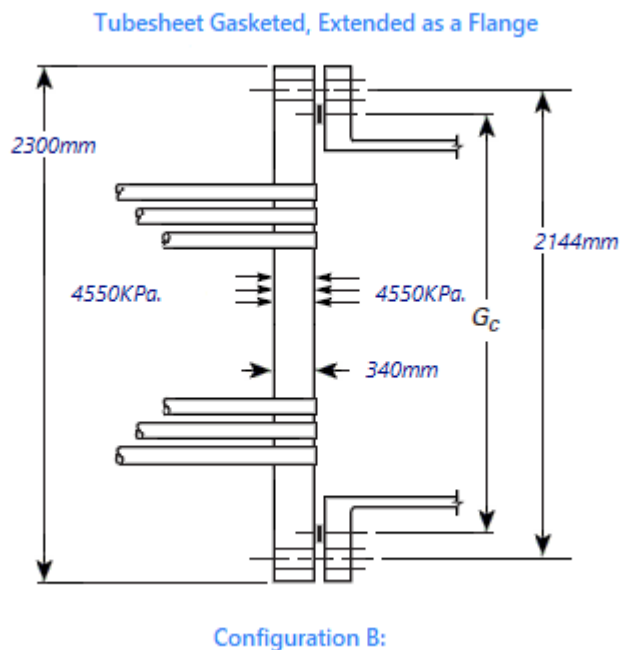
PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 77 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group D



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 78 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$$
 per Appendix 1-1 (a)(1)

$$= (5915.0 \cdot (25.4/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 5915.0)$$

$$= 0.6241 + 0.0000 = 0.6241 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	18920.00	12.04	50.0000	0.0075908	122.73

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.731) / (3 \cdot 12.0379) = 13593.0391 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0852	25.40	18920.00	23.41	50.0000	0.0020080	103.85

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 103.8465) / (3 \cdot 23.405) = 5915.562 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.0852 mm
Allowable Internal Pressure at Corroded thickness	20998.74 KPa.
Required Internal Design Pressure	5915.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	5915.00 KPa.
Required Thickness due to Shell Side pressure	1.0852 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2033.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

= lt_x / h (modified for corrosion if present)
= 0/340.0
= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d*]:

= $\max(dt - 2 \cdot tt(Et/E), (StT/S) \cdot (\rho), dt - 2 \cdot tt)$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 79 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

$$= \max(25.4 - 2 \cdot 2.11 (0.21097E+09 / 21097E+09) \cdot (117/147) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 987.701 + 25.4$$

$$= 2000.802 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2000.802 / 2$$

$$= 1000.4011 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(0 \cdot 100.0; 4 \cdot 2000.802 \cdot 31.75) / (\pi \cdot 2000.802^2)}$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1016.5778 / 1000.4011$$

$$= 1.0162$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1016.5778 / 1000.4011$$

$$= 1.0162$$

Compute parameter [xt]:

$$= 1 - Nt((d_t - 2 \cdot t_t) / (2 \cdot a_o))^2$$

$$= 1 - 3250 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1000.4011))^2$$

$$= 0.6358$$

Determine parameter [xs]:

$$= 1 - Nt(d_t / (2 \cdot a_o))^2$$

$$= 1 - 3250 (25.4 / (2 \cdot 1000.4011))^2$$

$$= 0.4762$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 10.708662; \mu^* = 0.200000; E^* = 32774746 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 80 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

$$E^*/E = 0.155349 ; v^* = 0.458572$$

Note: As h/p (10.709) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$\begin{aligned} &= \left(\left(24(1 - v^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4} \\ &= \left(\left(24(1 - 0.459^2)3250 \cdot 0.21097E+09 \cdot 2.11 \right. \right. \\ &\quad \left. \left. (25.4 - 2.11)1000.4011^2 \right) / (32774746 \cdot 18920.0 \cdot 340.0^3) \right)^{1/4} \\ &= 2.2629 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.117981 ; Z_v = 0.137663 ; Z_m = 0.586475 \\ Z_a &= 0.109820E+01 ; Z_w = 0.137663 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$\begin{aligned} &= A / Do \\ &= 2300.0/2000.8021 \\ &= 1.1495 \end{aligned}$$

Compute coefficient $[F]$:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 32774746 \cdot (0 + 0 + \\ &\quad 0.21097E+09 \cdot \ln(1.15)) \\ &= 0.4857 \end{aligned}$$

Compute parameter $[\phi]$:

$$\begin{aligned} &= (1 + v^*)F \\ &= (1 + 0.4586) 0.4857 \\ &= 0.7084 \end{aligned}$$

Compute parameter $[Q_1]$:

$$\begin{aligned} &= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.0162 - 1 - 0.7084 \cdot 0.1377) / (1 + 0.7084 \cdot 0.5865) \\ &= -0.057475 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$\begin{aligned} &= (G_c - C) / Do \text{ (config b)} \\ &= (2033.1555 - 2144.0) / 2000.8021 = -0.0554 \end{aligned}$$

Calculate parameter $[\omega_s]$:

$$\begin{aligned} &= ps \cdot ks \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s) \\ &= 1.0162 \cdot 0 \cdot 0 \cdot 0(1 + 340.0 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter $[\omega^*s]$:

$$\begin{aligned} &= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s \\ &= 1000.401^2(1.016^2 - 1) \cdot (1.016 - 1) / 4 - 0 \\ &= 131.9013 \text{ mm}^2 \end{aligned}$$

Parameter $[\omega_c]$:

$$\begin{aligned} &= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\ &= 1.016 \cdot 0 [0(1.0 + 0 \cdot 1.0 \cdot 340.0) - \\ &\quad 0(1.0/340.0 + 1.0 \cdot 0/2)) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter $[\omega^*c]$:

$$\begin{aligned} &= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c \\ &= 1000.40106^2[(1.01617^2 + 1) \cdot (1.01617 - 1) / 4 - (1.01617 - 1) / 2] - 0 \\ &= 131.9009 \text{ mm}^2 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 81 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 5915.0 - 5915.0 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega_s * P * s - \omega_c * P * c)) + W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$$

$$= [((131.901 * 5915.0 - 131.901 * 5915.0) -$$

$$(0 * 0 - 0 * 0)) * 0. + 2381345 * -0.055 / (2 * \pi)] / (1 + 0.70844 * 0.58648)$$

$$= -14833.6572 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -14833.657 / (0.2 * (340.0 - 0)^2)$$

$$= -37.7520 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 * \beta T * S_y = 1.5 * 0.95 * 309.99 = 441.74 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -17664.369 / (0.2 * (108.467 - 0)^2)$$

$$= -441.7258 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 108.467 + 0 + 0$$

$$= 108.4670 \text{ mm}$$

$$H \text{ reqd.} = H_{ReqB} = 108.4670 \text{ mm}$$

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	37.8 <=	441.7 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	177.059	59.995	233	442	Ps+Pt-Th	D1 Ok
D2uc	145.469	...	209	442	Ps+Pt-Th	D2 Ok
D3uc	108.467	...	38	442	Ps+Pt-Th	D3 Ok
Max:	177.0593	59.995 mm	0.528	...	(Str. Ratio)	

Load Case Definitions:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 82 of 214

ASME Fl-TS Calc: Pressure Test

Case: 1 9:23a Sep 9, 2026

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	177.059	340.000	Ok
Tubesheet thickness flanged extension :	59.995	324.000	Ok
Tube thickness :	1.085	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 83 of 214

Floating Head: Operating

Case: 1

9:23a

Sep 9, 2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	4550.00	KPa.
Shell Side (External) Design Pressure	Pss	4550.00	KPa.
Design Temperature for Spherical Head	Temp	454.00	°C
Head Material		SA-387 22	
Head Material UNS Number		K21590	
Head Allowable Stress at Temperature	Sho	148.00	N./mm ²
Head Allowable Stress at Ambient	Sha	148.00	N./mm ²
Crown Radius for Spherical Head	L	1812.000	mm
Head Thickness	Th	137.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-336 F22	
Flange Material UNS Number		K21590	
Flange Allowable Stress at Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	148.00	N./mm ²
Flange Outside Diameter	Fod	2300.000	mm
Flange Inside Diameter	Fid	2014.000	mm
Flange Thickness	Tf	327.0000	mm
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2144.000	mm
Nominal Bolt Diameter	dB	76.2000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	

Full face Gasket Flange Option Program Selects

Flange Face Outside Diameter	Ffod	2049.000	mm
Flange Face Inside Diameter	Ffid	2014.000	mm
Gasket Outside Diameter	Go	2046.000	mm
Gasket Inside Diameter	Gi	2020.000	mm
Gasket Factor,	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		3.0000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		15.8414	mm

The Flange is not Slotted.

This unit does not have a Backing Ring.

Use Full Bolt Load in Flange Design Yes



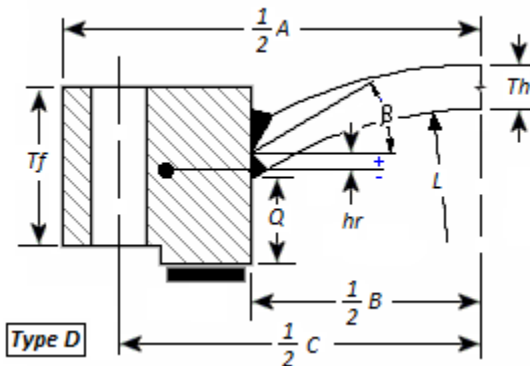
PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 84 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:

$$= 5PL / 6Sho$$

$$= (5 \cdot 4550.0 \cdot 1812.0) / (6 \cdot 147)$$

$$= 46.4258 \text{ mm}$$

Maximum Allowable Working Pressure at Given Thickness[Pa]:

$$= 6Sho \cdot (Th - Cass - Cats) / 5L$$

$$= (6 \cdot 147 \cdot 137.0) / (5 \cdot 1812.0)$$

$$= 13426.79 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold[Pnc]:

$$= 6Sha / 5L$$

$$= (6 \cdot 147 \cdot 137.0) / (5 \cdot 1812.0)$$

$$= 13426.79 \text{ KPa.}$$

Actual Stress at given Pressure and Thickness[Sact]:

$$= 5PL / 6(Th - Cass - Cats)$$

$$= (5 \cdot 4550.0 \cdot 1812.0) / (6 \cdot 137.0)$$

$$= 50. \text{ N./mm}^2$$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 5.464 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart CS-2 at 454.00 °C
Elastic Modulus for Material 150421952.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	137.0000	mm
Outside Crown Radius of Shell	OD/2	1949.0000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	14.2263	
Geometry Factor, A	f(D/T, LD)	0.0087866	
Materials Factor, B	f(A, Chart)	69.8878	N./mm ²
Maximum Allowable Working Pressure		4912.30	KPa.
EMAWP = $B / ((OD/T)/2) = 69.8878 / (28.4526/2.0) = 4912.3008$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	128.1332	mm
Outside Diameter of Shell	ODCA	3898.000	mm
Diameter / Thickness Ratio	(D/T)	30.4215	
Geometry Factor, A	f(DT, LD)	0.0082179	
Materials Factor, B	f(A, Chart)	69.2215	N./mm ²
Maximum Allowable Working Pressure		4550.57	KPa.
EMAWP = $B / ((D/T)/2) = 69.2215 / (30.4215/2) = 4550.5713$			



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 85 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	2033.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsm_{ax}]:

$= 2a + 6t / (m + 0.5)$
 $= 2 * 76.2 + 6 * 327.0 / (4.25 + 0.5)$
 $= 565.453 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:

$= C * \sin(\pi / n)$
 $= 2144.0 * \sin(\pi / 36)$
 $= 186.862 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{186.862 / (2 * 76.2 + 327.0)}, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

Total Area of Bolts 1468.797 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	73.025	78.000	
Circ. Spacing between the Bolts:	158.750	186.862	565.453

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1506395.5	Kgf
Contact Load on Gasket Surfaces	Hp	161783.2	Kgf
Hydrostatic End Load at Flange ID	Hd	1478144.0	Kgf
Pressure Force on Flange Face	Ht	28251.5	Kgf
Radial Component of Head Membrane Force	Hr	2323492.2	Kgf
Operating Bolt Load:	Wm1	1668178.8	Kgf
Gasket Seating Bolt Load	Wm2	291295.4	Kgf
Required Bolt Area	Am	1028.921	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$= A_b * S_b / (y * \pi * (G_o + G_i))$
 $= 1468.797 * 159.0 / (69.64 * 3.14 * (2046.0 + 2020.0))$
 $= 26.253 \text{ mm}$ [Note: Exceeds actual gasket width, 13.000]

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$= 38.100 \text{ mm}$ [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	2381345.5	Kgf
Gasket Seating Force (Operating)	Hg	161783.2	Kgf
Distance to Gasket Load Reaction	hg	55.4222	mm
Distance to Face Pressure Reaction	ht	60.2111	mm
Distance to End Pressure Reaction	hd	65.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	1478144.	65.0000	1.0000 96081. Kg-m.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 86 of 214

Floating Head: Operating

Case: 1

9:23a

Sep 9, 2026

Face Pressure,	Mt	28251.	60.2111	1.0000	1701. Kg-m.
Gasket Load,	Mg	161783.	55.4222	1.0000	8967. Kg-m.
Floating Hd. Load,	Mh	2323492.	15.8414	1.0000	-36808. Kg-m.
Gasket Seating,	Ma	2381346.	55.4222	1.0000	131982. Kg-m.

Total Moment for Operation (Internal Pressure) 69940.820 Kg-m.
Total Moment for Gasket Seating (Int. Pressure) 131982.016 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1506395.5	Kgf
Hydrostatic End Load at Flange ID	Hd	1478144.0	Kgf
Pressure Force on Flange Face	Ht	28251.5	Kgf
Radial Component of Head Membrane Force	Hr	2323492.2	Kgf
Gasket Seating Bolt Load	Wm2	291295.4	Kgf
Required Bolt Area	Am	179.669	cm ²

Flange Design Bolt Load (Seating)	W	2381345.5	Kgf
Distance to Gasket Load Reaction	hg	55.4222	mm
Distance to Face Pressure Reaction	ht	60.2111	mm
Distance to End Pressure Reaction	hd	65.0000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	1478144.	9.5778	1.0000 14158. Kg-m.
Face Pressure,	Mt	28251.	4.7889	1.0000 135. Kg-m.
Floating Hd. Load,	Mh	2323492.	15.8414	1.0000 -36808. Kg-m.
Gasket Seating,	Ma	2381346.	55.4222	1.0000 131982. Kg-m.

Total Moment for Operation (External Pressure) 22515.143 Kg-m.
Total Moment for Gasket Seating (Ext. Pressure) 131982.016 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 4550.0 \cdot 2014.0 \cdot \sqrt{(4 \cdot 1812.0 \cdot 1812.0 - 2014.0 \cdot 2014.0) / (8 \cdot 147.997(2300.0 - 2014.0))}$$

$$= 81.5383 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 69940(2300.0 + 2014.0) / (147.997 \cdot 2014.0)(2300.0 - 2014.0)$$

$$= 34709.7383 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 81.538 + \sqrt{(81.538 \cdot 81.538 + 34709.738)}$$

$$= 284.9055 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 131982(2300.0 + 2014.0) / (147.997 \cdot 2014.0)(2300.0 - 2014.0)$$

$$= 65499.1094 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 0 + \sqrt{(0 \cdot 0 + 65499.109)}$$

$$= 255.9279 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 4550.0 \cdot 2014.0 \cdot \sqrt{(4 \cdot 1812.0 \cdot 1812.0 - 2014.0 \cdot 2014.0) / (8 \cdot 147.997(2300.0 - 2014.0))}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 87 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

= 81.5383 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $22515(2300.0+2014.0)/(147.997 \cdot 2014.0)(2300.0-2014.0)$

= 11173.6572 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $81.538 + \sqrt{(81.538 \cdot 81.538 + 11173.657)}$

= 215.0378 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $131982(2300.0+2014.0)/(147.997 \cdot 2014.0)(2300.0-2014.0)$

= 65499.1094 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 65499.109)}$

= 255.9279 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	46.4258	284.9055
Shellside (External) Pressure	128.1332	215.0378
Tubeside Gasket Seating Load		255.9279
Shellside Gasket Seating Load		255.9279
Maximum + Corrosion Allowance	128.1332	284.9055
Actual Thickness as Given	137.0000	327.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	5518.90	5518.90	7487.54	7549.51
Floating head Stress	13426.69	13426.69	4912.20	8626.46
Bolt Stress	6495.08	6495.08	...	...
Minimum MAWP & MAPnc	5518.90	5518.90	4912.20	7549.51

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	292	No Calc	No Calc
20	30	No Calc	...	352	No Calc	No Calc
30	Tsht	2499.97	2416	45	0.0033454	131.013
Tsht	40	No Calc	...	45	No Calc	No Calc
40	50	18920	2416	45	0.00042691	42.6815
50	60	No Calc	...	354	No Calc	No Calc
60	70	No Calc	...	354	No Calc	No Calc
70	80	1243.33	2416	45	0.0072869	136.51
80	90	No Calc	2415.2	44.6	0.0025648	129.137

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	295	173.838	...	No Calc
20	30	355	348.082	...	No Calc
30	Tsht	48	No Calc	...	3253.46
Tsht	40	48	No Calc	...	No Calc
40	50	48	No Calc	...	1059.91
50	60	357	353.212	...	No Calc
60	70	357	353.212	...	No Calc
70	80	48	No Calc	...	3389.94
80	90	47.6	No Calc	...	2649.51

Minimum

1060

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	2499.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	18920	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	1243.33	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 89 of 214

Element and Detail Weights:

Step: 16

9:23am

Sep 9, 2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	15298.5	...	15298.5	...	1529.85
20	30	6782.25	1965552	6703.69	1967741	678.225
30	40	6918.99	10570212	6494.77	10624956	691.899
40	50	52362.9	48832348	49152.4	49246656	5236.29
50	60	6772.25	1931771	6695.04	1933742	677.225
60	70	6772.25	1931771	6695.04	1933742	677.225
70	80	2767.6	3308177	2597.91	3330075	276.76
80	90	3094.58	1846263	2928.8	1860075	309.458
Total		100769	70386088.00	96566	70896984.00	10076



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
20	Liqd	1964.35	227.5	-73	...	Liquid: 20
30	Liqd	10405.4	1250	50	...	Liquid: 30
30	Insl	1065.34	1250	...	...	Hot insulation
30	Nozl	911.94	1000	1447.35	...	Tube inlet
30	Nozl	889.944	1000	-1541	...	Tube outlet
40	Sadl	2191.28	4000	1473	...	Left Saddle
40	Sadl	2191.28	15080	1473	...	New Saddle
40	Liqd	48802.6	9460	50	...	Liquid: 40
40	Insl	8062.5	9460	...	...	Ins: 40
40	Nozl	1827.69	1500	1541	...	Shell inlet
40	Nozl	991.254	17000	-1541	...	N4- Shell outle
40	Nozl	17.6813	1000	-1181.42	...	Drain
40	Wght	550	4000	-1500	...	Fireproofing
40	Wght	550	14800	-1500	...	Wght:[2 of 2]
70	Liqd	3306.16	500	125	...	Liquid: 70
70	Insl	426.136	500	...	...	Ins: 70
80	Liqd	1842.6	243.333	25	...	Liquid: 80
80	Insl	480.419	308.75	...	...	Ins: 80
30	FTsh	8515.23	2670	...	...	Tubesheet
30	Tube	83842.4	12300	...	...	Tubes
30	FLTs	7269.79	22100	...	...	
30	FLHd	6736.06	22263.5	...	...	
40	Baf1	175.024	590	...	...	Baffles
40	Baf1	175.024	1180	...	...	Baffles
40	Baf1	175.024	1770	...	...	Baffles
40	Baf1	175.024	2360	...	...	Baffles
40	Baf1	175.024	2950	...	...	Baffles
40	Baf1	175.024	3540	...	...	Baffles
40	Baf1	175.024	4130	...	...	Baffles
40	Baf1	175.024	4720	...	...	Baffles
40	Baf1	175.024	5310	...	...	Baffles
40	Baf1	175.024	5900	...	...	Baffles
40	Baf1	175.024	6490	...	...	Baffles
40	Baf1	175.024	7080	...	...	Baffles
40	Baf1	175.024	7670	...	...	Baffles
40	Baf1	175.024	8260	...	...	Baffles



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 90 of 214

Element and Detail Weights:

Step: 16

9:23am

Sep 9, 2026

40 Baf1	175.024	8850	...	...	Baffles
40 Baf1	175.024	9440	...	...	Baffles
40 Baf1	175.024	10030	...	...	Baffles
40 Baf1	175.024	10620	...	...	Baffles
40 Baf1	175.024	11210	...	...	Baffles
40 Baf1	175.024	11800	...	...	Baffles
40 Baf1	175.024	12390	...	...	Baffles
40 Baf1	175.024	12980	...	...	Baffles
40 Baf1	175.024	13570	...	...	Baffles
40 Baf1	175.024	14160	...	...	Baffles
40 Baf1	175.024	14750	...	...	Baffles
40 Baf1	175.024	15340	...	...	Baffles
40 Baf1	175.024	15930	...	...	Baffles
40 Baf1	175.024	16520	...	...	Baffles
40 Baf1	175.024	17110	...	...	Baffles
40 Baf1	175.024	17700	...	...	Baffles
40 Baf1	175.024	18290	...	...	Baffles
40 Baf1	175.024	18880	...	...	Baffles
40 Baf1	176.521	19470	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	4382.6
Liquid	66321.0
Insulation	10034.4
Nozzles	4638.5
Weights	1100.0
Exchanger Components	112140.8
Liquid in Tubes	22433.3

Sum of the Detail Weights 221050.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	110846.2	110846.2	110846.2	110846.2	110846.2	110846.2
Saddles	4382.6	4382.6	4382.6	4382.6	4382.6	4382.6
Nozzles	4638.5	4638.5	4638.5	4638.5	4638.5	4638.5
Exchanger	112140.8	112140.8	112140.8	112140.8	112140.8	112140.8
Insulation	...	...	...	...	10034.4	10034.4 *
Empty Weights	...	...	...	1100.0	1100.0	...
Ope Weights	...	...	...	...	...	1100.0
Ope. Liquid	...	...	...	...	...	66321.0
Tube Ope Lqd	...	...	...	...	...	22433.3
Test Liquid	...	70343.1	...	...	...	...
Tube Tst Lqd	...	22433.3	...	...	...	...
Totals	232008.1	324784.6	232008.1	233108.1	243142.5	331896.8

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	232008.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	324784.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	232008.1 kg.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 91 of 214

Element and Detail Weights: Step: 16 9:23am Sep 9,2026

Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	233108.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	243142.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	331896.8 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	327273.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	334818.9 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	22447036.0 cm ³
Weight of Ope Liq in tubes :	22433.3 kg.
Weight of Water in tubes :	22433.3 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	92558.9	...
20	30	63455.2	33745.7
30	40	189752	182212
40	50	1436044	1378983
50	60	62839.4	33162.7
60	70	62839.4	33162.7
70	80	75900.9	72885
80	90	67933.8	61988.7

Total	2051324.5	1796140.6 cm ²
Total	2208.0	1933.3 ft ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 92 of 214

Nozzle Flange MAWP:

Step: 17

9:23am

Sep 9, 2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Tube inlet	7073.3	10340.0	454	600	GR 1.10	1985.2	7073.3	5088.1	7073.3

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 7073.296 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10340.004 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
Tube inlet	657.225	0.50	10380	5507	4579.8	7073.3	Pass

Evaluating Flange on Nozzle [Tube inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(10380) + 4 \cdot \text{abs}(5507) \cdot 657.225 \leq \pi \cdot 657.225^3 [(7073.296 - 4579.784) + .500 \cdot 7073.296]$$

$$= 0.18055E+06 \leq 0.54844E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Tube inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7073.296, 0.500 \cdot 7073.296 + 7073.296 - 180553.8 / (\pi \cdot 657.225^3))$$

$$= \min(7073.296, 8624.721)$$

$$= 7073.296 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.974)/(1 + 1.7 \cdot 3.4 \cdot 0.228))) \\
 &= \min(0.85, 0.911) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.614
Structure Height to Diameter ratio	9.556

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1850.0 + 300.0 = 2150.0 \text{ mm} \\
 &= 7.054 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (7.054 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900.0)^{2/9.5} \\
 &= 0.849
 \end{aligned}$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 94 of 214

Wind Load Calculation:

Step: 18

9:23am

Sep 9, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.614 • 10.974

= 91.7 lbs. [41.6] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Ch. cover	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Ch.flange	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Ch.shell	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell girth fln	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell girth fln	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell -2	2150.0	0.000	0.000	0.000	0.849	1.000	78.120
head- front	2150.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	300.00	78.12	0.00
Wght:[2 of 2]	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	2150	3456	10195.2	78.12	45.7385
20	30	2150	2784	12667.2	78.12	56.8286
30	40	2150	3259.2	81480	78.12	365.542
40	50	2150	3259.2	616641	78.12	2766.42
50	60	2150	2784	12444.5	78.12	55.8294
60	70	2150	2784	12444.5	78.12	55.8294
70	80	2150	3259.2	32592	78.12	146.217
80	90	2150	3258.24	18211.5	78.12	81.702

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.794^{0.75})$
 = 0.119 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.119$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 327514 \cdot 1.25$
 = 19257.838 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 327514$
 = 10270.8 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1440	32751.4	2334.28



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 96 of 214

Earthquake Load Calculation:

Step: 19

9:23am

Sep 9, 2026

20	30	1160	32751.4	1880.39
30	40	1160	32751.4	1880.39
40	Sadl	1160	32751.4	1880.39
Sadl	50	1160	32751.4	1880.39
40	50	1160	32751.4	1880.39
50	60	1160	32751.4	1880.39
60	70	1160	32751.4	1880.39
70	80	1160	32751.4	1880.39
80	90	1160	32751.4	1880.39

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 97 of 214

Center of Gravity Calculation: Step: 20 9:23am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	13134.499 mm
Center of Gravity of the Liquid	11813.413 mm
Center of Gravity of the Insulation	12901.063 mm
Center of Gravity of the Nozzles	7107.495 mm
Center of Gravity of the Added Weights (Operating)	12994.499 mm
Center of Gravity of the Added Weights (Empty)	12994.499 mm
Center of Gravity of the Tubesheet(s)	12372.985 mm
Center of Gravity of the Tubes	13054.500 mm
Center of Gravity of the Floating Head	23018.002 mm
Center of Gravity of the Baffles	13626.939 mm
Center of Gravity of Bare Shell New and Cold	11483.061 mm
Center of Gravity of Bare Shell Corroded	11418.409 mm
Vessel CG in the Operating Condition	12366.156 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	12446.094 mm
Vessel CG in the Test Condition	12479.345 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 98 of 214

Saddle Calcs: Road Transport:

Step: 21

9:23am

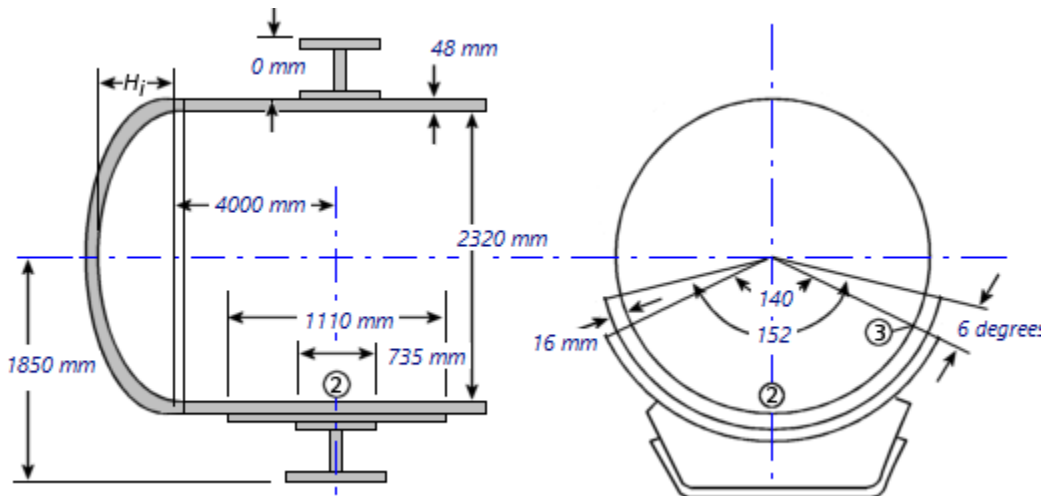
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1184.00	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		127.78	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		145769.83	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

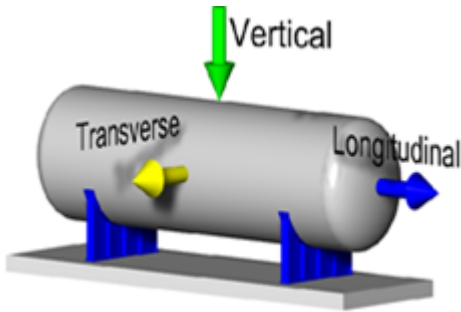
Page 99 of 214

Saddle Calcs: Road Transport:

Step: 21

9:23am

Sep 9, 2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-8.45	-294.49
Long. Stress at Top of Midspan	8.45	127.78
Long. Stress at Bottom of Midspan	8.45	127.78
Long. Stress at Top of Saddles	35.13	127.78
Long. Stress at Bottom of Saddles	-19.95	-294.49
Long. Stress at Bottom of Saddles	19.95	127.78
Tangential Shear in Shell	13.96	102.22
Circ. Stress at Horn of Saddle	24.88	191.67
Circ. Compressive Stress in Shell	1.41	127.78

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 857.53 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 7284.1 \text{ lbf} \\
 &= 3304.076 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.927^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 503.0 \text{ lbf} \\
 &= 228.2 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 227625.55 \cdot 1.0 + 3304.08 \\
 &= 26066.6 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 227625.55 \cdot 1.0 + 228.16 \\
 &= 22990.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 26066.6/2 \cdot 1850.0/2300.3677 \\
 &= 10481.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 22991 \cdot 1850.0/11080.0 \\
 &= 3838.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 128845.9 \cdot 0.05
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 100 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9, 2026

= 6442.294 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 128846 + 6442 + max(3839, 10482)
 = 145769.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	147961.09	Kgf
Transverse Shear Load	13033.32	Kgf
Longitudinal Shear Load	22990.71	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(2320.0/2 + 48.0 + 16.0)sin(140.0/2), 2•1184.0)
 = 2300.368 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -145770•4000.0[1-(1-4000.0/20864.0+(1184.0²-0.0²)/
 (2•4000.0•20864.0))/(1+(4•0.0)/(3•20864.0))]
 = -106891.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 145770•20864/4((1+2(1184²-0²)/(20864²))/(1+(4•0)/
 (3•20864))-4•4000/20864)
 = 182157.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0.0 • 1184.0/(2•48.0) - 182157.0/(π•1184.0²•48.0)
 = -8.45 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0.0 • 1184.0/(2 • 48.0) + 182157.0/(π • 1184.0² • 48.0)
 = 8.45 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0.0•1184.0/(2•48.0)--106891.6/(0.1412•π•1184.0²•48.0)
 = 35.13 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0.0•1184.0/(2•48.0)+-106891.6/(0.2486•π•1184.0²•48.0)
 = -19.95 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 145770(20864.0 - 2 • 4000.0)/(20864.0 + (4 • 0.0/3))
 = 89876.5 Kgf



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 101 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.9003 \cdot 89876.49 / (1184.0 \cdot 48.0)$$

$$= 13.96 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1184.0 \cdot 48.0)}$$

$$= 185.948 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(735.0 + 1.56 \cdot \sqrt{(1184.0 \cdot 48.0)}, 2 \cdot 4000.0)$$

$$= 1106.90 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6971 \cdot 145770 \cdot 0.1 / (1106.896(48.0 + 1.0 \cdot 16.0))$$

$$= -1.41 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -145770 / (4(48.0 + 1.0 \cdot 16.0)1106.896) -$$

$$3 \cdot 0.038 \cdot 145770 / (2(48.0 + 1.0 \cdot 16.0)^2)$$

$$= -24.88 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1160.0 \cdot 48.0)} + 1110.0)48.0^3) / 12$$

$$= 1362.222 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1362.222 + 709.491(250.659 - 24.0)^2$$

$$= 365858.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1478.1	48.0	24.0	709.5	1702779.4	0.366E+06
Wearplate	1110.0	16.0	56.0	177.6	994560.0	0.673E+05
Web	16.0	566.0	347.0	90.6	3142432.0	0.326E+05
BasePlate	760.0	60.0	660.0	456.0	30096002.0	0.765E+06
Totals	...	...	...	1433.7	35935772.0	0.123E+07

Distance to Centroid [C1]:

$$= AY / A$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 102 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9, 2026

$$= 35935772/1433.651$$

$$= 250.659 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140.0/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110.0) - 0.5 \cdot \sin(110.0)^2) / (\pi - 1.92 + \sin(110.0) \cos(110.0))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 145769.828$$

$$= 35047.5664 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 4.7463 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1850.0 - 1160.0 \cdot \sin(140.0/2) / 1.2217$$

$$= 957.787 \text{ mm}$$

Bending Moment, M = Fh * d = 33568.7773 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 6.7020 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(145770 + 2191)760.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 45.061 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140.0/2) (1160.0 + 48.0 + 16.0)$$

$$= 2300.367 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2300.3674 / (5 - 1)$$

$$= 575.092 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 575.0919 \cdot 760.0/2$$

$$= 2185.349 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 147961.094 / (2350.0 \cdot 760.0)$$

$$= 8.284 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2185.3 \cdot 8.28$$

$$= 18104.516 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 103 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

$$= 115.04 + 46.007$$

$$= 161.047 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 18104.5/161.0473$$

$$= 11.025 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 735.0 - 16.0$$

$$= 719.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/161.047$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 735.0/2$$

$$= 367.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(52942.0/161.047)}$$

$$= 181.311 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(140.0/2) (1160.0 + 48.0 + 16.0) - 60.0$$

$$= 1371.367 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1371.367/181.311$$

$$= 7.564$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 22990.7 / (2 \cdot 2350.0) \cdot 575.092 \cdot 1371.37/2$$

$$= 1928.964 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.5636 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.56)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.56) / (8 \cdot 123.21) - (7.56^3) / (8 \cdot 123.21^3))$$

$$= 153.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.02/153.58 + (1928.96 \cdot 367.5 / 52942\text{E}+09) / 173.33$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 104 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9, 2026

= 0.148

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/207.055

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 735.0/2

= 367.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1850.0 - cos(1224.0/2)(60.0 + 0.0 + 0.0) - 0.0

= 566.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(52961.1/207.055)}$

= 159.932 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 566.0/159.932

= 3.539

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 22990.713/(2 • 2350.0)

= 4.892 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 4.892 • 575.092 • 566.0

= 1592.271 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 1592.27/1441119

= 10.835 N./mm²

Compressive Allowable, KL/r < Cc (3.539 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (3.54)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (3.54) / (8 \cdot 123.21) - (3.54)^3 / (8 \cdot 123.21^3))$

= 154.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 17.16/154.93 + (1592.27 • 367.5/52961.082)/173.33

= 0.173

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 105 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

Vessel Mean Radius	Rm	1184.00	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		127.78	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		115355.28	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-6.69	-294.49
Long. Stress at Top of Midspan		6.69	127.78
Long. Stress at Bottom of Midspan		6.69	127.78
Long. Stress at Top of Saddles		27.80	127.78
Long. Stress at Bottom of Saddles		-15.79	-294.49
Long. Stress at Bottom of Saddles		15.79	127.78
Tangential Shear in Shell		11.05	102.22
Circ. Stress at Horn of Saddle		19.69	191.67
Circ. Compressive Stress in Shell		1.11	127.78

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 857.53 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 7284.1 \text{ lbf} \\
 &= 3304.076 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.927^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 503.0 \text{ lbf} \\
 &= 228.2 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 227625.55 \cdot 1.0 + 3304.08
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 106 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9, 2026

= 26066.6 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 227625.55 \cdot 1.0 + 228.16$

= 22990.7 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $26066.6/2 \cdot 1850.0/2300.3677$

= 10481.6 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $22991 \cdot 1850.0/11080.0$

= 3838.7 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $99879.7 \cdot 0.05$

= 4993.983 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $99880 + 4994 + \max(3839, 10482)$

= 115355.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	117546.56	Kgf
Transverse Shear Load	13033.32	Kgf
Longitudinal Shear Load	22990.71	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(2320.0/2 + 48.0 + 16.0) \sin(140.0/2), 2 \cdot 1184.0)$

= 2300.368 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 = $-115355 \cdot 4000.0 [1 - (1 - 4000.0/20864.0 + (1184.0^2 - 0.0^2) / (2 \cdot 4000.0 \cdot 20864.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 20864.0))]$
 = -84588.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 \cdot ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/3L)) - 4a/L$
 = $115355 \cdot 20864/4 \cdot ((1 + 2(1184^2 - 0^2)/(20864^2)) / (1 + (4 \cdot 0)/(3 \cdot 20864))) - 4 \cdot 4000/20864$
 = 144150.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 = $0.0 \cdot 1184.0/(2 \cdot 48.0) - 144150.3/(\pi \cdot 1184.0^2 \cdot 48.0)$
 = -6.69 N./mm²



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 107 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1184.0 / (2 \cdot 48.0) + 144150.3 / (\pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 6.69 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1184.0 / (2 \cdot 48.0) - 84588.9 / (0.1412 \cdot \pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 27.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1184.0 / (2 \cdot 48.0) + 84588.9 / (0.2486 \cdot \pi \cdot 1184.0^2 \cdot 48.0)$$

$$= -15.79 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 115355(20864.0 - 2 \cdot 4000.0) / (20864.0 + (4 \cdot 0.0 / 3))$$

$$= 71124.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9003 \cdot 71123.97 / (1184.0 \cdot 48.0)$$

$$= 11.05 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1184.0 \cdot 48.0}$$

$$= 185.948 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(735.0 + 1.56 \cdot \sqrt{1184.0 \cdot 48.0}, 2 \cdot 4000.0)$$

$$= 1106.90 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot r))$$

$$= -0.6971 \cdot 115355 \cdot 0.1 / (1106.896(48.0 + 1.0 \cdot 16.0))$$

$$= -1.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot r)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot r)^2)$$

$$= -115355 / (4(48.0 + 1.0 \cdot 16.0)1106.896) -$$

$$3 \cdot 0.038 \cdot 115355 / (2(48.0 + 1.0 \cdot 16.0)^2)$$

$$= -19.69 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 108 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1160.0 \cdot 48.0)} + 1110.0)48.0^3)/12$$

$$= 1362.222 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1362.222 + 709.491(250.659 - 24.0)^2$$

$$= 365858.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1478.1	48.0	24.0	709.5	1702779.4	0.366E+06
Wearplate	1110.0	16.0	56.0	177.6	994560.0	0.673E+05
Web	16.0	566.0	347.0	90.6	3142432.0	0.326E+05
BasePlate	760.0	60.0	660.0	456.0	30096002.0	0.765E+06
Totals	...	...	...	1433.7	35935772.0	0.123E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 35935772/1433.651$$

$$= 250.659 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140.0/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(110.0) - 0.5 \cdot \sin(110.0)^2) / (\pi - 1.92 + \sin(110.0)\cos(110.0))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 115355.281$$

$$= 27734.9688 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.7560 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1160.0 \cdot \sin(140.0/2) / 1.2217$$

$$= 957.787 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26564.7266 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.3036 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(115355 + 2191)760.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 40.164 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140.0/2) (1160.0 + 48.0 + 16.0)$$

$$= 2300.367 \text{ mm}$$

Distance between Ribs [e]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 109 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 2300.3674 / (5 - 1) \\
 &= 575.092 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 575.0919 \cdot 760.0 / 2 \\
 &= 2185.349 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 117546.555 / (2350.0 \cdot 760.0) \\
 &= 6.582 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 2185.3 \cdot 6.58 \\
 &= 14382.994 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 115.04 + 46.007 \\
 &= 161.047 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 14383.0 / 161.0473 \\
 &= 8.758 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 735.0 - 16.0 \\
 &= 719.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0.0 / 161.047 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 735.0 / 2 \\
 &= 367.500 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 52942.0 / 161.047 } \\
 &= 181.311 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1850.0 - \cos(140.0 / 2) (1160.0 + 48.0 + 16.0) - 60.0 \\
 &= 1371.367 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0 }
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 110 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9,2026

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1371.367/181.311$$

$$= 7.564$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 22990.7 / (2 \cdot 2350.0) \cdot 575.092 \cdot 1371.37 / 2$$

$$= 1928.964 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.5636 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.56)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.56) / (8 \cdot 123.21) - (7.56^3) / (8 \cdot 123.21^3))$$

$$= 153.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.76 / 153.58 + (1928.96 \cdot 367.5 / .52942E+09) / 173.33$$

$$= 0.133$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 207.055$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 735.0 / 2$$

$$= 367.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1850.0 - \cos(1224.0 / 2) (60.0 + 0.0 + 0.0) - 0.0$$

$$= 566.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (52961.1 / 207.055) }$$

$$= 159.932 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.0 / 159.932$$

$$= 3.539$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 22990.713 / (2 \cdot 2350.0)$$

$$= 4.892 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.892 \cdot 575.092 \cdot 566.0$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 111 of 214

Saddle Calcs: Road Transport: Step: 21 9:23am Sep 9, 2026

= 1592.271 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B , base, c]:

= Bending Moment / Section Modulus

= 1592.27/1441119

= 10.835 N./mm²

Compressive Allowable, $KL/r < C_c$ (3.539 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(3.54)²/(2 • 123.21²))260/

(5/3+3•(3.54)/(8• 123.21)-(3.54³)/(8•123.21³)

= 154.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 13.63/154.93 + (1592.27 • 367.5/52961.082)/173.33

= 0.150

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 112 of 214

Saddle Calcs: Operating Case:

Step: 22

9:23am

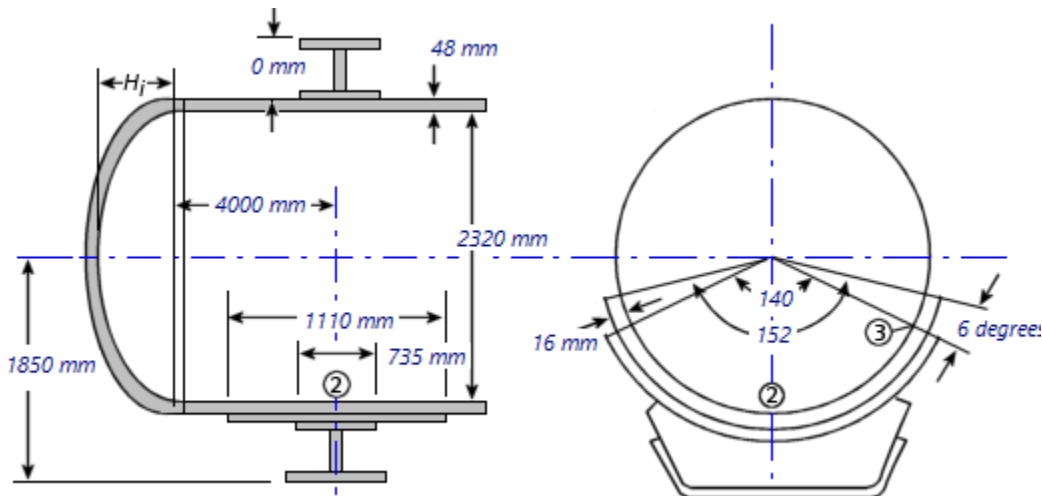
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1185.50	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		127.78	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	214887.89	Kgf
Pressure used in Analysis	P 4560.390	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	46.82	127.78
Long. Stress at Bottom of Midspan	73.33	127.78
Long. Stress at Top of Saddles	115.17	127.78
Long. Stress at Bottom of Saddles	28.79	127.78
Tangential Shear in Shell	21.93	102.22
Circ. Stress at Horn of Saddle	40.07	159.72
Circ. Compressive Stress in Shell	2.20	127.78



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 113 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3574.1/2 + 0) \cdot 1850.0/2300.3677$$

$$= 4311.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(567.29, 0) \cdot 1850.0/11080.0$$

$$= 94.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(19257.84, 0) \cdot 1850.0/11080.0$$

$$= 3215.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(19258/2 + 0) \cdot 1850.0/2300.3677$$

$$= 23231.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 191657 + \max(95, 4312, 3215, 23231)$$

$$= 214887.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.416 \cdot 1.2)^2)$$

$$= 5563.092 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	217079.16	Kgf
Transverse Shear Load	9628.92	Kgf
Longitudinal Shear Load	19257.84	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2320.0/2 + 48.0 + 16.0)\sin(140.0/2), 2 \cdot 1185.5)$$

$$= 2300.368 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -214888 \cdot 4000.0 [1 - (1 - 4000.0/20864.0 + (1185.5^2 - 0.0^2)/(2 \cdot 4000.0 \cdot 20864.0))/(1 + (4 \cdot 0.0)/(3 \cdot 20864.0))]$$

$$= -157556.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 214888 \cdot 20864/4((1 + 2(1186^2 - 0^2)/(20864^2))/(1 + (4 \cdot 0)/$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 114 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

$$(3 \cdot 20864) - 4 \cdot 4000 / 20864 \\ = 268546.6 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 4560.39 \cdot 1185.5 / (2 \cdot 45.0) - 268546.6 / (\pi \cdot 1185.5^2 \cdot 45.0) \\ = 46.82 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 4560.39 \cdot 1185.5 / (2 \cdot 45.0) + 268546.6 / (\pi \cdot 1185.5^2 \cdot 45.0) \\ = 73.33 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma_3]: \\ = P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 4560.39 \cdot 1185.5 / (2 \cdot 45.0) - 157556.9 / (0.1412 \cdot \pi \cdot 1185.5^2 \cdot 45.0) \\ = 115.17 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma_4]: \\ = P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 4560.39 \cdot 1185.5 / (2 \cdot 45.0) + 157556.9 / (0.2486 \cdot \pi \cdot 1185.5^2 \cdot 45.0) \\ = 28.79 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 214888(20864.0 - 2 \cdot 4000.0) / (20864.0 + (4 \cdot 0.0 / 3)) \\ = 132492.2 \text{ Kgf}$$

$$\text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ = K_2 \cdot T / (R_m \cdot t) \\ = 0.9003 \cdot 132492.23 / (1185.5 \cdot 45.0) \\ = 21.93 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{1185.5 \cdot 45.0} \\ = 180.157 \text{ mm}$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(735.0 + 1.56 \cdot \sqrt{1185.5 \cdot 45.0}, 2 \cdot 4000.0) \\ = 1095.31 \text{ mm}$$

$$\text{Wear Plate/Shell Stress ratio (4.15.26) } [\eta]: \\ = \min(S_r / S, 1) \\ = \min(127.779 / 127.779, 1) \\ = 1.0000$$

$$\text{Circumferential Stress at Saddle Base with Wear Plate (4.15.25) } [\sigma_6, r]: \\ = -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.6971 \cdot 214888 \cdot 0.1 / (1095.314(45.0 + 1.0 \cdot 16.0)) \\ = -2.20 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L \geq 8R_m \text{ (4.15.28) } [\sigma_7, r]: \\ = -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -214888 / (4(45.0 + 1.0 \cdot 16.0)1095.314) - \\ 3 \cdot 0.038 \cdot 214888 / (2(45.0 + 1.0 \cdot 16.0)^2) \\ = -40.07 \text{ N./mm}^2$$

$$\text{Distance between Saddle Supports } [L_s]: \\ = 11080.0 \text{ mm}$$

$$\text{Free Un-Restrained Thermal Expansion between the Saddles } [\text{Exp}]: \\ = \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000014 \cdot 11080.0 (454.0 - 21.1)$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 115 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

= 68.184 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1163.0 \cdot 45.0)} + 1110.0)45.0^3)/12$$

$$= 1113.910 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1113.91 + 660.095(256.462 - 22.5)^2$$

$$= 362439.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1466.9	45.0	22.5	660.1	1485214.9	0.362E+06
Wearplate	1110.0	16.0	53.0	177.6	941280.1	0.736E+05
Web	16.0	566.0	344.0	90.6	3115264.2	0.311E+05
BasePlate	760.0	60.0	657.0	456.0	29959200.0	0.733E+06
Totals	...	...	...	1384.3	35500960.0	0.120E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 35500960/1384.256$$

$$= 256.462 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140.0/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(110.0) - 0.5 \cdot \sin(110.0)^2) / (\pi - 1.92 + \sin(110.0)\cos(110.0))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 214887.891$$

$$= 51665.6797 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 6.9967 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1850.0 - 1163.0 \cdot \sin(140.0/2) / 1.2217$$

$$= 955.480 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 49366.4922 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 116 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Bending Stress, $S_b = (M * C_1 / I) = 10.3461 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 153.5988 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(214888 + 2191)760.0 / (4 * 2350.0 * 153.599))}$
 $= 57.981 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 140.0/2) (1160.0 + 48.0 + 16.0)$
 $= 2300.367 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2300.3674 / (5 - 1)$
 $= 575.092 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 575.0919 * 760.0/2$
 $= 2185.349 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 217079.156 / (2350.0 * 760.0)$
 $= 12.154 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 2185.3 * 12.15$
 $= 26561.799 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 115.04 + 46.007$
 $= 161.047 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 26561.8/161.0473$
 $= 16.175 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 735.0 - 16.0$
 $= 719.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0/161.047$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 117 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

= Saddle Width / 2

= 735.0/2

= 367.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{52942.0/161.047}$

= 181.311 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1850.0 - $\cos(140.0/2)$ (1160.0 + 48.0 + 16.0) - 60.0

= 1371.367 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4}$

= 130.885

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1371.367/181.311

= 7.564

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 19257.8 / (2 • 2350.0) • 575.092 • 1371.37/2

= 1615.769 Kg-m.

Compressive Allowable, $KL/r < Cc$ (7.5636 < 130.8854) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.56)^2 / (2 \cdot 130.89^2)) 230 /$

$(5/3 + 3 \cdot (7.56) / (8 \cdot 130.89) - (7.56^3) / (8 \cdot 130.89^3))$

= 136.2 N/mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$

= 16.17/136.24 + (1615.77 • 367.5 / 52942E+09) / 153.6

= 0.190

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/207.055

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 735.0/2

= 367.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1850.0 - $\cos(1224.0/2)$ (60.0 + 0.0 + 0.0) - 0.0

= 566.000 mm

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 118 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{52961.1 / 207.055}$$

$$= 159.932 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.0 / 159.932$$

$$= 3.539$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 19257.84 / (2 \cdot 2350.0)$$

$$= 4.097 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.097 \cdot 575.092 \cdot 566.0$$

$$= 1333.743 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1333.743 / 1441119$$

$$= 9.076 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.539 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.54)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (3.54) / (8 \cdot 130.89) - (3.54)^3 / (8 \cdot 130.89^3))$$

$$= 137.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 25.17 / 137.35 + (1333.74 \cdot 367.5 / 52961.082) / 153.6$$

$$= 0.242$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	7.1217 cm ²
Saddle Load Q0 (Weight)	Q0	193847.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3215.4 Kg
Maximum Transverse Force	Ft	9628.9 Kg
Maximum Longitudinal Force	F1	19257.8 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 44495.9 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 44495.9 / (7.122 \cdot 8)$$

$$= 76.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9628.9 / (7.122 \cdot 8)$$

$$= 16.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 119 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 19257.84/(82.8 • 8.0)
= 2.8511 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1850.0 • 9628.92 + 0.0
= 17813.85 Kg-m.

Eccentricity (e):
= Rmom / QO
= 17813.85/193847.88
= 91.89 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 2.8511, 0.0]
= 2.8511 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.832, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1185.50	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		127.78	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 120 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Coefficient of Friction mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 169359.78 Kgf
Pressure used in Analysis P 4560.390 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	49.63	127.78
Long. Stress at Bottom of Midspan	70.52	127.78
Long. Stress at Top of Saddles	103.49	127.78
Long. Stress at Bottom of Saddles	35.42	127.78
Tangential Shear in Shell	17.28	102.22
Circ. Stress at Horn of Saddle	31.58	159.72
Circ. Compressive Stress in Shell	1.73	127.78

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(3574.1/2 + 0) • 1850.0/2300.3677
= 4311.5 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(567.29, 0) • 1850.0/11080.0
= 94.7 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(19257.84, 0) • 1850.0/11080.0
= 3215.4 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(19258/2 + 0) • 1850.0/2300.3677
= 23231.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 146128 + max(95, 4312, 3215, 23231)
= 169359.8 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766.08(0.0 + $\pi/4$ (2.416 • 1.2)²)
= 5563.092 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	171551.05	Kgf
Transverse Shear Load	9628.92	Kgf
Longitudinal Shear Load	19257.84	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(2320.0/2 + 48.0 + 16.0)sin(140.0/2), 2•1185.5)
= 2300.368 mm

The Computed K values from Table 4.15.1:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 121 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)] \\
 &= -169360 \cdot 4000.0 [1 - (1 - 4000.0/20864.0 + (1185.5^2 - 0.0^2)/(2 \cdot 4000.0 \cdot 20864.0))/(1 + (4 \cdot 0.0)/(3 \cdot 20864.0))] \\
 &= -124175.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L] \\
 &= 169360 \cdot 20864/4 [(1 + 2(1186^2 - 0^2)/(20864^2))/(1 + (4 \cdot 0)/(3 \cdot 20864)) - 4 \cdot 4000/20864] \\
 &= 211649.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 4560.39 \cdot 1185.5/(2 \cdot 45.0) - 211649.9/(\pi \cdot 1185.5^2 \cdot 45.0) \\
 &= 49.63 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 4560.39 \cdot 1185.5/(2 \cdot 45.0) + 211649.9/(\pi \cdot 1185.5^2 \cdot 45.0) \\
 &= 70.52 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 4560.39 \cdot 1185.5/(2 \cdot 45.0) - 124175.5/(0.1412 \cdot \pi \cdot 1185.5^2 \cdot 45.0) \\
 &= 103.49 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 4560.39 \cdot 1185.5/(2 \cdot 45.0) + 124175.5/(0.2486 \cdot \pi \cdot 1185.5^2 \cdot 45.0) \\
 &= 35.42 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 169360(20864.0 - 2 \cdot 4000.0)/(20864.0 + (4 \cdot 0.0/3)) \\
 &= 104421.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 0.9003 \cdot 104421.22/(1185.5 \cdot 45.0) \\
 &= 17.28 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1185.5 \cdot 45.0)} \\
 &= 180.157 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(735.0 + 1.56 \cdot \sqrt{(1185.5 \cdot 45.0)}, 2 \cdot 4000.0) \\
 &= 1095.31 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(127.779/127.779, 1) \\
 &= 1.0000
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 122 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6971 \cdot 169360 \cdot 0.1 / (1095.314(45.0 + 1.0 \cdot 16.0))$$

$$= -1.73 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -169360 / (4(45.0 + 1.0 \cdot 16.0)1095.314) -$$

$$3 \cdot 0.038 \cdot 169360 / (2(45.0 + 1.0 \cdot 16.0)^2)$$

$$= -31.58 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1163.0 \cdot 45.0) + 1110.0})^3) / 12$$

$$= 1113.910 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1113.91 + 660.095(256.462 - 22.5)^2$$

$$= 362439.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1466.9	45.0	22.5	660.1	1485214.9	0.362E+06
Wearplate	1110.0	16.0	53.0	177.6	941280.1	0.736E+05
Web	16.0	566.0	344.0	90.6	3115264.2	0.311E+05
BasePlate	760.0	60.0	657.0	456.0	29959200.0	0.733E+06
Totals	...	...	...	1384.3	35500960.0	0.120E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 35500960 / 1384.256$$

$$= 256.462 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 140.0 / 2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(110.0) - 0.5 \cdot \sin(110.0)^2) / (\pi - 1.92 + \sin(110.0)\cos(110.0))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 169359.781$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 123 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

= 40719.3164 Kg

Tension Stress, $St = (Fh/As)$ = 5.5144 N./mm²
 Allowed Stress, $Sa = 0.6 * Yield Str$ = 138.2389 N./mm²

Saddle Splitting Dimension [d]:

= $B - R * \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $1850.0 - 1163.0 * \sin(140.0/2) / 1.2217$
 = 955.480 mm

Bending Moment, $M = Fh * d$ = 38907.2578 Kg-m.

Bending Stress, $Sb = (M * C1 / I)$ = 8.1541 N./mm²
 Allowed Stress, $Sa = 2/3 * Yield Str$ = 153.5988 N./mm²

Minimum Thickness of Baseplate per Moss:

= $\sqrt{(3 * (Q + SaddleWt) * BasePlateWidth) / (4 * BasePlateLength * AllStress)}$
 = $\sqrt{(3 * (169360 + 2191) * 760.0) / (4 * 2350.0 * 153.599)}$
 = 51.543 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 * \cos(90 - Saddle Angle/2) * (Inside Radius + Shell Thk + Wear Plate Thk)$
 = $2 * \cos(90 - 140.0/2) * (1160.0 + 48.0 + 16.0)$
 = 2300.367 mm

Distance between Ribs [e]:

= $Web Length / (Nribs - 1)$
 = $2300.3674 / (5 - 1)$
 = 575.092 mm

Baseplate Pressure Area [Ap]:

= $e * Bpwid / 2$
 = $575.0919 * 760.0 / 2$
 = 2185.349 cm²

Bearing Pressure [Bp]:

= $(Q + SaddleWt) / (BasePlateLength * BasePlateWidth)$
 = $171551.047 / (2350.0 * 760.0)$
 = 9.605 Kg/cm²

Axial Load [P]:

= $Ap * Bp$
 = $2185.3 * 9.61$
 = 20990.980 Kg

Area of the Rib and Web [Ar]:

= $Rib Area + Web Area$
 = $115.04 + 46.007$
 = 161.047 cm²

Compressive Stress [Sc]:

= P/Ar
 = $20991.0 / 161.0473$
 = 12.782 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 124 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Rib dimension [D]:

= Saddle Width - Web Thickness
= 735.0 - 16.0
= 719.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/161.047
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 735.0/2
= 367.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{52942.0/161.047}$
= 181.311 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk
= 1850.0 - $\cos(140.0/2)$ (1160.0 + 48.0 + 16.0) - 60.0
= 1371.367 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4}$
= 130.885

Slenderness ratio [KL/r]:

= KL/r
= 1 • 1371.367/181.311
= 7.564

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 19257.8/ (2 • 2350.0) • 575.092 • 1371.37/2
= 1615.769 Kg-m.

Compressive Allowable, $KL/r < Cc$ (7.5636 < 130.8854) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2/(2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r)^3/(8 \cdot Cc^3))$
= $(1 - (7.56)^2/(2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (7.56)/(8 \cdot 130.89) - (7.56^3)/(8 \cdot 130.89^3))$
= 136.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 12.78/136.24 + (1615.77 • 367.5/.52942E+09)/153.6
= 0.165

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/207.055
= 0.000 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 125 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9,2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 735.0 / 2$$

$$= 367.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1224.0/2)(60.0 + 0.0 + 0.0) - 0.0$$

$$= 566.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(52961.1 / 207.055)}$$

$$= 159.932 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.0 / 159.932$$

$$= 3.539$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 19257.84 / (2 \cdot 2350.0)$$

$$= 4.097 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.097 \cdot 575.092 \cdot 566.0$$

$$= 1333.743 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1333.743 / 1441119$$

$$= 9.076 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.539 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.54)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (3.54) / (8 \cdot 130.89) - (3.54)^3 / (8 \cdot 130.89^3))$$

$$= 137.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 19.89 / 137.35 + (1333.74 \cdot 367.5 / 52961.082) / 153.6$$

$$= 0.204$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	7.1217 cm ²
Saddle Load Q0 (Weight)	Q0	148319.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3215.4 Kg
Maximum Transverse Force	Ft	9628.9 Kg
Maximum Longitudinal Force	F1	19257.8 Kg
Saddle Bolted to Steel Foundation	No	No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 126 of 214

Saddle Calcs: Operating Case: Step: 22 9:23am Sep 9, 2026

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $19257.8 / (7.122 \cdot 8)$
 = 33.1 N./mm^2 . Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $9628.9 / (7.122 \cdot 8)$
 = 16.6 N./mm^2 . Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $19257.84 / (82.8 \cdot 8.0)$
 = 2.8511 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = $1850.0 \cdot 9628.92 + 0.0$
 = 17813.85 Kg-m .

Eccentricity (e):

= R_{mom} / QO
 = $17813.85 / 148319.75$
 = $120.10 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 2.8511, 0.0]$
 = 2.8511 cm^2

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 16.0, c = 3.0 mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.832$, Temp. Reduction = 9 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 127 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

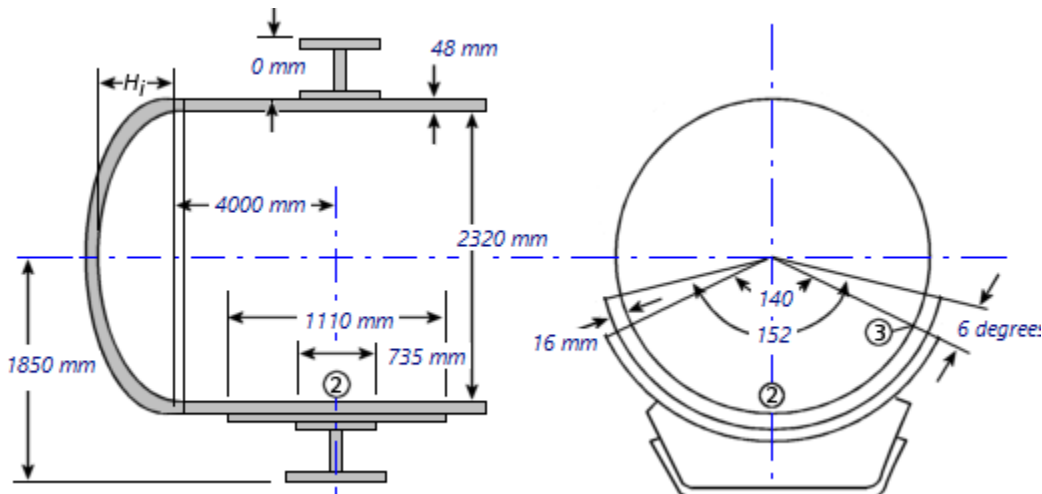
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1184.00	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		294.49	N./mm ²
Head Allowable Stress used in Calculation		294.49	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	182176.16	Kgf
Pressure used in Analysis	P 5926.605	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.54	294.49
Long. Stress at Bottom of Midspan	83.66	294.49
Long. Stress at Top of Saddles	117.00	294.49
Long. Stress at Bottom of Saddles	48.17	294.49
Tangential Shear in Shell	17.45	165.23
Circ. Stress at Horn of Saddle	31.09	441.74
Circ. Compressive Stress in Shell	1.76	294.49



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 128 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1787.1/2 + 0) \cdot 1850.0/2300.3677$$

$$= 2155.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(283.64, 0) \cdot 1850.0/11080.0$$

$$= 47.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 180020 + \max(47, 2156, 0, 0)$$

$$= 182176.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.416 \cdot 1.2)^2)$$

$$= 5563.092 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	184367.44	Kgf
Transverse Shear Load	893.53	Kgf
Longitudinal Shear Load	283.64	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2320.0/2 + 48.0 + 16.0)\sin(140.0/2), 2 \cdot 1184.0)$$

$$= 2300.368 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -182176 \cdot 4000.0 [1 - (1 - 4000.0/20864.0 + (1184.0^2 - 0.0^2)/(2 \cdot 4000.0 \cdot 20864.0))/(1 + (4 \cdot 0.0)/(3 \cdot 20864.0))]$$

$$= -133588.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 182176 \cdot 20864/4((1 + 2(1184^2 - 0^2)/(20864^2))/(1 + (4 \cdot 0)/(3 \cdot 20864)) - 4 \cdot 4000/20864)$$

$$= 227651.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 5926.605 \cdot 1184.0/(2 \cdot 48.0) - 227651.1/(\pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 62.54 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 129 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

$$= 5926.605 \cdot 1184.0 / (2 \cdot 48.0) + 227651.1 / (\pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 83.66 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5926.605 \cdot 1184.0 / (2 \cdot 48.0) - 133588.0 / (0.1412 \cdot \pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 117.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5926.605 \cdot 1184.0 / (2 \cdot 48.0) + 133588.0 / (0.2486 \cdot \pi \cdot 1184.0^2 \cdot 48.0)$$

$$= 48.17 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 182176(20864.0 - 2 \cdot 4000.0) / (20864.0 + (4 \cdot 0.0 / 3))$$

$$= 112323.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9003 \cdot 112323.34 / (1184.0 \cdot 48.0)$$

$$= 17.45 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1184.0 \cdot 48.0)}$$

$$= 185.948 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(735.0 + 1.56 \cdot \sqrt{(1184.0 \cdot 48.0)}, 2 \cdot 4000.0)$$

$$= 1106.90 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6971 \cdot 182176 \cdot 0.1 / (1106.896(48.0 + 1.0 \cdot 16.0))$$

$$= -1.76 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -182176 / (4(48.0 + 1.0 \cdot 16.0)1106.896) -$$

$$3 \cdot 0.038 \cdot 182176 / (2(48.0 + 1.0 \cdot 16.0)^2)$$

$$= -31.09 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1160.0 \cdot 48.0)} + 1110.0)^3) / 12$$

$$= 1362.222 \text{ cm}^4$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 130 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

Resolved Inertia for the Shell [I,shell]:

= ShellInertia + AShell(C1 - Y)²

= 1362.222 + 709.491(250.659 - 24.0)²

= 365858.406 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1478.1	48.0	24.0	709.5	1702779.4	0.366E+06
Wearplate	1110.0	16.0	56.0	177.6	994560.0	0.673E+05
Web	16.0	566.0	347.0	90.6	3142432.0	0.326E+05
BasePlate	760.0	60.0	660.0	456.0	30096002.0	0.765E+06
Totals	...	...	...	1433.7	35935772.0	0.123E+07

Distance to Centroid [C1]:

= AY / A

= 35935772/1433.651

= 250.659 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 140.0/2

= 110.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(110.0) - 0.5•sin(110.0)²)/(π - 1.92 + sin(110.0)cos(110.0))

= 0.2404

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.24 • 182176.156

= 43800.7695 Kgf

Tension Stress, St = (Fh/As) = 5.9317 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1850.0 - 1160.0 • sin(140.0/2)/1.2217

= 957.787 mm

Bending Moment, M = Fh * d = 41952.6523 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 8.3758 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(182176 + 2191)760.0/(4 • 2350.0 • 173.33))

= 50.301 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 140.0/2)(1160.0 + 48.0 + 16.0)

= 2300.367 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2300.3674/(5 - 1)

= 575.092 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 131 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 575.0919 \cdot 760.0 / 2$$

$$= 2185.349 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 184367.438 / (2350.0 \cdot 760.0)$$

$$= 10.323 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2185.3 \cdot 10.32$$

$$= 22559.193 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 115.04 + 46.007$$

$$= 161.047 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 22559.2 / 161.0473$$

$$= 13.737 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 735.0 - 16.0$$

$$= 719.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 161.047$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 735.0 / 2$$

$$= 367.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(52942.0 / 161.047)}$$

$$= 181.311 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1850.0 - \cos(140.0 / 2) (1160.0 + 48.0 + 16.0) - 60.0$$

$$= 1371.367 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994E+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 132 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

$$= 1 \cdot 1371.367 / 181.311$$

$$= 7.564$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 283.6 / (2 \cdot 2350.0) \cdot 575.092 \cdot 1371.37 / 2$$

$$= 23.798 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($7.5636 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.56)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.56) / (8 \cdot 123.21) - (7.56^3) / (8 \cdot 123.21^3))$$

$$= 153.6 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 13.74 / 153.58 + (23.8 \cdot 367.5 / .52942E+09) / 173.33$$

$$= 0.090$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 207.055$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 735.0 / 2$$

$$= 367.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1850.0 - \cos(1224.0 / 2) (60.0 + 0.0 + 0.0) - 0.0$$

$$= 566.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (52961.1 / 207.055) }$$

$$= 159.932 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 566.0 / 159.932$$

$$= 3.539$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 283.643 / (2 \cdot 2350.0)$$

$$= 0.060 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.06 \cdot 575.092 \cdot 566.0$$

$$= 19.644 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 133 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

$$= 19.644/1441119$$

$$= 0.134 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.539 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.54)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.54) / (8 \cdot 123.21) - (3.54^3) / (8 \cdot 123.21^3))$$

$$= 154.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 21.38/154.93 + (19.64 \cdot 367.5/52961.082) / 173.33$$

$$= 0.139$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	7.1217 cm ²
Saddle Load QO (Weight)	QO	182211.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	47.4 Kgf
Maximum Transverse Force	Ft	893.5 Kgf
Maximum Longitudinal Force	F1	283.6 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 283.6 / (7.122 \cdot 8)$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 893.5 / (7.122 \cdot 8)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 283.64 / (103.2 \cdot 8.0)$$

$$= 0.0337 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1850.0 \cdot 893.53 + 0.0$$

$$= 1653.06 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1653.06/182211.66$$

$$= 9.07 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 134 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0337, 0.0]

= 0.0337 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1184.00	mm
Shell Thickness used in this Case	t	48.000	mm
Stiffened Vessel Length per 4.15.6	L	20864.00	mm
Distance from Saddle to Vessel tangent	a1 or a	4000.00	mm
Saddle Width	b1 or b	735.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	1110.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	127.78	N./mm ²
Shell Allowable Stress used in Calculation		294.49	N./mm ²
Head Allowable Stress used in Calculation		294.49	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	143637.42	Kgf
Pressure used in Analysis	P	5926.605 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.77	294.49
Long. Stress at Bottom of Midspan	81.43	294.49
Long. Stress at Top of Saddles	107.72	294.49
Long. Stress at Bottom of Saddles	53.44	294.49
Tangential Shear in Shell	13.76	165.23
Circ. Stress at Horn of Saddle	24.51	441.74
Circ. Compressive Stress in Shell	1.39	294.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(1787.1/2 + 0) • 1850.0/2300.3677

= 2155.8 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(283.64, 0) • 1850.0/11080.0

= 47.4 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 141482 + max(47, 2156, 0, 0)



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 135 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

= 143637.4 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08(0.0 + $\pi/4$ (2.416 • 1.2)²)

= 5563.092 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 145828.70 Kgf

Transverse Shear Load 893.53 Kgf

Longitudinal Shear Load 283.64 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(2320.0/2 + 48.0 + 16.0)sin(140.0/2), 2•1184.0)

= 2300.368 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412 K2 = 0.9003 K3 = 0.5922 K4 = 0.3269

K5 = 0.6971 K6 = 0.0379 K7 = 0.0379 K8 = 0.3167

K9 = 0.2372 K10 = 0.0432 K1* = 0.2486 K6p = 0.0305

K7p = 0.0305

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -143637•4000.0[1 - (1 - 4000.0/20864.0 + (1184.0²-0.0²)/

(2•4000.0•20864.0))/(1+(4•0.0)/(3•20864.0))]

= -105327.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)

= 143637•20864/4((1+2(1184²-0²)/(20864²))/(1+(4•0)/

(3•20864)) - 4•4000/20864)

= 179492.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 5926.605 • 1184.0/(2•48.0) - 179492.3/(π •1184.0²•48.0)

= 64.77 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 5926.605 • 1184.0/(2 • 48.0) + 179492.3/(π • 1184.0² • 48.0)

= 81.43 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 5926.605•1184.0/(2•48.0)--105327.9/(0.1412• π •1184.0²•48.0)

= 107.72 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 5926.605•1184.0/(2•48.0)+-105327.9/(0.2486• π •1184.0²•48.0)

= 53.44 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 143637(20864.0 - 2 • 4000.0)/(20864.0 + (4 • 0.0/3))

= 88561.7 Kgf



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 136 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9003 \cdot 88561.73 / (1184.0 \cdot 48.0)$$

$$= 13.76 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1184.0 \cdot 48.0)}$$

$$= 185.948 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(735.0 + 1.56 \cdot \sqrt{(1184.0 \cdot 48.0)}, 2 \cdot 4000.0)$$

$$= 1106.90 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.6971 \cdot 143637 \cdot 0.1 / (1106.896(48.0 + 1.0 \cdot 16.0))$$

$$= -1.39 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -143637 / (4(48.0 + 1.0 \cdot 16.0)1106.896) -$$

$$3 \cdot 0.038 \cdot 143637 / (2(48.0 + 1.0 \cdot 16.0)^2)$$

$$= -24.51 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	760.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	720.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1160.0 \cdot 48.0)} + 1110.0)48.0^3) / 12$$

$$= 1362.222 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1362.222 + 709.491(250.659 - 24.0)^2$$

$$= 365858.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1478.1	48.0	24.0	709.5	1702779.4	0.366E+06
Wearplate	1110.0	16.0	56.0	177.6	994560.0	0.673E+05
Web	16.0	566.0	347.0	90.6	3142432.0	0.326E+05
BasePlate	760.0	60.0	660.0	456.0	30096002.0	0.765E+06
Totals	...	...	...	1433.7	35935772.0	0.123E+07

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 137 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

$$\begin{aligned}
 &= AY / A \\
 &= 35935772 / 1433.651 \\
 &= 250.659 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 140.0 / 2 \\
 &= 110.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(110.0) - 0.5 \cdot \sin(110.0)^2) / (\pi - 1.92 + \sin(110.0) \cos(110.0)) \\
 &= 0.2404
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.24 \cdot 143637.422 \\
 &= 34534.8711 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 4.6768 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1850.0 - 1160.0 \cdot \sin(140.0/2) / 1.2217 \\
 &= 957.787 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 33077.7148 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 6.6040 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(143637 + 2191) 760.0 / (4 \cdot 2350.0 \cdot 173.33) } \\
 &= 44.735 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 140.0 / 2) (1160.0 + 48.0 + 16.0) \\
 &= 2300.367 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2300.3674 / (5 - 1) \\
 &= 575.092 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 575.0919 \cdot 760.0 / 2 \\
 &= 2185.349 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 145828.688 / (2350.0 \cdot 760.0) \\
 &= 8.165 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 2185.3 \cdot 8.17 \\
 &= 17843.594 \text{ Kgf}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 138 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

= Rib Area + Web Area
= 115.04 + 46.007
= 161.047 cm²

Compressive Stress [Sc]:

= P/Ar
= 17843.6/161.0473
= 10.866 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	735.0	...	117.6	...	0.529E+05

Rib dimension [D]:

= Saddle Width - Web Thickness
= 735.0 - 16.0
= 719.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/161.047
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 735.0/2
= 367.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{52942.0/161.047}$
= 181.311 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1850.0 - cos(140.0/2)(1160.0 + 48.0 + 16.0) - 60.0
= 1371.367 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$
= 123.211

Slenderness ratio [KL/r]:

= KL/r
= 1 • 1371.367/181.311
= 7.564

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 283.6 / (2 • 2350.0) • 575.092 • 1371.37/2
= 23.798 Kg-m.

Compressive Allowable, KL/r < Cc (7.5636 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (7.56)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (7.56) / (8 \cdot 123.21) - (7.56^3) / (8 \cdot 123.21^3))$
= 153.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 139 of 214

Saddle Calcs: Test Case:

Step: 23

9:23am

Sep 9, 2026

$$= 10.87/153.58 + (23.8 \cdot 367.5/52942E+09)/173.33$$

$$= 0.072$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	719.0	0.0	115.0	0.0	0.529E+05
Web	575.1	16.0	0.0	92.0	0.0	19.6
Totals	...	...	...	207.1	...	0.530E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/207.055$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 735.0/2$$

$$= 367.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1224.0/2)(60.0 + 0.0 + 0.0) - 0.0$$

$$= 566.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{52961.1/207.055}$$

$$= 159.932 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.0/159.932$$

$$= 3.539$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 283.643/(2 \cdot 2350.0)$$

$$= 0.060 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.06 \cdot 575.092 \cdot 566.0$$

$$= 19.644 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 19.644/1441119$$

$$= 0.134 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.539 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2/(2 \cdot C_c^2))F_y/(5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$$

$$= (1 - (3.54)^2/(2 \cdot 123.21^2))260/$$

$$(5/3 + 3 \cdot (3.54)/(8 \cdot 123.21) - (3.54)^3/(8 \cdot 123.21^3))$$

$$= 154.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I)/Sba$$

$$= 16.91/154.93 + (19.64 \cdot 367.5/52961.082)/173.33$$

$$= 0.110$$

Input Data for Base Plate Bolting Calculations:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 140 of 214

Saddle Calcs: Test Case:

Step: 23 9:23am

Sep 9, 2026

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	8.6100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	7.1217 cm ²
Saddle Load QO (Weight)	QO	143672.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	47.4 Kg
Maximum Transverse Force	Ft	893.5 Kg
Maximum Longitudinal Force	Fl	283.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $283.6 / (7.122 \cdot 8)$

= 0.5 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $893.5 / (7.122 \cdot 8)$

= 1.5 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $283.64 / (103.2 \cdot 8.0)$

= 0.0337 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1850.0 \cdot 893.53 + 0.0$

= 1653.06 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $1653.06 / 143672.92$

= 11.51 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0.0, 0.0337, 0.0]$

= 0.0337 cm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 141 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026

Input, Nozzle Desc: Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	4550.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2320.00	mm
Shell Finished (Minimum) Thickness	t	48.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1754.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-182 F22	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	42.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.10	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

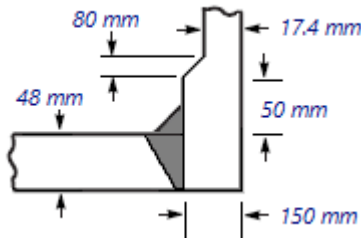
Page 142 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026



Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	22.626 in.
Actual Thickness Used in Calculation	0.687 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550.0 \cdot 1163.0 / (128 \cdot 1.0 - 0.6 \cdot 4550.0)$
 $= 42.3192 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550.0 \cdot 290.35 / (128 \cdot 1.0 - 0.6 \cdot 4550.0)$
 $= 10.5652 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 132.55 / 0.5773$
 $= 229.5838 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.4008	mm
Parallel to Vessel Wall, opening length	d	580.7004	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	112.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 127.8 / 127.8)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 127.8 / 127.8)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

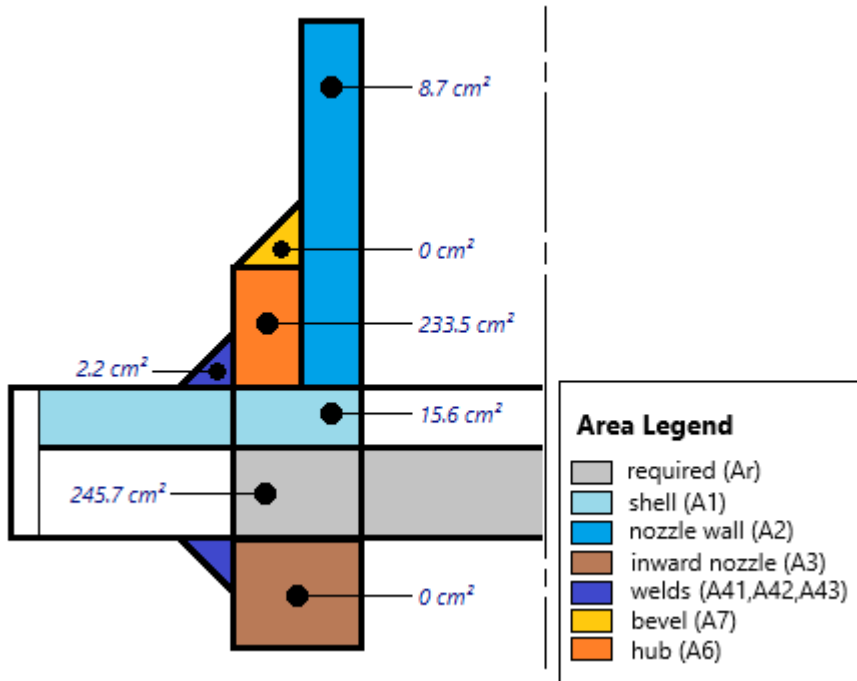
Page 143 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	245.748	NA	NA
Area in Shell	A1	15.567	NA	NA
Area in Nozzle Wall	A2	8.740	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.250	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	233.516	NA	NA
TOTAL AREA AVAILABLE	Atot	260.074	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (580.7004 \cdot 42.3192 \cdot 1.0 + 2 \cdot 147.0 \cdot 42.3192 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 245.748 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 580.7(1.0 \cdot 45.0 - 1.0 \cdot 42.319) - 2 \cdot 147.0 \\
 &\quad (1.0 \cdot 45.0 - 1.0 \cdot 42.3192) \cdot (1 - 1.0) \\
 &= 15.567 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 112.5)(14.45 - 10.57)1.0 \\
 &= 8.740 \text{ cm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 144 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 15.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 2.250 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(112.5, 400.0, 50.0)) \cdot (150.0 - 17.4) \cdot 1.0$$

$$= 233.516 \text{ cm}^2 \text{ Includes Bevel Area of } 100.966 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 13.5652 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 45.3192 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 45.3192 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(45.3192, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(13.5652, 11.3312)$$

$$= 13.5652 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	68.5,	Allowable	:	127.8 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	276.2 N./mm ²	Passed
Occasional	:	44.6,	Allowable	:	169.9 N./mm ²	Passed
Shear	:	15.0,	Allowable	:	89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 15.269$, $t_r = 9.095$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.741$, Temp. Reduction = $14 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-16 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-30 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 48.0$, $t_r = 36.429$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.81$, Temp. Reduction = $11 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $35 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $25 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $25 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 145 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11 9:23am

Sep 9, 2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4550.00/10340.00 = 0.440$

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (245.7477 - 15.5675 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 45.0 - 42.3192)) \cdot 128)$
 $= 310184.34 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can}) / 707)^2 \cdot f_r2 \cdot S_v$
 $= (242.2564 + 2.25 - 0.0 \cdot 1.0) \cdot 128$
 $= 318581.41 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (242.2564 + 0.0 + 2.25 + (132.3)) \cdot 128$
 $= 490962.69 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (242.2564 + 0.0 + 2.25 + 0.0 + (132.3)) \cdot 128$
 $= 490962.69 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 874.7004 \cdot 15.0 \cdot 0.49 \cdot 128$
 $= 131582. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 363.8502) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 128$
 $= 1532562. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 874.7004 \cdot (42.0 - 0.0) \cdot 0.74 \cdot 128$
 $= 556404. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (131582 + 1532562) = 1664144 \text{ Kgf}$

Failure Path [Path 2-2]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 146 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (131582 + 0 + 556404 + 0) = 687986 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (131582 + 556404 + 0) = 687986 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 1664144 Kg, must exceed W = 310184 Kg or W1 = 318581 Kg

Path 2-2 = 687986 Kg, must exceed W = 310184 Kg or W2 = 490962 Kg

Path 3-3 = 687986 Kg, must exceed W = 310184 Kg or W3 = 490962 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4674 KPa.

The Drop for this Nozzle is : 85.6049 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 533.6049 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2320.000	mm	
Shell Thickness	48.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 127.78	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	574.700	mm	
Nozzle Thickness	150.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 127.78	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	4550.00	KPa.
Radial Load	P	-5506.84	Kgf
Circumferential Shear	Vc	3977.16	Kgf
Longitudinal Shear	Vl	3977.16	Kgf
Circumferential Moment	Mc	7339.62	Kg-m.
Longitudinal Moment	Ml	7339.62	Kg-m.
Torsional Moment	Mt	8440.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.831) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 147 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026

$$= 874.7 + 2 \cdot 1.65 \cdot \sqrt{1185.5(48.0 - 3.0)}$$

$$= 1636.904 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2371.000 mm
 Nozzle Outside Diameter (d) = 874.700 mm
 Vessel Thickness used (T) = 45.000 mm
 Nozzle Thickness used (t) = 147.000 mm
 T / t = 0.306
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.678$$

Nr/P = 0.061
 Mr/P = 0.115
 M0/P = 0.035
 N0/P = 0.074
 MrD/Mc = 0.273
 NrDL/Mc = 0.095
 M0d/Mc = 0.082
 N0DL/Mc = 0.084
 MrD/Ml = 0.094
 NrDL/Ml = 0.065
 M0D/Ml = 0.027
 N0DL/Ml = 0.091

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	115	119	115	119
Outplane Stress Summary	112	119	133	113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	52	61	103	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 148 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11 9:23am

Sep 9,2026

Shear Stress Summary	2	2	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	112	119	133	113
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-20	20	20	-20
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	41	70	88	37
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-66	66	66	-66
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	119	115	119

Inplane Stress Summary	65	165	205	40
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	65	165	205	41
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 149 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11 9:23am

Sep 9, 2026

Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	14	8	14

Hoop Stress Summary	6	12	14	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	2	3	8	3
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	6	12	14	19
------------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	14	8	14

Hoop Stress Summary	7	12	13	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 150 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11 9:23am

Sep 9, 2026

Axial Pressure Stress	4	4	4	4
-----------------------	---	---	---	---

Axial Stress Summary	0	6	12	0
----------------------	---	---	----	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---

Shear due to (Vl)	0	0	0	0
-------------------	---	---	---	---

Shear due to Torsion	0	0	0	0
----------------------	---	---	---	---

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	7	12	14	20
------------------------	---	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	119	115	119	115	119	115	119
Circ. Pl (SUS)	-2	-2	4	4	-2	-2	4	4
Circ. Q (SUS)	-1	1	11	-11	-48	48	84	-84
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	3	3	-2	-2	4	4
Long. Q (SUS)	-4	4	40	-40	-15	15	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	115.0	119.0	115.0	119.0	115.0	119.0	115.0	119.0
Pm+Pl (SUS)	113.0	117.0	119.0	123.0	113.0	117.0	119.0	123.0
Pm+Pl+Q (Total)	112.0	118.0	130.0	112.0	65.0	165.0	203.0	39.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.00	127.78	Passed
Pm+Pl (SUS)	123.00	191.67	Passed
Pm+Pl+Q (TOTAL)	203.01	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 151 of 214

Nozzle Calcs.: Tube inlet

Nozl: 11

9:23am

Sep 9, 2026

Circ. Pm (SUS)	8	14	8	14	8	14	8	14
Circ. Pl (SUS)	-2	-2	4	4	-2	-2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	8.0	14.0	8.0	14.0	8.0	14.0	8.0	14.0
Pm+Pl (SUS)	6.0	12.0	12.0	18.0	6.0	12.0	12.0	18.0
Pm+Pl+Q (Total)	6.0	12.0	12.0	18.0	7.0	12.0	12.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	127.78	Passed
Pm+Pl (SUS)	18.00	191.67	Passed
Pm+Pl+Q (TOTAL)	19.00	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 152 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12

9:23am

Sep 9, 2026

Input, Nozzle Desc: Tube outlet

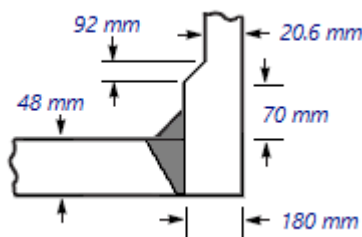
From: 30

Pressure for Reinforcement Calculations	P	4571.761	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2320.00	mm
Shell Finished (Minimum) Thickness	t	48.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1754.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6200	mm
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	92.0180	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	42.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Tube outlet



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 153 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12

9:23am

Sep 9, 2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.812 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4571.76 \cdot 1163.0 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
 $= 42.5260$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4571.76 \cdot 384.0 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
 $= 14.0413$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 159.38 / 0.5773$
 $= 276.0544$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000	mm
Parallel to Vessel Wall, opening length	d	768.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	112.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 127.8 / 127.8)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 127.8 / 127.8)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



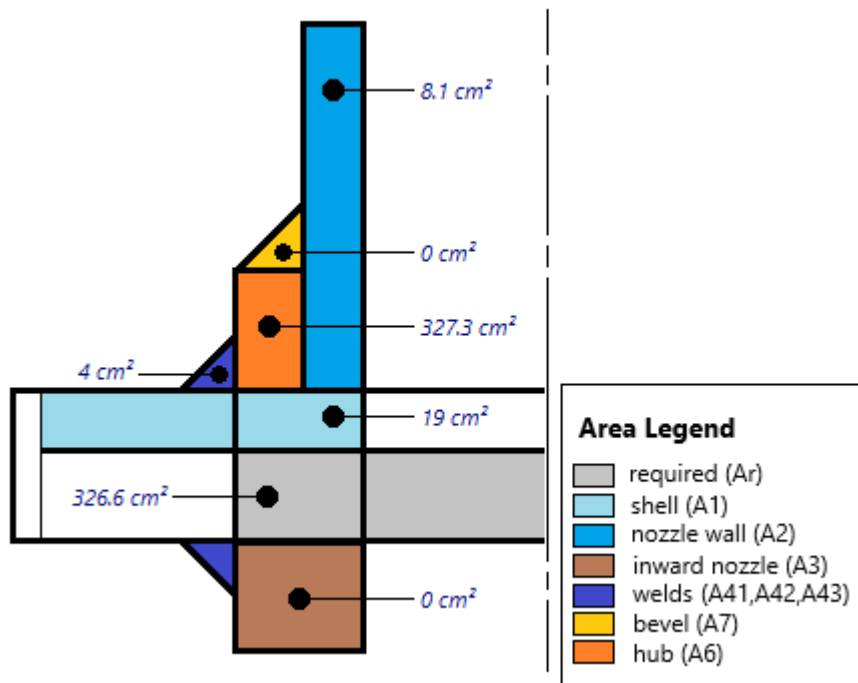
PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 154 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am Sep 9,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	326.600	NA	NA
Area in Shell	A1	19.000	NA	NA
Area in Nozzle Wall	A2	8.052	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	327.320	NA	NA
TOTAL AREA AVAILABLE	Atot	358.372	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768.0 \cdot 42.526 \cdot 1.0 + 2 \cdot 177.0 \cdot 42.526 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 326.600 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768.0(1.0 \cdot 45.0 - 1.0 \cdot 42.526) - 2 \cdot 177.0$$

$$(1.0 \cdot 45.0 - 1.0 \cdot 42.526) \cdot (1 - 1.0)$$

$$= 19.000 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 112.5)(17.62 - 14.04)1.0$$

$$= 8.052 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 155 of 214

Nozzle Calcs.: Tube outlet

Noz1: 12

9:23am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(112.5, 500.0, 70.0)) \cdot (180.0 - 20.6) \cdot 1.0$$

$$= 327.320 \text{ cm}^2 \text{ Includes Bevel Area of } 104.188 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.0413 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 45.5260 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5260 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(45.526, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.0413, 11.3312)$$

$$= 17.0413 \text{ mm}$$

Available Nozzle Neck Thickness = 20.6200 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 60.7,	Allowable	: 127.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 284.0 N./mm ²	Passed
Occasional	: 48.7,	Allowable	: 169.9 N./mm ²	Passed
Shear	: 9.4,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govern. thk, tg = 48.0, tr = 36.607, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.813, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 156 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12

9:23am

Sep 9, 2026

$$= \max(0, (326.5999 - 19.0002 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 45.0 - 42.526)) / 128)$$

$$= 412200.41 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / 707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (335.372 + 4.0 - 0.0 \cdot 1.0) \cdot 128$$

$$= 442187.25 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (335.372 + 0.0 + 4.0 + (159.3)) \cdot 128$$

$$= 649748.38 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (335.372 + 0.0 + 4.0 + 0.0 + (159.3)) \cdot 128$$

$$= 649748.38 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot 20.0 \cdot 0.49 \cdot 128$$

$$= 225045. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 472.5) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 128$$

$$= 2396366. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot (42.0 - 0.0) \cdot 0.74 \cdot 128$$

$$= 713714. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (225045 + 2396367) = 2621411 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 0 + 713714 + 0) = 938758 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 713714 + 0) = 938758 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2621411 Kgf, must exceed W = 412200 Kgf	or W1 = 442187 Kgf
Path 2-2 = 938758 Kgf, must exceed W = 412200 Kgf	or W2 = 649748 Kgf
Path 3-3 = 938758 Kgf, must exceed W = 412200 Kgf	or W3 = 649748 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4757 KPa.

The Drop for this Nozzle is : 144.6779 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 692.6779 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 157 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am Sep 9, 2026

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2320.000	mm	
Shell Thickness	48.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 127.78	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	180.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 127.78	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4571.76	KPa.
Radial Load	P	-5736.27	Kgf
Circumferential Shear	Vc	3824.19	Kgf
Longitudinal Shear	Vl	3824.19	Kgf
Circumferential Moment	Mc	8027.71	Kg-m.
Longitudinal Moment	Ml	8027.71	Kg-m.
Torsional Moment	Mt	11468.14	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.233) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 1122.0 + 2 \cdot 1.65 \cdot \sqrt{1185.5(48.0 - 3.0)}$$

$$= 1884.204 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2371.000 mm
Nozzle Outside Diameter	(d) =	1122.000 mm
Vessel Thickness used	(T) =	45.000 mm
Nozzle Thickness used	(t) =	177.000 mm
T / t	=	0.254
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.435$$

Nr/P	=	0.055
Mr/P	=	0.094
M0/P	=	0.029



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 158 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am Sep 9,2026

NØ/P = 0.063
 MrD/Mc = 0.261
 NrDL/Mc = 0.092
 MØd/Mc = 0.078
 NØDL/Mc = 0.086
 MrD/Ml = 0.078
 NrDL/Ml = 0.054
 MØD/Ml = 0.023
 NØDL/Ml = 0.071

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	115	120	115	120
Outplane Stress Summary	115	119	129	115

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-16	16	16	-16
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	57	58	93	29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	115	119	129	115

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 159 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am

Sep 9, 2026

Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	45	68	83	41
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-54	54	54	-54
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	120	115	120

Inplane Stress Summary	75	157	190	55
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	75	157	190	55
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15

Hoop Stress Summary	8	14	13	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 160 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am

Sep 9,2026

Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	4	4	7	4
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	8	14	13	19
------------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15

Hoop Stress Summary	8	13	14	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	2	6	9	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 161 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12 9:23am

Sep 9, 2026

Two * Max Shear Stress 8 13 14 19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	120	115	120	115	120	115	120
Circ. Pl (SUS)	-1	-1	3	3	-2	-2	4	4
Circ. Q (SUS)	0	0	8	-8	-39	39	69	-69
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	-1	1	31	-31	-12	12	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	115.0	120.0	115.0	120.0	115.0	120.0	115.0	120.0
Pm+Pl (SUS)	114.0	119.0	118.0	123.0	113.0	118.0	119.0	124.0
Pm+Pl+Q (Total)	114.0	119.0	126.0	115.0	74.0	157.0	188.0	55.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	120.00	127.78	Passed
Pm+Pl (SUS)	124.00	191.67	Passed
Pm+Pl+Q (TOTAL)	188.01	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	15	9	15	9	15	9	15
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	9.0	15.0	9.0	15.0	9.0	15.0	9.0	15.0
Pm+Pl (SUS)	8.0	14.0	12.0	18.0	8.0	14.0	12.0	18.0
Pm+Pl+Q (Total)	8.0	14.0	12.0	18.0	8.0	14.0	12.0	18.0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 162 of 214

Nozzle Calcs.: Tube outlet

Nozl: 12

9:23am

Sep 9, 2026

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	127.78	Passed
Pm+Pl (SUS)	18.00	191.67	Passed
Pm+Pl+Q (TOTAL)	18.00	383.34	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 163 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026

Input, Nozzle Desc: Shell inlet

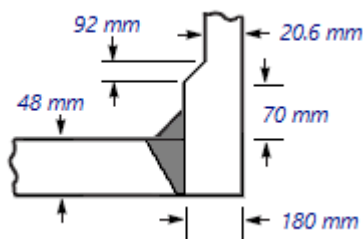
From: 40

Pressure for Reinforcement Calculations	P	4550.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2320.00	mm
Shell Finished (Minimum) Thickness	t	48.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5094.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6200	mm
Hub Height of Integral Nozzle	h	69.9820	mm
Height of Beveled Transition	L'	92.0180	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	42.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Shell inlet



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 164 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 0.812 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550.0 \cdot 1163.0 / (128 \cdot 1.0 - 0.6 \cdot 4550.0)$
= 42.3192 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550.0 \cdot 384.0 / (128 \cdot 1.0 - 0.6 \cdot 4550.0)$
= 13.9730 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 159.38/0.5773
= 276.0544 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000	mm
Parallel to Vessel Wall, opening length	d	768.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	112.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 127.8/127.8)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 127.8/127.8)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

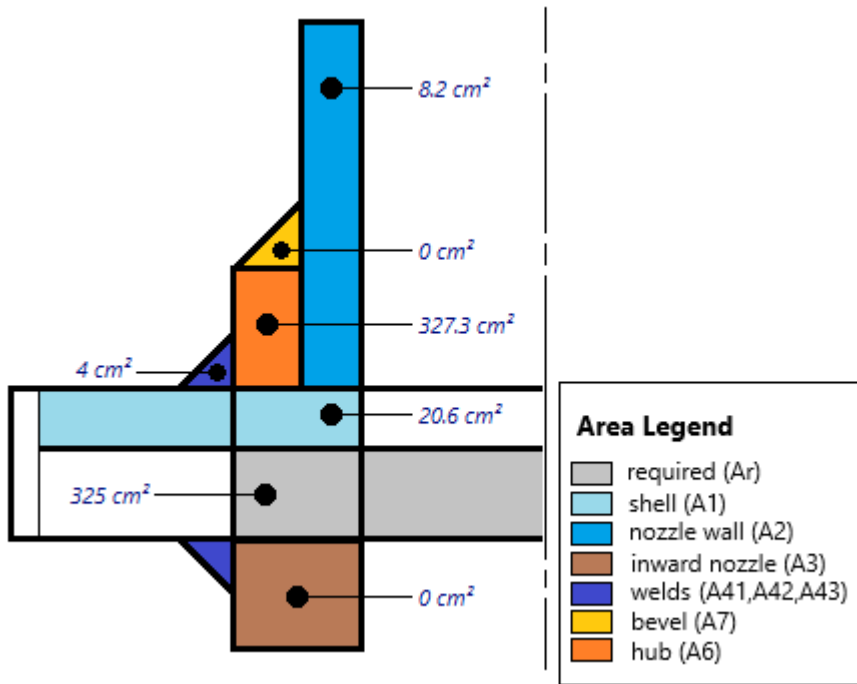
Page 165 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	325.011	NA	NA
Area in Shell	A1	20.589	NA	NA
Area in Nozzle Wall	A2	8.206	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	327.293	NA	NA
TOTAL AREA AVAILABLE	Atot	360.088	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768.0 \cdot 42.3192 \cdot 1.0 + 2 \cdot 177.0 \cdot 42.3192 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 325.011 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768.0(1.0 \cdot 45.0 - 1.0 \cdot 42.319) - 2 \cdot 177.0$$

$$(1.0 \cdot 45.0 - 1.0 \cdot 42.3192) \cdot (1 - 1.0)$$

$$= 20.589 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 112.5)(17.62 - 13.97)1.0$$

$$= 8.206 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 166 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(112.5, 500.0, 70.0)) \cdot (180.0 - 20.6) \cdot 1.0$$

$$= 327.293 \text{ cm}^2 \text{ Includes Bevel Area of } 104.219 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 16.9730 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 45.3192 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 45.3192 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(45.3192, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(16.973, 11.3312)$$

$$= 16.9730 \text{ mm}$$

Available Nozzle Neck Thickness = 20.6200 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 65.9,	Allowable	: 127.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 278.8 N./mm ²	Passed
Occasional	: 48.5,	Allowable	: 169.9 N./mm ²	Passed
Shear	: 11.2,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 48.0, tr = 36.429, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.81, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Weld Size Calculations, Description: Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 167 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026

$$= \max(0, (325.0114 - 20.5887 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 45.0 - 42.3192)) \cdot 128)$$

$$= 409015.03 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / 707)^2 \cdot \text{fr2}) \cdot \text{Sv}$$

$$= (335.4992 + 4.0 - 0.0 \cdot 1.0) \cdot 128$$

$$= 442352.97 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr1})) \cdot \text{Sv}$$

$$= (335.4992 + 0.0 + 4.0 + (159.3)) \cdot 128$$

$$= 649914.12 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr1})) \cdot \text{S}$$

$$= (335.4992 + 0.0 + 4.0 + 0.0 + (159.3)) \cdot 128$$

$$= 649914.12 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot 20.0 \cdot 0.49 \cdot 128$$

$$= 225045. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo})/4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn}$$

$$= (3.1416 \cdot 472.5) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 128$$

$$= 2396366. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgnvi} \cdot 0.74 \cdot \text{Sng}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot (42.0 - 0.0) \cdot 0.74 \cdot 128$$

$$= 713714. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (225045 + 2396367) = 2621411 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 0 + 713714 + 0) = 938758 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 713714 + 0) = 938758 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2621411 Kgf, must exceed W = 409015 Kgf or W1 = 442352 Kgf
 Path 2-2 = 938758 Kgf, must exceed W = 409015 Kgf or W2 = 649914 Kgf
 Path 3-3 = 938758 Kgf, must exceed W = 409015 Kgf or W3 = 649914 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4779 KPa.

The Drop for this Nozzle is : 144.6779 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 692.6779 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 168 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13 9:23am Sep 9, 2026

Input Echo, WRC297 Item 1, Description: Shell inlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2320.000	mm	
Shell Thickness	48.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 127.78	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	180.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 127.78	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4550.00	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.14	Kg-m.
Longitudinal Moment	Ml	11468.14	Kg-m.
Torsional Moment	Mt	13188.37	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.233) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 1122.0 + 2 \cdot 1.65 \cdot \sqrt{1185.5(48.0 - 3.0)}$$

$$= 1884.204 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2371.000 mm
Nozzle Outside Diameter	(d) =	1122.000 mm
Vessel Thickness used	(T) =	45.000 mm
Nozzle Thickness used	(t) =	177.000 mm
T / t	=	0.254
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.435$$

Nr/P	=	0.055
Mr/P	=	0.094
M0/P	=	0.029



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 169 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13 9:23am Sep 9,2026

NØ/P = 0.063
 MrD/Mc = 0.261
 NrDL/Mc = 0.092
 MØd/Mc = 0.078
 NØDL/Mc = 0.086
 MrD/Ml = 0.078
 NrDL/Ml = 0.054
 MØD/Ml = 0.023
 NØDL/Ml = 0.071

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	115	119	115	119
Outplane Stress Summary	113	119	133	112

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	52	61	103	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	113	119	133	112

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 170 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13 9:23am

Sep 9, 2026

Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	38	72	93	34
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-77	77	77	-77
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	119	115	119

Inplane Stress Summary	53	175	217	30
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	54	175	217	35
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15

Hoop Stress Summary	8	13	15	20
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 171 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13 9:23am

Sep 9, 2026

Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	3	4	7	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	8	13	15	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15
Hoop Stress Summary	7	12	15	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	0	7	10	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 172 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13 9:23am

Sep 9, 2026

Two * Max Shear Stress 7 12 15 21

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	119	115	119	115	119	115	119
Circ. Pl (SUS)	-1	-1	5	5	-3	-3	5	5
Circ. Q (SUS)	-1	1	11	-11	-59	59	95	-95
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	3	3	-2	-2	6	6
Long. Q (SUS)	-5	5	41	-41	-18	18	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	115.0	119.0	115.0	119.0	115.0	119.0	115.0	119.0
Pm+Pl (SUS)	114.0	118.0	120.0	124.0	112.0	116.0	120.0	124.0
Pm+Pl+Q (Total)	113.0	119.0	131.0	113.0	53.1	175.0	215.0	35.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.00	127.78	Passed
Pm+Pl (SUS)	124.00	191.67	Passed
Pm+Pl+Q (TOTAL)	215.01	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	15	9	15	9	15	9	15
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-4	4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	9.0	15.0	9.0	15.0	9.0	15.0	9.0	15.0
Pm+Pl (SUS)	8.0	14.0	14.0	20.0	7.0	13.0	15.0	21.0
Pm+Pl+Q (Total)	8.0	14.0	14.0	20.0	8.0	13.0	15.0	21.0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 173 of 214

Nozzle Calcs.: Shell inlet

Nozl: 13

9:23am

Sep 9, 2026

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	127.78	Passed
Pm+Pl (SUS)	21.00	191.67	Passed
Pm+Pl+Q (TOTAL)	21.00	383.34	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 174 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

Input, Nozzle Desc: N4- Shell outle

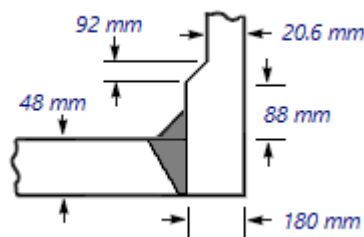
From: 40

Pressure for Reinforcement Calculations	P	4571.761	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2320.00	mm
Shell Finished (Minimum) Thickness	t	48.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		20594.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6200	mm
Hub Height of Integral Nozzle	h	87.9820	mm
Height of Beveled Transition	L'	92.0180	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	42.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4- Shell outle



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 175 of 214

Nozzle Calcs.: N4- Shell outle Nozl: 14 9:23am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 0.812 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4571.76 \cdot 1163.0 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
= 42.5260 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4571.76 \cdot 384.0 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
= 14.0413 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 159.38/0.5773
= 276.0544 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000	mm
Parallel to Vessel Wall, opening length	d	768.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	112.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 127.8/127.8)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 127.8/127.8)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

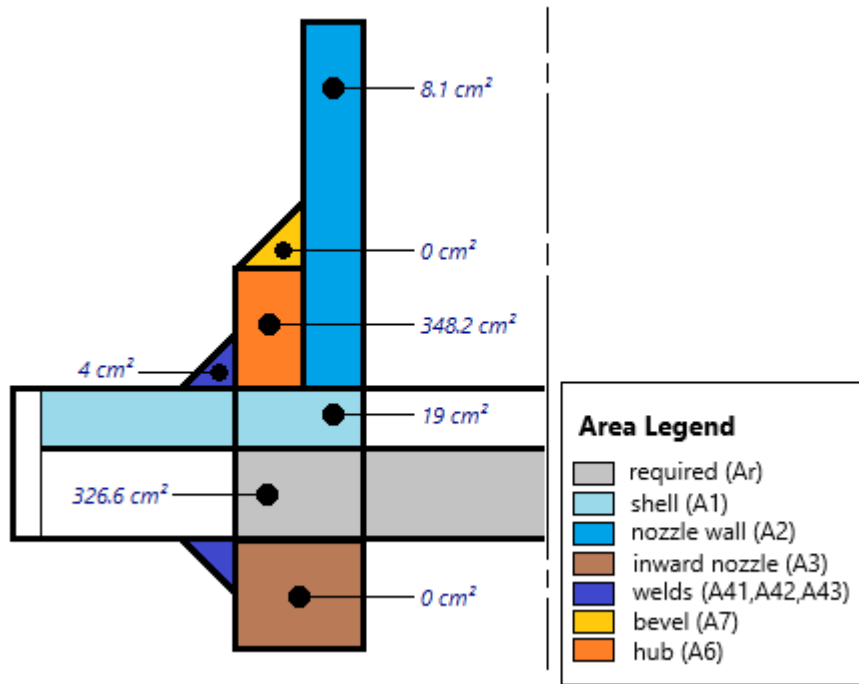
Page 176 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	326.600	NA	NA
Area in Shell	A1	19.000	NA	NA
Area in Nozzle Wall	A2	8.052	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	348.193	NA	NA
TOTAL AREA AVAILABLE	Atot	379.246	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768.0 \cdot 42.526 \cdot 1.0 + 2 \cdot 177.0 \cdot 42.526 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 326.600 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768.0(1.0 \cdot 45.0 - 1.0 \cdot 42.526) - 2 \cdot 177.0$$

$$(1.0 \cdot 45.0 - 1.0 \cdot 42.526) \cdot (1 - 1.0)$$

$$= 19.000 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 112.5)(17.62 - 14.04)1.0$$

$$= 8.052 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 177 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(112.5, 500.0, 88.0)) \cdot (180.0 - 20.6) \cdot 1.0$$

$$= 348.193 \text{ cm}^2 \text{ Includes Bevel Area of } 67.742 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.0413 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 45.5260 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5260 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(45.526, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.0413, 11.3312)$$

$$= 17.0413 \text{ mm}$$

Available Nozzle Neck Thickness = 20.6200 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 66.2,	Allowable	: 127.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 278.5 N./mm ²	Passed
Occasional	: 48.7,	Allowable	: 169.9 N./mm ²	Passed
Shear	: 11.2,	Allowable	: 89.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govern. thk, tg = 48.0, tr = 36.607, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.813, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Weld Size Calculations, Description: N4- Shell outle

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 178 of 214

Nozzle Calcs.: N4- Shell outle Nozl: 14 9:23am Sep 9, 2026

$$= \max(0, (326.5999 - 19.0002 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 45.0 - 42.526)) / 128)$$

$$= 412200.41 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr2}) \cdot S_v$$

$$= (356.2454 + 4.0 - 0.0 \cdot 1.0) \cdot 128$$

$$= 469384.28 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v$$

$$= (356.2454 + 0.0 + 4.0 + (159.3)) \cdot 128$$

$$= 676945.44 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S$$

$$= (356.2454 + 0.0 + 4.0 + 0.0 + (159.3)) \cdot 128$$

$$= 676945.44 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot 20.0 \cdot 0.49 \cdot 128$$

$$= 225045. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 472.5) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 128$$

$$= 2396366. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 1122.0 \cdot (42.0 - 0.0) \cdot 0.74 \cdot 128$$

$$= 713714. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (225045 + 2396367) = 2621411 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 0 + 713714 + 0) = 938758 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (225045 + 713714 + 0) = 938758 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2621411 Kgf, must exceed W = 412200 Kgf or W1 = 469384 Kgf
 Path 2-2 = 938758 Kgf, must exceed W = 412200 Kgf or W2 = 676945 Kgf
 Path 3-3 = 938758 Kgf, must exceed W = 412200 Kgf or W3 = 676945 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4810 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 179 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

The Drop for this Nozzle is : 144.6779 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 692.6779 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 263.87 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm

Input Echo, WRC297 Item 1, Description: N4- Shell outl:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2320.000	mm	
Shell Thickness	48.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 127.78	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	180.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 127.78	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4571.76	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.14	Kg-m.
Longitudinal Moment	Ml	11468.14	Kg-m.
Torsional Moment	Mt	13188.37	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.233) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 1122.0 + 2 \cdot 1.65 \cdot \sqrt{1185.5(48.0 - 3.0)} \\
 &= 1884.204 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2371.000	mm
Nozzle Outside Diameter	(d) =	1122.000	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	177.000	mm
T / t	=	0.254	
d / t	=	10.000	



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 180 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9,2026

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.435$$

Nr/P	=	0.055
Mr/P	=	0.094
M0/P	=	0.029
N0/P	=	0.063
MrD/Mc	=	0.261
NrDL/Mc	=	0.092
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.078
NrDL/Ml	=	0.054
M0D/Ml	=	0.023
N0DL/Ml	=	0.071

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	115	120	115	120
Outplane Stress Summary	113	119	134	113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	52	61	104	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	113	119	134	113

Vessel Stresses (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 181 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	38	73	93	35
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-77	77	77	-77
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	120	115	120

Inplane Stress Summary	54	176	218	30
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	54	176	218	36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15

Hoop Stress Summary	8	13	15	20
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 182 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	3	4	7	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	8	13	15	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	15	9	15
Hoop Stress Summary	7	13	15	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	0	7	10	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 183 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

Shear Stress Summary 0 0 0 0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7	13	16	21

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	120	115	120	115	120	115	120
Circ. Pl (SUS)	-1	-1	5	5	-3	-3	5	5
Circ. Q (SUS)	-1	1	11	-11	-59	59	95	-95
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	3	3	-2	-2	6	6
Long. Q (SUS)	-5	5	41	-41	-18	18	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	115.0	120.0	115.0	120.0	115.0	120.0	115.0	120.0
Pm+Pl (SUS)	114.0	119.0	120.0	125.0	112.0	117.0	120.0	125.0
Pm+Pl+Q (Total)	113.0	120.0	131.0	114.0	53.1	176.0	215.0	35.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	120.00	127.78	Passed
Pm+Pl (SUS)	125.00	191.67	Passed
Pm+Pl+Q (TOTAL)	215.01	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	15	9	15	9	15	9	15
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-4	4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 184 of 214

Nozzle Calcs.: N4- Shell outle

Nozl: 14

9:23am

Sep 9, 2026

Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	9.0	15.0	9.0	15.0	9.0	15.0	9.0	15.0
Pm+Pl (SUS)	8.0	14.0	14.0	20.0	7.0	13.0	15.0	21.0
Pm+Pl+Q (Total)	8.0	14.0	14.0	20.0	8.0	13.0	15.0	21.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	127.78	Passed
Pm+Pl (SUS)	21.00	191.67	Passed
Pm+Pl+Q (TOTAL)	21.00	383.34	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 185 of 214

Nozzle Calcs.: Drain

Nozl: 15

9:23am

Sep 9, 2026

Input, Nozzle Desc: Drain

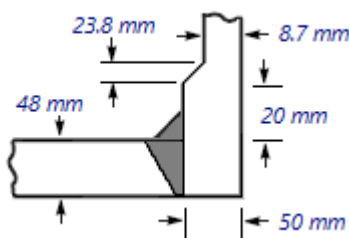
From: 40

Pressure for Reinforcement Calculations	P	4571.761	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2320.00	mm
Shell Finished (Minimum) Thickness	t	48.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4594.50	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F22	
Hub Height of Integral Nozzle	h	20.0000	mm
Height of Beveled Transition	L'	23.8228	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		42.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.





PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 186 of 214

Nozzle Calcs.: Drain

Nozl: 15

9:23am

Sep 9, 2026

Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
Actual Thickness Used in Calculation 0.344 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4571.76 \cdot 1163.0 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
= 42.5260 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4571.76 \cdot 24.42 / (128 \cdot 1.0 - 0.6 \cdot 4571.76)$
= 0.8931 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 41.262/0.5773
= 71.4686 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.3250	mm
Parallel to Vessel Wall	Rn+tn+t	75.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	58.1668	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8931	mm
Wall Thickness per UG16(b),	tr16b = 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 45.5260	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 45.5260	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000	mm
Wall Thickness per table UG-45	tb3 = 6.4200	mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[6.42, max(45.526, 5.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.8931, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm -> OK



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 187 of 214

Nozzle Calcs.: Drain

Nozl: 15 9:23am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.645, tr = 0.769, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.165$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 48.0, tr = 36.607, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.813$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.2010 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 300.2010 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 188 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	4550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4550.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	48.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2320.000 mm
Shell Temperature for Internal Pressure	Ts	454.00 °C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	127.78 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4550.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2320.000 mm
Channel Design Temperature	TEMPC	454.00 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	127.78 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3250
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	19600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18920.00 mm
Design Temperature of the Tubes		454.00 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		111.78 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.52 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	987.701 mm
Largest Center-to-Center Tube Distance	U1	31.7500 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 189 of 214

ASME TS Calc: DESIGN CASE

Case: 1

9:23a

Sep 9, 2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Integral both Sides, Conf. a
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	454.00 °C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

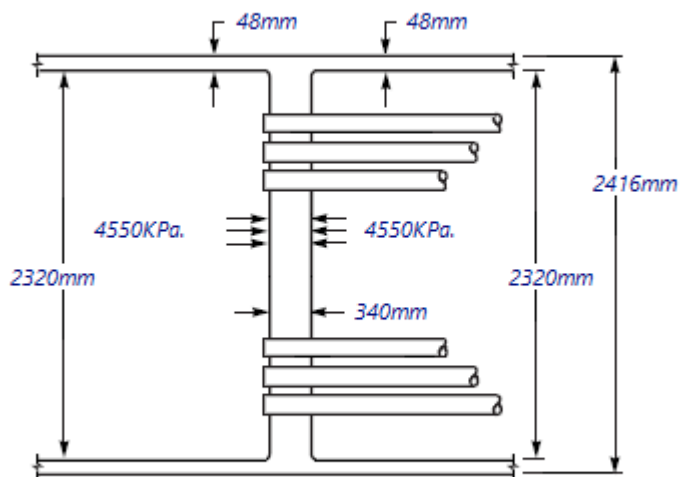
Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	127.78 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2416.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No

Tubesheet Integral w/Shell and Channel



Configuration a:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 190 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.21221E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.21221E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4550.0 \cdot (25.4/2 - 0)) / (111.78 \cdot 1.0 + 0.4 \cdot 4550.0)$$

$$= 0.5087 + 0.0000 = 0.5087 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 454.00 °C
Elastic Modulus for Material 150421952.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	18920.00	12.04	50.0000	0.0075908	68.42
EMAWP = (4*B)/(3*(D/T)) = (4 *68.4243)/(3 *12.0379) = 7578.3105 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.4372	25.40	18920.00	17.67	50.0000	0.0035219	60.32



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 191 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 60.3198) / (3 \cdot 17.673) = 4550.5435 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4372 mm
Allowable Internal Pressure at Corroded thickness	19892.05 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	7578.31 KPa.
Required External Design Pressure	4550.00 KPa.
Required Thickness due to Shell Side pressure	1.4372 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Stationary Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/340.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.17896E+09 / 1.7896E+09) \cdot (111/127) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 987.701 + 25.4$$

$$= 2000.802 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2000.802 / 2$$

$$= 1000.4011 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 2000.802 \cdot 31.75) / (\pi \cdot 2000.802^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

Ratio [ps]:

$$= as / ao$$

$$= 1160.0 / 1000.4011$$

$$= 1.1595$$

Ratio [pc]:

$$= ac / ao$$

$$= 1160.0 / 1000.4011$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 192 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

= 1.1595

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 3250 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1000.4011))^2$$

$$= 0.6358$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 3250 (25.4 / (2 \cdot 1000.4011))^2$$

$$= 0.4762$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2320.0 + 48.0) 48.0)$$

$$= 0.0054 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 0.17896E+09 \cdot 48.0^3 / (6(1 - 0.3^2))$$

$$= 19544446976.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2320.0 + 48.0) \cdot 0.19544E+11 / 340.0^3 \cdot (1 + 340.0 \cdot 0.005 + (340.0 \cdot 0.005)^2 / 2)$$

$$= 31889618.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot v_s / (2(Ds + ts)))$$

$$= 2320.0^2 / (4 \cdot 0.17896E+09 \cdot 48.0) \cdot (1 - 2320.0 \cdot 0.3 / (2(2320.0 + 48.0)))$$

$$= 0.13362E+00 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / ((Dc + tc) tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2320.0 + 48.0) 48.0)$$

$$= 0.0054 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - v_c^2))$$

$$= 0.005 \cdot 0.17896E+09 \cdot 48.0^3 / (6(1 - 0.3^2))$$

$$= 19544446976.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2320.0 + 48.0 / \cos(0)) \cdot 0.19544E+11 / 340.0^3 \cdot (1.0 + 340.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (340.0 \cdot 0.005)^2 / 2 \cdot 1.0)$$

$$= 31889618.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot E_c \cdot Tc) \cdot (1 - Dc \cdot v_c / (2(Dc + tc)))$$

$$= 2320.0^2 / (4 \cdot 0.17896E+09 \cdot 48.0) \cdot (1 - 2320.0 \cdot 0.3 / (2(2320.0 + 48.0)))$$

$$= 0.13362E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 193 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

zeta_c = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 10.708662$; $\mu^* = 0.200000$; $E^* = 27800778$ KPa.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (10.709) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= \left(\left(\frac{24(1 - \nu^{*2})N_t \cdot E_t \cdot t_t (d_t - t_t) a_o^2}{E^* \cdot L \cdot H^3} \right) \right)^{1/4}$$

$$= \left(\left(\frac{24(1 - 0.459^2)3250 \cdot 0.17896E+09 \cdot 2.11}{(25.4 - 2.11)1000.4011^2} \right) \right)^{1/4}$$

$$= 2.2629$$

Values from Table 4.18.3

$$Z_d = 0.117981 ; Z_v = 0.137663 ; Z_m = 0.586475$$

$$Z_a = 0.109820E+01 ; Z_w = 0.137663$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2416.0/2000.8021$$

$$= 1.2075$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 27800778 \cdot (31889618 + 31889618 + 0.17896E+09 \cdot \ln(1.21))$$

$$= 1.8993$$

Compute parameter [ϕ]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.4586) 1.8993$$

$$= 2.7703$$

Compute parameter [Q₁]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.1595 - 1 - 2.7703 \cdot 0.1377) / (1 + 2.7703 \cdot 0.5865)$$

$$= -0.084516$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.1595 \cdot 0.19544E+11 \cdot 0.0054 \cdot 0.13362 (1 + 340.0 \cdot 0.0054)$$

$$= 46262.9688 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1000.401^2 (1.16^2 - 1) \cdot (1.16 - 1) / 4 - 46262.969$$

$$= -32511.1309 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta + h \cdot \beta C / 2)]$$

$$= 1.16 \cdot 0.19544E+11 [0.13362 (1.0 + 0.005 \cdot 1.0 \cdot 340.0) - 0 (1.0/340.0 + 1.0 \cdot 0.005/2)]$$

$$= 46262.9688 \text{ mm}^2$$

Parameter [ω^*c]:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 194 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

$$\begin{aligned}
 &= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - wc \\
 &= 1000.40106^2 [(1.15954^2 + 1) \cdot (1.15954 - 1)/4 - (1.15954 - 1)/2] - 46262.96875 \\
 &= -32511.1309 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$Pe = Ps - Pt$$

$$= 4550.0 - 4550.0 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega * s \cdot Ps - \omega * c \cdot Pt) - (\omega s \cdot P * s - \omega c \cdot P * c)) + \\
 &\quad W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\
 &= [((-32511.131 \cdot 4550.0 - -32511.131 \cdot 4550.0) - \\
 &\quad (46262.969 \cdot 0 - 46262.969 \cdot 0)) \cdot 0. + \\
 &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 2.77028 \cdot 0.58648) \\
 &= 0.0000 \text{ Kg}
 \end{aligned}$$

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot Q2 / (\mu * (H - h'g)^2) \\
 &= 6 \cdot 0 / (0.2 \cdot (340.0 - 0)^2) \\
 &= 0.0000 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 127.78 = 191.67 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 0.0000 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 3.81 + 0 + 0 \\
 &= 3.8100 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.2 \cdot 340.0 / 1000.401) \cdot 102.223 \\
 &= 13.8968 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned}
 &= (1 / (2 \cdot \mu)) \cdot (ao/h) \cdot Pe \\
 &= (1 / (2 \cdot 0.2)) \cdot (1000.401 / 3.81) \cdot 0 \\
 &= 0.0000 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned}
 &= H + Cats + Catc \\
 &= 3.81 + 0 + 0 \\
 &= 3.8100 \text{ mm}
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(3.81; 3.81) \\
 &= 3.8100 \text{ mm}
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 195 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -1.6503, F_{tmax} = 1.805$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmin} / a_o^2) / (X_t - X_s) \\ &= ((4550.0 \cdot 0.4762 - 4550.0 \cdot 0.6358) - 2 \cdot 0 \cdot -1.65 / 1000.401^2) / (0.6358 - 0.4762) \\ &= -4.5503 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmax} / a_o^2) / (X_t - X_s) \\ &= ((4550.0 \cdot 0.4762 - 4550.0 \cdot 0.6358) - 2 \cdot 0 \cdot 1.805 / 1000.401^2) / (0.6358 - 0.4762) \\ &= -4.5503 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 4.55 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 111.7791 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as } P_e \text{ is 0.0)}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ (d_t^2 + (d_t - 2 \cdot t_t)^2) }) / 4 \\ &= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4 \\ &= 8.2680 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot E / S_{yt}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.17896E+09 / 180) } \\ &= 139.8902 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0 \cdot 0 / 8.268 \\ &= 0.0000 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ &= (180 / 1.25) \cdot (1 - 0 / (2 \cdot 139.89)) \\ &= 111.779 \text{ N./mm}^2 \text{ (Never greater than } S_{ot}) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 4.55 \cdot 1.5438 \\ &= 71.63 \text{ Kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.11) 2.11 \\ &= 1.544 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 196 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

$$= \pi \cdot 2.11 (25.4 - 2.11) 111.779$$

$$= 1759.680 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 111.78$$

$$= 1722.0270 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 111.779 / 111.779$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 111.78$$

$$= 1114.2529 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1759.68 is $\geq$ tube strength 1759.68 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.9.4 Step 10:

Note:

For a given Shell thickness of 48.0 mm, the minimum Shell length adjacent to the tubesheet should be 600.672 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1000.401^2 / ((2320.0 + 48.0) 48.0) [0 + (1.16^2 - 1) (4550.0 - 4550.0)] + 1160.0^2 \cdot 4550.0 / ((2320.0 + 48.0) 48.0)$$

$$= 53.8678 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 / (E_s \cdot t_s) \cdot P \cdot s] + 6(1 - \nu^*) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.19544 \text{E}+11 / 48.0^2 \{ 0.005 [0.13362 \cdot 4550.0 + 1160.0^2 \cdot 0 / (0.85899 \text{E}+1)] + 6(1 - 0.459) / (27800778) \cdot (1000.4 / 340.0)^3 (1 + 340.0 \cdot 0.01 / 2) \cdot [0 (0.138 + 0.586 \cdot -0.085) + 2 / 1000.401^2 \cdot 0.586 \cdot 0] \}$$

$$= 166.8665 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$:: |53.87| + |166.87| \leq 191.67 \text{ N./mm}^2$$

$$:: 220.73, \text{ must be } \leq 191.67 \text{ N./mm}^2 \text{ (See Elastic-Plastic iteration below)}$$

Note:

For a given Channel thickness of 48.0 mm, the minimum Channel length adjacent to the tubesheet should be 600.672 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 197 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= ac^2 \cdot Pt / ((Dc + tc)tc \cdot \cos(\theta))$$

$$= 1160.0^2 \cdot 4550.0 / ((2320.0 + 48.0)48.0 \cdot \cos(0))$$

$$= 53.868 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot kc/tc^2 \cdot \{ (\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt$$

$$- 6((1 - \nu^2)/E) \cdot (ao/h)^3 \cdot (Cm\theta +$$

$$h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot [Pe(Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 0.19544E+11/48.0^2 \cdot \{ (0.13362 \cdot 1.0 \cdot 0.005 - 0 \cdot 1.0) \cdot 4550.0$$

$$- 6((1 - 0.459^2)/27800778) \cdot$$

$$(1000.401/340.0)^3 \cdot (1.0 + 340.0 \cdot 0.005/2 \cdot 1.0) \cdot [0(0.13766 + 0.58648$$

$$\cdot -0.085) + 2/1000.401^2 \cdot 0.58648 \cdot 0] \}$$

$$= 166.8665 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot Sc$$

$$:: |53.87| + |166.87| \leq 191.67 \text{ N./mm}^2$$

$$:: 220.73 \text{ must be } \leq 191.67 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 <=	191.7 N./mm ²	Pass
Tubesheet Shear Stress	0.0 <=	102.2 N./mm ²	Pass
Maximum Tube Stress	4.6 <=	111.8 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-4.6 <=	-111.8 N./mm ²	Pass
Maximum Force on any one Tube	71.6 <=	1759.7 Kgf	Pass
Shell Stress (jt. inter.)	220.7 <=	191.7 N./mm ²	See below
Channel Stress (jt. inter.)	220.7 <=	191.7 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

$$= \min(\text{Sys}, \text{SPS}, s/2)$$

$$= \min(230, 461/2) = 230.9 \text{ N./mm}^2$$

Compute Allowable [S*c]:

$$= \min(\text{Syc}, \text{SPS}, c/2)$$

$$= \min(230, 461/2) = 230.9 \text{ N./mm}^2$$

Compute Factor [facts]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_s, b) / \text{Sstars}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(166.87) / 230.9], 1) = 1.0$$

Compute Factor [factc]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_c, b) / \text{Sstarc}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(166.87) / 230.9], 1) = 1.0$$

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	220.7 <=	461.8 N./mm ²	Pass
Channel Stress (jt. inter.)	220.7 <=	461.8 N./mm ²	Pass



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 198 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9,2026

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	340.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	111.327 ...	142	192	33 102	Ps+Pt-Th D1	Ok
D2uc	111.327 ...	142	192	33 102	Ps+Pt-Th D2	Ok
D3uc	3.810 ...	...	192	... 102	Ps+Pt-Th D3	Ok
D4uc	3.810 ...	...	192	... 102	Ps+Pt-Th D4	Ok
D1c	117.327 ...	149	192	34 102	Ps+Pt-Th-c D1	Ok
D2c	117.327 ...	149	192	34 102	Ps+Pt-Th-c D2	Ok
D3c	9.810 ...	...	192	... 102	Ps+Pt-Th-c D3	Ok
D4c	9.810 ...	...	192	... 102	Ps+Pt-Th-c D4	Ok
Max:	117.3273 ... mm	0.777	0.333	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(-+)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	Ten	---- Tube Stresses Allwd Cmp Allwd	Tube Loads Ld Allwd	Shell Stress Stress Allwd	Channel Stress Stress Allwd	Pass Fail
D1uc	20	111 -2 -90	320 1759	142 461	363 461	Ok
D2uc	25	111 -25 -90	392 1759	363 461	142 461	Ok
D3uc	5	111 -5 -111	72 1759	221 461	221 461	Ok
D4uc	...	111	... 1759	... 191	... 191	Ok
D1c	21	111 -3 -90	335 1759	154 461	391 461	Ok
D2c	26	111 -26 -90	406 1759	391 461	154 461	Ok
D3c	5	111 -5 -111	72 1759	236 461	236 461	Ok
D4c	...	111	... 1759	... 191	... 191	Ok



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 199 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

Max RATIO 0.231 0.286 0.231 0.846 0.846

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	117.327	340.000	Ok
Tube thickness :	1.437	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (48.000) adj. to tubesheet: 600.672 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 600.672 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5790.34	1.000	5790.33	1.000
Tubesheet Shear Stress	13650.72	1.000	13650.72	1.000
Tube Tensile Stress	23929.49	1.000	19709.77	1.000
Tube Compressive Stress	122842.95	1.000	15915.45	1.000
Tube-Tubesheet Joint load	23929.42	1.000	19709.75	1.000
Shell Stress (Axial, Junction)	5790.34	1.000	5790.33	1.076
Tubesheet-Channel Junction	5790.34	1.076	5790.33	1.000
Tube Pressure Stress	19892.04	1.000	7578.30	1.000
Minimum MAWP	5790.34		5790.33	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7065.17	0.999	7070.31	1.000
Tubesheet Shear Stress	16094.72	1.000	16094.71	1.000
Tube Tensile Stress	26374.89	1.000	21556.34	1.000
Tube Compressive Stress	195626.67	1.000	18724.79	1.000
Tube-Tubesheet Joint load	26374.88	1.000	21556.34	1.000
Shell Stress (Axial, Junction)	7065.17	0.999	7070.31	1.000
Tubesheet-Channel Junction	7065.17	0.999	7070.31	1.000
Tube Pressure Stress	20998.73	1.000	13593.03	1.000
Minimum MAPnc	7065.17		7070.31	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(340.0/4, \min(340.0, 48.0)) \\
 &= 85.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(4550.0/7070.31, 1)
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 200 of 214

ASME TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

= 0.644

Thickness Ratio = 0.644, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	26 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(340.0/4, min(340.0, 48.0))
= 85.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(4550.0/7065.17, 1)
= 0.644

Thickness Ratio = 0.644, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	26 °C

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 201 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9,2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	4550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4550.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	48.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2320.000 mm
Shell Temperature for Internal Pressure	Ts	454.00 °C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	127.78 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4550.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2320.000 mm
Channel Design Temperature	TEMPC	454.00 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	127.78 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3250
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	19600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18920.00 mm
Design Temperature of the Tubes		454.00 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		111.78 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.52 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	987.701 mm
Largest Center-to-Center Tube Distance	U1	31.7500 mm



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 202 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1

9:23a

Sep 9, 2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	454.00 °C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	127.78 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2300.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2300.000 mm
Flange Inside Diameter	B	2014.000 mm
Flange Face Outside Diameter	Fod	2049.000 mm
Flange Face Inside Diameter	Fid	2014.000 mm
Gasket Outside Diameter	Go	2046.000 mm
Gasket Inside Diameter	Gi	2020.000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.0000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2144.000 mm
Nominal Bolt Diameter	dB	76.2000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	113.19 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	324.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes



PV Elite 28 Licensee: SL User

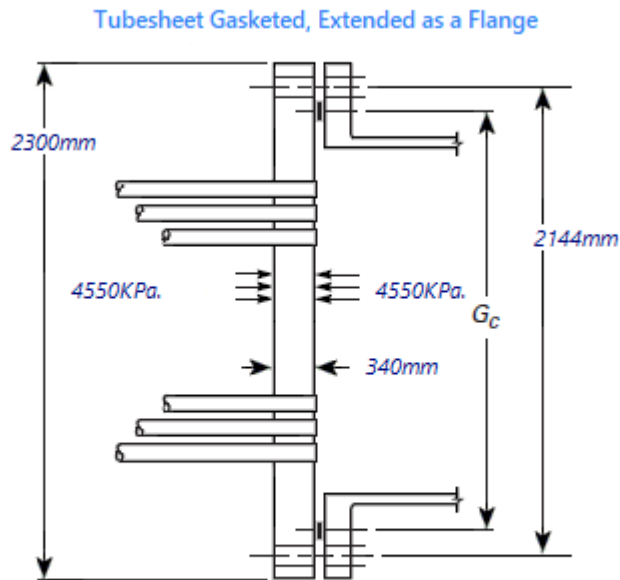
FileName : 114-C-Pass

Page 203 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9,2026

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1

Shell - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.21221E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1

Channel - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1

Tubes - TM-1 Material Group D

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.21221E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 204 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature 454.0 °C 0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.21097E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(4550.0 \cdot (25.4/2 - 0)) / (111.78 \cdot 1.0 + 0.4 \cdot 4550.0)$
= $0.5087 + 0.0000 = 0.5087$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 454.00 °C
Elastic Modulus for Material 150421952.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	18920.00	12.04	50.0000	0.0075908	68.42

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 68.4243) / (3 \cdot 12.0379) = 7578.3105$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.4372	25.40	18920.00	17.67	50.0000	0.0035219	60.32

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 60.3198) / (3 \cdot 17.673) = 4550.5435$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4372 mm
Allowable Internal Pressure at Corroded thickness	19892.05 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	7578.31 KPa.
Required External Design Pressure	4550.00 KPa.
Required Thickness due to Shell Side pressure	1.4372 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2033.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:
= ltx / h (modified for corrosion if present)
= $0/340.0$
= 0.0000 (must be $0 \leq \rho \leq 1$)



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 205 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt (EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.17896E+09 / 0.17896E+09) \cdot (111/127) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 987.701 + 25.4$$

$$= 2000.802 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2000.802 / 2$$

$$= 1000.4011 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 31.75 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 2000.802 \cdot 31.75) / (\pi \cdot 2000.802^2) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ *]:

$$= (p* - d*) / p*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

Ratio [ps]:

$$= as / ao$$

$$= 1016.5778 / 1000.4011$$

$$= 1.0162$$

Ratio [pc]:

$$= ac / ao$$

$$= 1016.5778 / 1000.4011$$

$$= 1.0162$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 3250 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1000.4011))^2$$

$$= 0.6358$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 3250 (25.4 / (2 \cdot 1000.4011))^2$$

$$= 0.4762$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 206 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 10.708662$; $\mu^* = 0.200000$; $E^* = 27800778 \text{ KPa}$.
 $E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (10.709) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.459^2)3250 \cdot 0.17896E+09 \cdot 2.11 \right. \right. \\ \left. \left. (25.4 - 2.11)1000.4011^2 \right) / (27800778 \cdot 18920.0 \cdot 340.0^3) \right)^{1/4}$$

$$= 2.2629$$

Values from Table 4.18.3

$$Z_d = 0.117981; \quad Z_v = 0.137663; \quad Z_m = 0.586475$$

$$Z_a = 0.109820E+01; \quad Z_w = 0.137663$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$= A / Do$$

$$= 2300.0/2000.8021$$

$$= 1.1495$$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 27800778 \cdot (0 + 0 + 0.17896E+09 \cdot \ln(1.15))$$

$$= 0.4857$$

Compute parameter $[\phi]$:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4586) \cdot 0.4857$$

$$= 0.7084$$

Compute parameter $[Q_1]$:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0162 - 1 - 0.7084 \cdot 0.1377) / (1 + 0.7084 \cdot 0.5865)$$

$$= -0.057475$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$= (G_c - C) / Do \text{ (config b)}$$

$$= (2033.1555 - 2144.0) / 2000.8021 = -0.0554$$

Calculate parameter $[\omega_s]$:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.0162 \cdot 0 \cdot 0 \cdot 0 (1 + 340.0 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1000.401^2(1.016^2 - 1) \cdot (1.016 - 1) / 4 - 0$$

$$= 131.9013 \text{ mm}^2$$

Parameter $[\omega_c]$:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.016 \cdot 0 [0 (1.0 + 0 \cdot 1.0 \cdot 340.0) - \\ 0 (1.0/340.0 + 1.0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter $[\omega^*c]$:

$$= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 207 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

$$= 1000.40106^2 [(1.01617^2 + 1) \cdot (1.01617 - 1)/4 - (1.01617 - 1)/2] - 0$$

$$= 131.9009 \text{ mm}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$

= 4550.0 - 4550.0 = 0 KPa.

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((131.901 \cdot 4550.0 - 131.901 \cdot 4550.0) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + 2381345 \cdot -0.055 / (2 \cdot \pi)] / (1 + 0.70844 \cdot 0.58648)$$

$$= -14833.6572 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 \cdot Q2 / (\mu * (H - h'g)^2)$$

$$= 6 \cdot -14833.657 / (0.2 \cdot (340.0 - 0)^2)$$

$$= -37.7520 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 127.78 = 191.67 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 6 \cdot Q2 / (\mu * (H - h'g)^2)$$

$$= 6 \cdot -16737.005 / (0.2 \cdot (160.289 - 0)^2)$$

$$= -191.6555 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 160.2888 + 0 + 0$$

$$= 160.2888 \text{ mm}$$

$$H_{reqd.} = H_{ReqB} = 160.2888 \text{ mm}$$

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	37.8 <=	191.7 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	329.745	84.521	185	192	Ps+Pt-Th	D1 Ok
D2uc	285.524	...	161	192	Ps+Pt-Th	D2 Ok
D3uc	160.289	...	38	192	Ps+Pt-Th	D3 Ok
D1c	335.688	84.521	189	192	Ps+Pt-Th-c	D1 Ok



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 208 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

D2c	291.429	...	164	192	...	...	Ps+Pt-Th-c D2	Ok
D3c	166.291	...	39	192	...	...	Ps+Pt-Th-c D3	Ok

Max: 335.6881 84.521 mm 0.986 ... (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	335.688	340.000	Ok
Tubesheet thickness flanged extension :	84.521	324.000	Ok
Tube thickness :	1.437	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	4805.87	1.000	5316.93	1.000
Tube Pressure Stress	19892.04	1.000	7578.30	1.000
Tubesheet Extension Stress	4624.08	...	No Calc	No Calc
Minimum MAWP	4624.08		5316.93	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	5682.47	1.000	6290.28	1.000
Tube Pressure Stress	20998.73	1.000	13593.03	1.000
Tubesheet Extension Stress	6495.17	...	No Calc	No Calc
Minimum MAPnc	5682.47		6290.28	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 209 of 214

ASME Fl-TS Calc: DESIGN CASE

Case: 1 9:23a Sep 9, 2026

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4

= 340.0/4

= 85.000 mm

Thickness Ratio = 0.801, Temperature Reduction per Fig. UCS 66.1 = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 28 °C

Min Metal Temp. at Required thickness (UCS 66.1) 17 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(p_t /Tubeside MAPnc, p_s /Shellside MAPnc), must be ≤ 1

= max(4550.0/5682.47, 4550.0/6290.28)

= 0.801

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 210 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	4550.00	KPa.
Shell Side (External) Design Pressure	Pss	4550.00	KPa.
Design Temperature for Spherical Head	Temp	454.00	°C
Head Material		SA-387 22	
Head Material UNS Number		K21590	
Head Allowable Stress at Temperature	Sho	127.78	N./mm ²
Head Allowable Stress at Ambient	Sha	148.00	N./mm ²
Crown Radius for Spherical Head	L	1812.000	mm
Head Thickness	Th	137.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	3.0000	mm
Shell Side (External) Corrosion Allowance	Cass	3.0000	mm
Flange Material		SA-336 F22	
Flange Material UNS Number		K21590	
Flange Allowable Stress at Temperature	Sfo	127.78	N./mm ²
Flange Allowable Stress at Ambient	Sfa	148.00	N./mm ²
Flange Outside Diameter	Fod	2300.000	mm
Flange Inside Diameter	Fid	2014.000	mm
Flange Thickness	Tf	327.0000	mm
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	113.19	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2144.000	mm
Nominal Bolt Diameter	dB	76.2000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	

Full face Gasket Flange Option Program Selects

Flange Face Outside Diameter	Ffod	2049.000	mm
Flange Face Inside Diameter	Ffid	2014.000	mm
Gasket Outside Diameter	Go	2046.000	mm
Gasket Inside Diameter	Gi	2020.000	mm
Gasket Factor,	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		3.0000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		15.8414	mm

The Flange is not Slotted.

This unit does not have a Backing Ring.

Use Full Bolt Load in Flange Design Yes



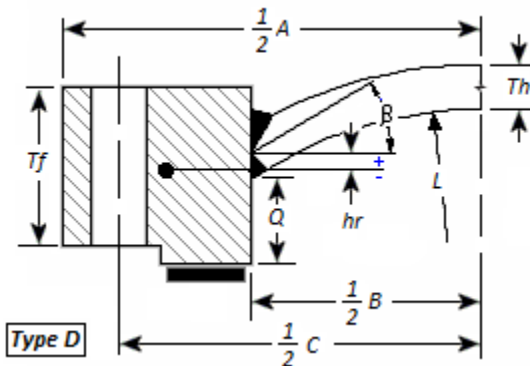
PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 211 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:

$$\begin{aligned} &= 5PL / 6Sho \\ &= (5 \cdot 4550.0 \cdot 1815.0) / (6 \cdot 127) \\ &= 53.8608 \text{ mm} \end{aligned}$$

Maximum Allowable Working Pressure at Given Thickness[Pa]:

$$\begin{aligned} &= 6Sho \cdot (Th - Cass - Cats) / 5L \\ &= (6 \cdot 127 \cdot 131.0) / (5 \cdot 1815.0) \\ &= 11066.48 \text{ KPa.} \end{aligned}$$

Maximum Allowable Pressure, New and Cold[Pnc]:

$$\begin{aligned} &= 6Sha / 5L \\ &= (6 \cdot 147 \cdot 137.0) / (5 \cdot 1812.0) \\ &= 13426.79 \text{ KPa.} \end{aligned}$$

Actual Stress at given Pressure and Thickness[Sact]:

$$\begin{aligned} &= 5PL / 6(Th - Cass - Cats) \\ &= (5 \cdot 4550.0 \cdot 1815.0) / (6 \cdot 131.0) \\ &= 53. \text{ N./mm}^2 \end{aligned}$$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 5.464 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart CS-2 at 454.00 °C
Elastic Modulus for Material 150421952.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	131.0000	mm
Outside Crown Radius of Shell	OD/2	1946.0000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	14.8550	
Geometry Factor, A	f(D/T, LD)	0.0084147	
Materials Factor, B	f(A, Chart)	69.4578	N./mm ²
Maximum Allowable Working Pressure		4675.46	KPa.
$EMAWP = B / ((OD/T)/2) = 69.4578 / (29.7099/2.0) = 4675.4614$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	127.9357	mm
Outside Diameter of Shell	ODCA	3892.000	mm
Diameter / Thickness Ratio	(D/T)	30.4215	
Geometry Factor, A	f(DT, LD)	0.0082179	
Materials Factor, B	f(A, Chart)	69.2215	N./mm ²
Maximum Allowable Working Pressure		4550.56	KPa.
$EMAWP = B / ((D/T)/2) = 69.2215 / (30.4215/2) = 4550.561$			



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 212 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width, $N = (Goc - Gic) / 2$ 13.000 mm
 Basic Gasket Width, $b0 = N / 2$ 6.500 mm
 Effective Gasket Width, $b = \text{sqrt}(b0) * 2.5$ 6.422 mm
 Gasket Reaction Diameter, $G = Go - 2.0 * b$ 2033.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 * 76.2 + 6 * 327.0 / (4.25 + 0.5)$$

$$= 565.453 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C * \sin(\pi / n)$$

$$= 2144.0 * \sin(\pi / 36)$$

$$= 186.862 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{186.862 / (2 * 76.2 + 327.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

Total Area of Bolts 1468.797 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	73.025	75.000	
Circ. Spacing between the Bolts:	158.750	186.862	565.453

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1506395.5	Kgf
Contact Load on Gasket Surfaces	Hp	161783.2	Kgf
Hydrostatic End Load at Flange ID	Hd	1486964.4	Kgf
Pressure Force on Flange Face	Ht	19431.1	Kgf
Radial Component of Head Membrane Force	Hr	2332665.5	Kgf
Operating Bolt Load:	Wm1	1668178.8	Kgf
Gasket Seating Bolt Load	Wm2	291295.4	Kgf
Required Bolt Area	Am	1445.264	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= Ab * Sb / (y * PI * (Go + Gi))$$

$$= 1468.797 * 159.0 / (69.64 * 3.14 * (2046.0 + 2020.0))$$

$$= 26.253 \text{ mm} \text{ [Note: Exceeds actual gasket width, 13.000]}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 38.100 \text{ mm} \text{ [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load (Seating)	W	2381345.5	Kgf
Gasket Seating Force (Operating)	Hg	161783.2	Kgf
Distance to Gasket Load Reaction	hg	55.4222	mm
Distance to Face Pressure Reaction	ht	58.7111	mm
Distance to End Pressure Reaction	hd	62.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	1486964.	62.0000	1.0000
				92194. Kg-m.



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 213 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

Face Pressure,	Mt	19431.	58.7111	1.0000	1141. Kg-m.
Gasket Load,	Mg	161783.	55.4222	1.0000	8967. Kg-m.
Floating Hd. Load,	Mh	2332666.	15.8414	1.0000	-36953. Kg-m.
Gasket Seating,	Ma	2381346.	55.4222	1.0000	131982. Kg-m.

Total Moment for Operation (Internal Pressure) 65347.590 Kg-m.

Total Moment for Gasket Seating (Int. Pressure) 131982.016 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1506395.5	Kgf
Hydrostatic End Load at Flange ID	Hd	1486964.4	Kgf
Pressure Force on Flange Face	Ht	19431.1	Kgf
Radial Component of Head Membrane Force	Hr	2332665.5	Kgf
Gasket Seating Bolt Load	Wm2	291295.4	Kgf
Required Bolt Area	Am	179.669	cm ²

Flange Design Bolt Load (Seating)	W	2381345.5	Kgf
Distance to Gasket Load Reaction	hg	55.4222	mm
Distance to Face Pressure Reaction	ht	58.7111	mm
Distance to End Pressure Reaction	hd	62.0000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment	
End Pressure,	Md	1486964.	6.5778	1.0000	9781. Kg-m.
Face Pressure,	Mt	19431.	3.2889	1.0000	64. Kg-m.
Floating Hd. Load,	Mh	2332666.	15.8414	1.0000	-36953. Kg-m.
Gasket Seating,	Ma	2381346.	55.4222	1.0000	131982. Kg-m.

Total Moment for Operation (External Pressure) 27108.369 Kg-m.

Total Moment for Gasket Seating (Ext. Pressure) 131982.016 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 4550.0 \cdot 2020.0 \cdot \sqrt{(4 \cdot 1815.0 \cdot 1815.0 - 2020.0 \cdot 2020.0) / (8 \cdot 127.779(2294.0 - 2020.0))}$$

$$= 98.9750 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 65347(2294.0 + 2020.0) / (127.779 \cdot 2020.0)(2294.0 - 2020.0)$$

$$= 39090.2852 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 98.975 + \sqrt{(98.975 \cdot 98.975 + 39090.285)}$$

$$= 320.0776 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 131982(2294.0 + 2020.0) / (147.997 \cdot 2020.0)(2294.0 - 2020.0)$$

$$= 68164.6172 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 0 + \sqrt{(0 \cdot 0 + 68164.617)}$$

$$= 261.0835 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 4550.0 \cdot 2020.0 \cdot \sqrt{(4 \cdot 1815.0 \cdot 1815.0 - 2020.0 \cdot 2020.0) / (8 \cdot 127.779(2294.0 - 2020.0))}$$



PV Elite 28 Licensee: SL User

FileName : 114-C-Pass

Page 214 of 214

Floating Head: Operating

Case: 1 9:23a Sep 9, 2026

= 98.9750 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $27108(2294.0+2020.0)/(127.779 \cdot 2020.0)(2294.0-2020.0)$

= 16215.9570 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $98.975 + \sqrt{(98.975 \cdot 98.975 + 16215.957)}$

= 260.2574 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $131982(2294.0+2020.0)/(147.997 \cdot 2020.0)(2294.0-2020.0)$

= 68164.6172 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 68164.617)}$

= 261.0835 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	53.8608	320.0776
Shellside (External) Pressure	127.9357	260.2574
Tubeside Gasket Seating Load		261.0835
Shellside Gasket Seating Load		261.0835
Maximum + Corrosion Allowance	133.9357	326.0776
Actual Thickness as Given	137.0000	327.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	4579.78	5518.90	5858.81	7549.51
Floating head Stress	11066.38	13426.69	4675.36	8626.46
Bolt Stress	4623.99	6495.08	...	...
Minimum MAWP & MAPnc	4579.78	5518.90	4675.36	7549.51

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: